```
{
"data":{
"id":"1",
"type":"Ticket",
"attributes":{
"subject":"Problem with Ticket",
"description":"Hello,\r\n\r\nI have a problem with my ticket.\r\n\r\nCan you please check it?\r\n\r\nThanks\r\n\r\n\r\n
```

Mr. Gabriel Sanguinetti
Director de Cuentas Corrientes Tarjeta Shopping

Mr. Gabriel Sanguinetti

Mr. Gabriel Sanguinetti
Gerente
Tarjeta Shopping
Mr.
Mr.

Mr. Gabriel Sanguinetti

Mr.
Mr. Gabriel Sanguinetti
Mr.
Mr.
Mr.

```
{    "data": [{
        "_id": "5f1d0f0f0f0f0f0f0f0f0f0f",
        "name": "Aaron Sutton",
        "gender": "male",
        "company": "ZILCHAR",
        "email": "aaronsutton@zilchar.com",
        "phone": "+1 (940) 564-3437",
        "address": "Ivy Place, 1 Ivy Place, Ivy Place, Ivy Place,
```

```
{"id":"0000-00-00T00:00:00.000+00:00",
"_id":"0000-00-00",
"_rev":"1-00000000000000000000000000000000",
"_attachments":{},
"_conflicts":["0000-00-00T00:00:00.000+00:00"],
"_deleted_conflicts":["0000-00-00T00:00:00.000+00:00"],
"_local_seq":"0000-00-00T00:00:00.000+00:00",
"_revisions":{"start":1,"ids":["00000000000000000000000000000000"]},
"_revs_info":[{"rev":"1-00000000000000000000000000000000","status":"available"}],
"_rev_tree":[["1-00000000000000000000000000000000",{"status":"available","children":{}}]],
"_id_conflicts":["0000-00-00T00:00:00.000+00:00"],
"_rev_conflicts":["1-00000000000000000000000000000000"],
"_id_deleted_conflicts":["0000-00-00T00:00:00.000+00:00"],
"_rev_deleted_conflicts":["1-00000000000000000000000000000000"],
"_local_seq_conflicts":["0000-00-00T00:00:00.000+00:00"],
"_local_seq_deleted_conflicts":["0000-00-00T00:00:00.000+00:00"],
"_attachment_stubs":{"0000-00-00":{"content_type":"application/octet-stream","digest":"md5-00000000000000000000000000000000","length":0,"revpos":1,"stub":true}},
"_attachment_data":{"0000-00-00":"AAAA
```

```
BEGIN:VCARD
VERSION:3.0
N:;Ruth Ann;;;
FN:Ruth Ann
ORG;CHARset=UTF-8;PHONETIC=katakana:Tsuchi Corporation
TITLE:President
TEL;TYPE=WORK,VOICE:
TEL;TYPE=CELL,VOICE:
END:VCARD

BEGIN:VCARD
VERSION:3.0
N:Lopez;Jose Luis;;;
FN:Jose Luis Lopez
ORG;CHARset=UTF-8;PHONETIC=katakana:Tsuchi Corporation
TITLE:Sales Manager
```

Chapter 1 Sample Chapter

1.1 Sample Section

1.1.1 Sample Subsection

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nunc ac ante ut ante ultrices ultrices. Nunc ac ante ut ante ultrices ultrices. Nunc ac ante ut ante ultrices ultrices

```
<!-- ~~~
```

```
<div class="page">

<div>
<span id="1" />
<span id="2" />
<span id="3" />
<span id="4" />
<span id="5" />
<span id="6" />
<span id="7" />
<span id="8" />
<span id="9" />
<span id="10" />
<span id="11" />
<span id="12" />
<span id="13" />
```

```
<?xml version="1.0"?>

<!DOCTYPE
  PLOT_SPECIFICATION SYSTEM "http://www.fr.com/fr/fr.dtd">

<PLOT_SPECIFICATION
  COMMENT "Generated by Gnuplot 4.4 patchlevel 0"
  VERSION "1.0"
  DATE "Wed Jan 13 00:00:00 2010"
  DEVICE "unknown"
  TERMINAL "unknown"
  OUTPUT "/tmp/tmpkbcb8Ia_.png"
>

<?xml version="1.0"?>

<!DOCTYPE
  PLOT_TEXT SYSTEM "http://www.fr.com/fr/fr.dtd">

<PLOT_TEXT
  ID "1"
  TEXT "Gnuplot"
  FONT "Helvetica"
  POINT_SIZE "12.000000"
  ROTATION "0.000000"
  ALIGNMENT "CENTER"
  COLOR "0.000000,0.000000,0.000000"
>

<?xml version="1.0"?>

<!DOCTYPE
  PLOT_TEXT SYSTEM "http://www.fr.com/fr/fr.dtd">

<PLOT_TEXT
  ID "2"
  TEXT "Gnuplot"
  FONT "Helvetica"
  POINT_SIZE "12.000000"
  ROTATION "0.000000"
  ALIGNMENT "CENTER"
  COLOR "0.000000,0.000000,0.000000"
>
```

Gnuplot is a command-line driven interactive function plotting program. Originally intended as a means to stimulate students' interest in mathematics, it has been used in scientific fields for over a decade. Gnuplot is capable of plotting 2D functions and data in various styles, and 3D functions and data in the form of surfaces and contour maps. Gnuplot can output to a variety of formats, including PostScript, PNG, and SVG.

Gnuplot is a portable command-line driven interactive data and function plotting utility for UNIX, IBM OS/2, MS Windows, DOS, and many other platforms. It was originally intended for students to explore mathematical functions and data in a friendly way, but has been used for many scientific and engineering applications since then. Gnuplot supports many types of plots in either 2D or 3D. It can plot functions and data read from ASCII files. Gnuplot can output to a variety of formats, including PNG, PostScript, and SVG.

```
<?xml version="1.0"?>
```

```
<div class="page">

<div>
<div>
<div>
```

Ad Clash Report

Ad Clash Report

Ad Clash Report

{
"id": "1707.07187",
"title": "Adversarial Examples for Semantic Segmentation and Object Detection",
"abstract": "Numerous works have recently shown that state-of-the-art classifiers are vulnerable to adversarial perturbations. We extend the study of adversarial examples to more complex vision tasks, namely semantic segmentation and object detection. We show that adversarial examples can also be easily generated for these tasks, and that they transfer across datasets and models. We also show that adversarial training can be used to defend against such examples, and that it leads to improved robustness.",
"authors": "Cihang Xie, Jianyu Wang, Zhishuai Zhang, Yuyin Zhou, Lingxi Xie, Alan Yuille",
"subjects": "Computer Vision and Pattern Recognition (cs.CV); Machine Learning (cs.LG); Machine Learning (stat.ML)",
"comments": "Technical report. Revised version of arXiv:1703.08603 and arXiv:1704.08784",
"doi": "10.48550/arXiv.1707.07187",
"licenses": "Creative Commons Attribution Non-Commercial Share Alike 4.0"
}

Adversarial Examples for Semantic Segmentation and Object Detection

Cihang Xie[1], Jianyu Wang[2], Zhishuai Zhang[1], Yuyin Zhou[1], Lingxi Xie[1], Alan Yuille[1]
[1]Johns Hopkins University, [2]Huazhong University of Science and Technology
{cihang.xie, zhishuai.zhang, yuyin.zhou, lingxi.xie, alan.yuille@jhu.edu, jianyuwang@hust.edu.cn}

Numerous works have recently shown that state-of-the-art classifiers are vulnerable to adversarial perturbations. We extend the study of adversarial examples to more complex vision tasks, namely semantic segmentation and object detection. We show that adversarial examples can also be easily generated for these tasks, and that they transfer across datasets and models. We also show that adversarial training can be used to defend against such examples, and that it leads to improved robustness.

Intro

東京都内の公園ベンチ設置状況調査報告書

1. 調査概要

調査対象は、東京都内の公園に設置されているベンチ（以下「ベンチ」）である。本調査は、ベンチの設置場所、種類、材質、座面の高さ、背もたれの有無、経過年数、および利用者からの苦情の有無を網羅的に把握することを目的とした。調査期間は2025年4月1日から2025年8月31日までであり、対象公園は23区内の都立公園および区立公園の合計412公園である。

2. 調査方法

現地調査は、調査員2名が1公園あたり平均3.5時間を目安に巡回し、ベンチの有無を確認した上で、1台ずつ写真を撮影し、所定の調査票に記入した。座面の高さは、メジャーを用いて地面から座面までの垂直距離を0.5cm単位で計測した。材質は、目視および触診により、木材、金属、樹脂、コンクリートの4区分に分類した。苦情の有無については、公園管理事務所に過去1年間に寄せられた苦情記録を照合した。

3. 調査結果

3.1 設置台数

412公園で合計1,847台のベンチが確認された。1公園あたりの平均設置台数は4.48台であった。最も多かったのは台東区の上野恩賜公園で38台、最も少なかったのは3台であった。ベンチを設置していない公園は7公園であった。

3.2 材質別内訳

材質	台数	構成比
木材	1,109	60.0%
金属	461	25.0%
樹脂	277	15.0%
コンクリート	0	0.0%

3.3 座面の高さ

座面の高さの平均値は43.2cmであった。最頻値は45cmであり、JIS S 1021 の推奨値（40cm）を3cm上回る。

```
# 座面高さ分布（cm）
import matplotlib.pyplot as plt
plt.hist(seat_height, bins=8)
plt.savefig('seat_height.png')
```

3.4 経過年数

経過年数の平均は6.7年であった。最も古いベンチは1974年設置とされ、50年経過していた。

3.5 苦情の有無

苦情記録のあるベンチは312台であった。内訳は、座面が硬い、背もたれがない、ささくれがある、の順であった。

4. 考察

木材が60%を占めるが、経年劣化による苦情は木材に集中する傾向がある。

5. 結論

本調査により、東京都内の公園ベンチの設置状況を網羅的に把握した。

```
{"id":"0000-00-00T00:00:00.000Z",
"_id":"5ffb0ffb0ffb0ffb0ffb0ffb",
"index":0,
"hash":"0000000000000000000000000000000000000000000000000000000000000000",
"nonce":"0000000000000000000000000000000000000000000000000000000000000000",
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"prev_block_hash":"0000000000000000000000000000000000000000000000000000000000000000",
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"next_block_hash":"",
"transactions":[]}

{"id":"0000-00-00T00:00:00.000Z",
"_id":"5ffb0ffb0ffb0ffb0ffb0ffb",
"index":0,
```

Cash Transactions

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Jan-2021 Impact Factor: 4.101 (SJIF) Peer Reviewed Journal International Advanced Research in Engineering and Technology
Page 1

Design of Electrical Generator operated on Low speed Wind Mill for Village Electrification

Prof. M. Ashok Kumar, Prof. Ch. Ramesh, Prof. Ch. Ramesh, Prof. Ch. Ramesh

```
<?xml version="1.0" ?>
<rss version="2.0" xmlns:dc="http://purl.org/dc/elements/1.1/" xmlns:content="http://purl.org/rss/1.0/modules/content/" xmlns:atom="http://www.w3.org/2005/Atom" xmlns:media="http://search.yahoo.com/mrss/" xmlns:wfw="http://wellformedweb.org/CommentAPI/" xmlns:itunes="http://www.itunes.com/dtd/2.0/" xmlns:sy="http://purl.org/rss/1.0/modules/syndication/" xmlns:slash="http://purl.org/rss/1.0/modules/slash/" xmlns:dsq="http://disqus.com/disqus-internal/" xmlns:opensearch="http://a9.com/-/spec/opensearchrss/1.0/" xmlns:geo="http://www.w3.org/2003/01/geo/wgs84_pos#" xmlns:georss="http://www.georss.org/georss" xmlns:gml="http://www.opengis.net/gml" xmlns:osi="http://rssnamespace.org/feedburner/extensions/1.0" xmlns:feedburner="http://rssnamespace.org/feedburner/extensions/1.0" xmlns:fb="http://www.facebook.com/2008/fbml" xmlns:media="http://search.yahoo.com/mrss/" xmlns:atom10="http://www.w3.org/2005/Atom
```

7.2.2 Equilibrium

Let's return to our model of monopoly power.

```
In [1]: import numpy as np
        import matplotlib.pyplot as plt
        %matplotlib inline
        from sympy import symbols, Eq, solve, lambdify, latex, init_printing, Rational, linsolve, diff, simplify
```

```
Out[1]:
```

Suppose the price-cost mark-up is 0.25 (25%), and the elasticity is 1.5. We know that a firm's profit-maximizing mark-up is $\frac{1}{\epsilon}$, so we can solve for the elasticity. This tells us that we can write the quantity demanded as a function of the price as:

$$q = p^{-\epsilon} = p^{-1.5} \Rightarrow p = q^{-\frac{2}{3}}$$

```
In [2]: q, p = symbols('q p')
        demand = Eq(q, p**(-Rational(3,2)))
        demand
```

```
Out[2]:
```

$$q = \frac{1}{p^{\frac{3}{2}}}$$

```
In [3]: inverse = solve(demand, p)
        inverse
```

```
Out[3]:
```

$$\left[\frac{1}{q^{\frac{2}{3}}} \right]$$

```
In [4]: inverse = inverse[0]
        inverse
```

```
Out[4]:
```

$$\frac{1}{q^{\frac{2}{3}}}$$

```
In [5]: inverse.subs(q, 1)
```

```
Out[5]:
```

$$1$$

```
In [6]: inverse.subs(q, 2)
```

```
Out[6]:
```

$$\frac{1}{2^{\frac{2}{3}}}$$

```
In [7]: inverse.subs(q, 3)
```

```
Out[7]:
```

$$\frac{1}{3^{\frac{2}{3}}}$$

```
In [8]: inverse.subs(q, 4)
```

```
Out[8]:
```

$$\frac{1}{4^{\frac{2}{3}}}$$

```
In [9]: inverse.subs(q, 5)
```

```
Out[9]:
```

$$\frac{1}{5^{\frac{2}{3}}}$$

```
In [10]: %matplotlib inline
         from sympy.plotting import plot, plot_implicit
         plot(inverse, (q, 0, 10))
```

```
Out[10]:
```



```
<sympy.plotting.plot.Plot object at 0x7f8a1a0a0a90>
```

Suppose the price-cost mark-up is 0.25 (25%), and the elasticity is 1.5. We know that a firm's profit-maximizing mark-up is $\frac{1}{\epsilon}$, so we can solve for the elasticity. This tells us that we can write the quantity demanded as a function of the price as:

$$q = p^{-\epsilon} = p^{-1.5} \Rightarrow p = q^{-\frac{2}{3}}$$

```
In [11]: q, p = symbols('q p')
         demand = Eq(q, p**(-Rational(3,2)))
         demand
```

```
Out[11]:
```

$$q = \frac{1}{p^{\frac{3}{2}}}$$

```
In [12]: inverse = solve(demand, p)
         inverse
```

```
Out[12]:
```

$$\left[\frac{1}{q^{\frac{2}{3}}} \right]$$

```
In [13]: inverse = inverse[0]
         inverse
```

```
Out[13]:
```

$$\frac{1}{q^{\frac{2}{3}}}$$

```
In [14]: inverse.subs(q, 1)
```

```
Out[14]:
```

$$1$$

```
In [15]: inverse.subs(q, 2)
```

```
Out[15]:
```

$$\frac{1}{2^{\frac{2}{3}}}$$

```
In [16]: inverse.subs(q, 3)
```

```
Out[16]:
```

$$\frac{1}{3^{\frac{2}{3}}}$$

```
In [17]: inverse.subs(q, 4)
```

```
Out[17]:
```

$$\frac{1}{4^{\frac{2}{3}}}$$

```
In [18]: inverse.subs(q, 5)
```

```
Out[18]:
```

$$\frac{1}{5^{\frac{2}{3}}}$$

```
In [19]: %matplotlib inline
         from sympy.plotting import plot, plot_implicit
         plot(inverse, (q, 0, 10))
```

```
Out[19]:
```



```
<sympy.plotting.plot.Plot object at 0x7f8a1a0a0a90>
```

Suppose the price-cost mark-up is 0.25 (25%), and the elasticity is 1.5. We know that a firm's profit-maximizing mark-up is $\frac{1}{\epsilon}$, so we can solve for the elasticity. This tells us that we can write the quantity demanded as a function of the price as:

$$q = p^{-\epsilon} = p^{-1.5} \Rightarrow p = q^{-\frac{2}{3}}$$

```
In [20]: q, p = symbols('q p')
         demand = Eq(q, p**(-Rational(3,2)))
         demand
```

```
Out[20]:
```

$$q = \frac{1}{p^{\frac{3}{2}}}$$

```
In [21]: inverse = solve(demand, p)
         inverse
```

```
Out[21]:
```

$$\left[\frac{1}{q^{\frac{2}{3}}} \right]$$

```
In [22]: inverse = inverse[0]
         inverse
```

```
Out[22]:
```

$$\frac{1}{q^{\frac{2}{3}}}$$

```
In [23]: inverse.subs(q, 1)
```

```
Out[23]:
```

$$1$$

```
In [24]: inverse.subs(q, 2)
```

```
Out[24]:
```

$$\frac{1}{2^{\frac{2}{3}}}$$

```
In [25]: inverse.subs(q, 3)
```

```
Out[25]:
```

$$\frac{1}{3^{\frac{2}{3}}}$$

```
In [26]: inverse.subs(q, 4)
```

```
Out[26]:
```

$$\frac{1}{4^{\frac{2}{3}}}$$

```
In [27]: inverse.subs(q, 5)
```

```
Out[27]:
```

$$\frac{1}{5^{\frac{2}{3}}}$$

```
In [28]: %matplotlib inline
         from sympy.plotting import plot, plot_implicit
         plot(inverse, (q, 0, 10))
```

```
Out[28]:
```



```
<sympy.plotting.plot.Plot object at 0x7f8a1a0a0a90>
```

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```
<?xml version="1.0" encoding="UTF-8"?>
<rss version="2.0" xmlns:dc="http://purl.org/dc/elements/1.1/" xmlns:sy="http://purl.org/rss/1.0/modules/syndication/" xmlns:slash="http://purl.org/rss/1.0/modules/slash/" xmlns:rdfs="http://www.w3.org/2000/01/rdf-schema#" xmlns:wfw="http://wellformedweb.org/CommentAPI/" xmlns:dcterms="http://purl.org/dc/terms/" xmlns:media="http://search.yahoo.com/mrss/" xmlns:atom="http://www.w3.org/2005/Atom" xmlns:opensearch="http://a9.com/-/spec/opensearchrss/1.0/" xmlns:geo="http://www.w3.org/2003/01/geo/wgs84_pos#" xmlns:georss="http://www.georss.org/georss" xmlns:gml="http://www.opengis.net/gml" xmlns:osm="http://www.openstreetmap.org/dtd/0.6" xmlns:fb="http://www.facebook.com/2008/fbml" xmlns:content="http://purl.org/rss/1.0/modules/content/" xmlns:wfw="http://wellformedweb.org/CommentAPI/" xmlns:dc="http://purl.org/dc/elements/1.1/" xmlns:atom="http://www.w3.org/2005/Atom" xmlns:sy="http://purl.org/rss/1.0/modules/syndication/" xmlns:slash="http://purl.org/rss/1.0/modules/slash/" xmlns:media="http://search.yahoo.com/mrss/
```

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Number 4
http://www.kvasy.sk
Print version: 1336-8672

```
{
  "name": "Sample",
  "version": "0.0.0",
  "description": "",
  "keywords": [],
  "author": "",
  "license": "ISC",
  "dependencies": {
    "@ffmpeg-installer/ffmpeg": "1.1.0",
    "@ffmpeg-installer/linux-arm": "1.1.0",
    "@ffmpeg-installer/linux-arm64": "1.1.0",
    "@ffmpeg-installer/linux-ia32": "1.1.0",
    "@ffmpeg-installer/linux-x64": "1.1.0",
    "@ffmpeg-installer/mac": "1.1.0",
    "@ffmpeg-installer/win32": "1.1.0",
```

Project Gutenberg's Alice's Adventures in Wonderland, by Lewis Carroll
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Release Date: [EBook #11]
Last Updated: [July 14, 2011]
Language: [English]
Character set encoding: [UTF-8]
Produced by: [Arthur DiBianca and David Widger]
Source: [Gutenberg]

```
<?xml version="1.0" encoding="UTF-8"?>
<?xml-stylesheet href="chrome://global/skin/" type="text/xsl"?>
<!--
  This file contains the default icons for Firefox.
  -->
<svg
  xmlns="http://www.w3.org/2000/svg"
  xmlns:xlink="http://www.w3.org/1999/xlink"
  xmlns:rdf="http://www.w3.org/22-rdf-syntax-ns#"
  xmlns:cc="http://creativecommons.org/ns#"
  xmlns:dc="http://purl.org/dc/elements/1.1/"
  xmlns:sodipodi="http://sodipodi.sourceforge.net/DTD/sodipodi-0.dtd"
  xmlns:inkscape="http://www.inkscape.org/namespaces/inkscape"
  id="svg"
  viewBox="0 0 16 16"
  height="16px"
  width="16px">
```

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Material and Methods 2
Observation 3
Discussion 4
Conclusion 5
References 6

Introduction

Cell Cycle

Cell Cycle Checkpoints: The Mammalian Perspective A Tumor Cell Cycle Controlling Checkpoint: A Tumor Cell Cycle Controlling Checkpoint: Cell Cycle Checkpoints

Cell Cycle Checkpoints

Cell Cycle Checkpoints

Cell Cycle Checkpoints: A Critical Point Cell Cycle Checkpoints

Introduction 1
Material and Methods 2
Observation 3
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Conclusion 5
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Material and Methods

Cell Cycle

Cell Cycle Checkpoints

Cell Cycle Checkpoints: Critical Points

Cell Cycle Checkpoints

Cell Cycle

Cell Cycle

Cell Cycle Checkpoints: Critical Points Cell Cycle Checkpoints: A Critical Point Cell Cycle Checkpoints: A Critical Point Cell Cycle Checkpoints: A Critical Point

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Sub-Interface - Layer3 Subinterface
Configure a Layer3 Subinterface on the GUI
Configure a Layer3 Subinterface on the CLI
Verify Layer3 Subinterface Configuration on the GUI
Verify Layer3 Subinterface Configuration on the CLI
Related Topics

Sub-Interface - Layer3 Subinterface

Palo Alto Networks firewalls support Layer 3 routing on subinterfaces to accommodate multiple virtual routers and security zones on a single physical interface. The ability to define subinterfaces is useful when you want to route traffic for multiple virtual routers or security zones through a single physical interface. Each subinterface can be assigned to a different virtual router and security zone. Each subinterface can be configured with its own IP address, static routes, RIP, OSPF, BGP, or multicast routing protocol. Subinterfaces are also useful if you have a firewall that is connected to a switch that is configured for multiple VLANs. You can define a subinterface for each VLAN and assign traffic from each VLAN to a different virtual router and security zone.

Configure a Layer3 Subinterface on the GUI

Step 1. Define the parent interface. Select Network > Interfaces and click the interface name. Select Interface Type Layer3 and configure the IPv4, IPv6 or both addresses, Management Profile, MTU, Link Speed, Link Duplex, and Link State as needed. Click OK.

Step 2. Define the subinterface. Select Network > Interfaces and click Add Subinterface. Enter the Interface Name in the format eth<slot>.<port_number>. Enter the Tag (VLAN ID) and Virtual Router. Select the Security Zone and configure the IPv4, IPv6 or both addresses, Management Profile, MTU, Link Speed, Link Duplex, and Link State as needed. Click OK.

Step 3. Commit the changes. Click Commit.

Verify Layer3 Subinterface Configuration on the GUI

Step 1. Verify the parent interface. Select Network > Interfaces and click the interface name. Select Interface Type Layer3 and configure the IPv4, IPv6 or both addresses, Management Profile, MTU, Link Speed, Link Duplex, and Link State as needed. Click OK.

Step 2. Verify the subinterface. Select Network > Interfaces and click Add Subinterface. Enter the Interface Name in the format eth<slot>.<port_number>. Enter the Tag (VLAN ID) and Virtual Router. Select the Security Zone and configure the IPv4, IPv6 or both addresses, Management Profile, MTU, Link Speed, Link Duplex, and Link State as needed. Click OK.

Step 3. Commit the changes. Click Commit.

Configure a Layer3 Subinterface on the CLI

```
set interfaces ethernet <slot>/<port_number>
set interfaces ethernet <slot>/<port_number> mtu <mtu_size>
set interfaces ethernet <slot>/<port_number> link-speed <speed_value>
set interfaces ethernet <slot>/<port_number> link-duplex <duplex_value>
set interfaces ethernet <slot>/<port_number> link-state <state_value>
set interfaces ethernet <slot>/<port_number> layer3 ip <ipv4_address>/<mask_length>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 <ipv6_address>/<mask_length>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 enable-duplicate-address-detection yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery router-advertisement enable yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery router-advertisement hop-limit <hop_limit_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery router-advertisement lifetime <lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery router-advertisement reachable-time <reachable_time_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery router-advertisement retransmit-time <retransmit_time_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery router-advertisement prefix <ipv6_prefix>/<mask_length> on-link yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery router-advertisement prefix <ipv6_prefix>/<mask_length> autonomous yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery router-advertisement prefix <ipv6_prefix>/<mask_length> preferred-lifetime <preferred_lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery router-advertisement prefix <ipv6_prefix>/<mask_length> valid-lifetime <valid_lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery router-advertisement dns-server <dns_server_address_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery router-advertisement dns-suffix <dns_suffix_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery router-advertisement enable yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery neighbor <ipv6_address_value> <mac_address_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery duplicate-address-detection clone-limit <clone_limit_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 neighbor-discovery duplicate-address-detection freeze-interface-id yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type prefix-only
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address identifier <identifier_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address identifier <identifier_value> eui-64 yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address identifier <identifier_value> randomize yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address lifetime <lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address lifetime <lifetime_value> no-expiry yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address lifetime <lifetime_value> on-link yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address lifetime <lifetime_value> preferred yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address lifetime <lifetime_value> autonomous yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address lifetime <lifetime_value> preferred-lifetime <preferred_lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address lifetime <lifetime_value> valid-lifetime <valid_lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool identifier <identifier_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool identifier <identifier_value> eui-64 yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool identifier <identifier_value> randomize yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool lifetime <lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool lifetime <lifetime_value> no-expiry yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool lifetime <lifetime_value> on-link yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool lifetime <lifetime_value> preferred yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool lifetime <lifetime_value> autonomous yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool lifetime <lifetime_value> preferred-lifetime <preferred_lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool lifetime <lifetime_value> valid-lifetime <valid_lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude identifier <identifier_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude identifier <identifier_value> eui-64 yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude identifier <identifier_value> randomize yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> no-expiry yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> on-link yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> preferred yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> autonomous yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> preferred-lifetime <preferred_lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> valid-lifetime <valid_lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude identifier <identifier_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude identifier <identifier_value> eui-64 yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude identifier <identifier_value> randomize yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> no-expiry yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> on-link yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> preferred yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> autonomous yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> preferred-lifetime <preferred_lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> valid-lifetime <valid_lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude identifier <identifier_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude identifier <identifier_value> eui-64 yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude identifier <identifier_value> randomize yes
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set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> on-link yes
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set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> autonomous yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> preferred-lifetime <preferred_lifetime_value>
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set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude identifier <identifier_value> eui-64 yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude identifier <identifier_value> randomize yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> no-expiry yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> on-link yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> preferred yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> autonomous yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> preferred-lifetime <preferred_lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> valid-lifetime <valid_lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude identifier <identifier_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude identifier <identifier_value> eui-64 yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude identifier <identifier_value> randomize yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> no-expiry yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> on-link yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> preferred yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> autonomous yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> preferred-lifetime <preferred_lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> valid-lifetime <valid_lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude identifier <identifier_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude identifier <identifier_value> eui-64 yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude identifier <identifier_value> randomize yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> no-expiry yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> on-link yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> preferred yes
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set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> valid-lifetime <valid_lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length>
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude yes
set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude identifier <identifier_value>
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set interfaces ethernet <slot>/<port_number> layer3 ipv6 inherited assign-address <name_value> type address pool-name <pool_name_value> prefix-pool prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude prefix <ipv6_prefix>/<mask_length> exclude lifetime <lifetime_value> valid-lifetime <valid_lifetime_value>
set interfaces ethernet <slot>/<port_number> layer3 
```

$\int \frac{1}{x^2 - 2x + 2}\,dx$ - Integral of $\frac{1}{x^2 - 2x + 2}$ with respect to x | Wolfram|Alpha Integral Calculator, Antiderivative

$\int \frac{1}{x^2 - 2x + 2}\,dx$ - Integral of $\frac{1}{x^2 - 2x + 2}$ with respect to x

Input interpretation:

$\int \frac{1}{x^2 - 2x + 2}\,dx$

Result:

$\tan^{-1}(x - 1)$

Indefinite integral:

$\int \frac{1}{x^2 - 2x + 2}\,dx = \tan^{-1}(x - 1) + \text{constant}$

Alternate form:

$\int \frac{1}{x^2 - 2x + 2}\,dx = \frac{1}{2} i \log(-i x + i + 1) - \frac{1}{2} i \log(i x - i + 1) + \text{constant}$

Alternate form assuming x is real:

$\int \frac{1}{x^2 - 2x + 2}\,dx = \tan^{-1}(x - 1) + \text{constant}$

Alternate form assuming x is complex:

$\int \frac{1}{x^2 - 2x + 2}\,dx = \frac{1}{2} i \log(-i x + i + 1) - \frac{1}{2} i \log(i x - i + 1) + \text{constant}$

Derivative:

$\frac{d}{dx}\left(\int \frac{1}{x^2 - 2x + 2}\,dx\right) = \frac{1}{x^2 - 2x + 2}$

Alternate form:

$\frac{d}{dx}\left(\int \frac{1}{x^2 - 2x + 2}\,dx\right) = \frac{1}{x^2 - 2x + 2}$

Alternate form assuming x is real:

$\frac{d}{dx}\left(\int \frac{1}{x^2 - 2x + 2}\,dx\right) = \frac{1}{x^2 - 2x + 2}$

Alternate form assuming x is complex:

$\frac{d}{dx}\left(\int \frac{1}{x^2 - 2x + 2}\,dx\right) = \frac{1}{x^2 - 2x + 2}$

Definite integral:

$\int_0^1 \frac{1}{x^2 - 2x + 2}\,dx = \frac{\pi}{4}$

Alternate form:

$\int_0^1 \frac{1}{x^2 - 2x + 2}\,dx = \frac{\pi}{4}$

Alternate form assuming x is real:

$\int_0^1 \frac{1}{x^2 - 2x + 2}\,dx = \frac{\pi}{4}$

Alternate form assuming x is complex:

$\int_0^1 \frac{1}{x^2 - 2x + 2}\,dx = \frac{\pi}{4}$

Alternate form:

$\int_0^1 \frac{1}{x^2 - 2x + 2}\,dx = \frac{1}{2} i \log(-i + i + 1) - \frac{1}{2} i \log(i - i + 1)$

Alternate form assuming x is real:

$\int_0^1 \frac{1}{x^2 - 2x + 2}\,dx = \frac{\pi}{4}$

Alternate form assuming x is complex:

$\int_0^1 \frac{1}{x^2 - 2x + 2}\,dx = \frac{1}{2} i \log(-i + i + 1) - \frac{1}{2} i \log(i - i + 1)$

Indefinite integral:

$\int \frac{1}{x^2 - 2x + 2}\,dx = \tan^{-1}(x - 1) + \text{constant}$

Alternate form:

$\int \frac{1}{x^2 - 2x + 2}\,dx = \frac{1}{2} i \log(-i x + i + 1) - \frac{1}{2} i \log(i x - i + 1) + \text{constant}$

Alternate form assuming x is real:

$\int \frac{1}{x^2 - 2x + 2}\,dx = \tan^{-1}(x - 1) + \text{constant}$

Alternate form assuming x is complex:

$\int \frac{1}{x^2 - 2x + 2}\,dx = \frac{1}{2} i \log(-i x + i + 1) - \frac{1}{2} i \log(i x - i + 1) + \text{constant}$

Derivative:

$\frac{d}{dx}\left(\int \frac{1}{x^2 - 2x + 2}\,dx\right) = \frac{1}{x^2 - 2x + 2}$

Alternate form:

$\frac{d}{dx}\left(\int \frac{1}{x^2 - 2x + 2}\,dx\right) = \frac{1}{x^2 - 2x + 2}$

Alternate form assuming x is real:

$\frac{d}{dx}\left(\int \frac{1}{x^2 - 2x + 2}\,dx\right) = \frac{1}{x^2 - 2x + 2}$

Alternate form assuming x is complex:

$\frac{d}{dx}\left(\int \frac{1}{x^2 - 2x + 2}\,dx\right) = \frac{1}{x^2 - 2x + 2}$

Definite integral:

$\int_0^1 \frac{1}{x^2 - 2x + 2}\,dx = \frac{\pi}{4}$

Alternate form:

$\int_0^1 \frac{1}{x^2 - 2x + 2}\,dx = \frac{\pi}{4}$

Alternate form assuming x is real:

$\int_0^1 \frac{1}{x^2 - 2x + 2}\,dx = \frac{\pi}{4}$

Alternate form assuming x is complex:

$\int_0^1 \frac{1}{x^2 - 2x + 2}\,dx = \frac{\pi}{4}$

Alternate form:

$\int_0^1 \frac{1}{x^2 - 2x + 2}\,dx = \frac{1}{2} i \log(-i + i + 1) - \frac{1}{2} i \log(i - i + 1)$

Alternate form assuming x is real:

$\int_0^1 \frac{1}{x^2 - 2x + 2}\,dx = \frac{\pi}{4}$

Alternate form assuming x is complex:

$\int_0^1 \frac{1}{x^2 - 2x + 2}\,dx = \frac{1}{2} i \log(-i + i + 1) - \frac{1}{2} i \log(i - i + 1)$

DOI: 10.1007/s00401-
Received: 5 July 2017
Revised: 14 November 2017
Accepted: 15 November 2017
Online First: 27 November 2017

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Section 1.10.32 of "de Finibus Bonorum et Malorum", written by Cicero in 45 BC

"Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo. Nemo enim ipsam voluptatem quia voluptas sit aspernatur aut odit aut fugit, sed quia consequuntur magni dolores eos qui ratione voluptatem sequi nesciunt. Neque porro quisquam est, qui dolorem ipsum quia dolor sit amet, consectetur, adipisci velit, sed quia non numquam eius modi tempora incidunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim ad minima veniam, quis nostrum exercitationem ullam corporis suscipit laboratoriosam, nisi ut aliquid ex ea commodi consequatur? Quis autem vel eum iure reprehenderit qui in ea voluptate velit esse quam nihil molestiae consequatur, vel illum qui dolorem eum fugiat, quo voluptas nulla pariatur?"

1914 translation by H. Rackham

"But I must explain to you how all this mistaken idea of denouncing pleasure and praising pain was born and I will give you a complete account of the system, and expound the actual teachings of the great explorer of the truth, the master-builder of human happiness. No one rejects, dislikes, or avoids pleasure itself, because it is pleasure, but because those who do not know how to pursue pleasure rationally encounter consequences that are extremely painful. Nor again is there anyone who loves or pursues or desires to obtain pain of itself, because it is pain, but because occasionally circumstances occur in which toil and pain can procure him some great pleasure. To take a trivial example, which of us ever undertakes laborious physical exercise, except to obtain some advantage from it? But who has any right to find fault with a man who chooses to enjoy a pleasure that has no annoying consequences, or one who avoids a pain that produces no resultant pleasure?"

1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 |

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一种电缆 专利说明书

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[57]摘要 [illegible]

[illegible]

이사회 의장 주주 제안서

제1조 (제안의 목적)

제1조 (제안의 목적) 본 제안서는 주주 김지훈이 이사회 의장 선임을 제안하기 위해 작성되었다. 의장 선임은 이사회 운영 전반에 대한 주주 권리 행사의 가장 직접적인 수단이다. 의장은 의제 설정, 회의 진행, 의결 정리, 이사회 의사록 작성 등 이사회 운영 전반에 대한 권한을 가진다. 의장 선임 제안은 이사회 구성 변경 없이 이사회 운영 방식을 바꾸는 가장 빠른 경로다. 이사 선임 제안은 시간이 걸리지만, 의장 선임은 다음 이사회 한 번이면 된다.

제2조 (의장 선임 제안의 근거)

제2조 (의장 선임 제안의 근거) 의장 선임 제안은 세 가지 근거에 기한다. 첫째, 현 의장 김민수는 이사회 의사록 작성 의무를 지키지 않았다. 둘째, 의장 선임 제안은 이사회 구성 변경 없이 이사회 운영 방식을 바꾸는 가장 빠른 경로다. 셋째, 의장은 의제 설정, 회의 진행, 의결 정리, 이사회 의사록 작성 등 이사회 운영 전반에 대한 권한을 가진다.

제3조 (의사록 미작성 사실)

제3조 (의사록 미작성 사실) 2024년 3월 이사회 의사록 미작성 사실은 이사회 의사록 작성 의무를 지키지 않았다는 것을 보여준다. 의사록 작성 의무는 이사회 의장의 의무다. 의사록 미작성 사실은 의장 김민수가 이사회 의사록 작성 의무를 지키지 않았다는 것을 보여준다.

제4조 (의장 김민수의 의사록 작성 의무 불이행)

제4조 (의장 김민수의 의사록 작성 의무 불이행) 의장 김민수는 이사회 의사록 작성 의무를 지키지 않았다. 의사록 작성 의무는 이사회 의장의 의무다. 의장 김민수는 이사회 의사록 작성 의무를 지키지 않았다.

제5조 (의장 선임 제안의 효과)

제5조 (의장 선임 제안의 효과) 의장 선임 제안은 이사회 구성 변경 없이 이사회 운영 방식을 바꾸는 가장 빠른 경로다. 이사 선임 제안은 시간이 걸리지만, 의장 선임은 다음 이사회 한 번이면 된다. 의장 선임 제안은 이사회 구성 변경 없이 이사회 운영 방식을 바꾸는 가장 빠른 경로다.

[illegible]

[illegible]

[illegible]

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256.000,
256.000,
```

1.1 Introduction

1.1.1 Why Study Differential Equations?

Differential equations are equations involving derivatives of unknown functions. For example, the equation

$$\frac{dy}{dx} = 2x + 3$$

is a differential equation. Here, y is a function of x, and $\frac{dy}{dx}$ represents the rate at which y changes with respect to x.

Differential equations are fundamental in mathematics because they describe how quantities change over time or in response to other variables. For example, Newton's Second Law, $F = ma$, can be rewritten as $F = m\frac{d^2x}{dt^2}$, where $x(t)$ is the position of an object at time t. This equation allows us to predict the motion of the object by solving for $x(t)$.

Differential equations are used in physics, engineering, economics, biology, and many other fields to model dynamic systems. For example, in electrical circuits, the current $I(t)$ in a circuit with resistance R, inductance L, and voltage $V(t)$ is described by the equation $L\frac{dI}{dt} + RI = V(t)$. Solving this equation allows us to predict how the current changes over time.

1.1.2 Ordinary Differential Equations (ODEs)

An ordinary differential equation (ODE) is a differential equation in which the unknown function depends on a single variable. For example, the equation $\frac{dy}{dx} = 2x + 3$ is an ODE because y depends only on x.

ODEs are used to model systems where the rate of change of a quantity depends on the quantity itself and other factors. For example, the population of a city may be modeled by the ODE $\frac{dP}{dt} = kP$, where $P(t)$ is the population at time t and k is a constant representing the growth rate.

ODEs are used in physics, engineering, economics, biology, and many other fields to model dynamic systems. For example, in electrical circuits, the current $I(t)$ in a circuit with resistance R, inductance L, and voltage $V(t)$ is described by the ODE $L\frac{dI}{dt} + RI = V(t)$. Solving this ODE allows us to predict how the current changes over time.

1.1.3 Partial Differential Equations (PDEs)

A partial differential equation (PDE) is a differential equation in which the unknown function depends on multiple variables. For example, the equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ is a PDE because u depends on both x and t.

PDEs are used to model systems where the rate of change of a quantity depends on the quantity itself and other factors. For example, the heat equation $\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2}$ describes how the temperature $u(x,t)$ changes over time and space, where k is a constant representing the thermal conductivity.

PDEs are used in physics, engineering, economics, biology, and many other fields to model dynamic systems. For example, in electrical circuits, the current $I(t)$ in a circuit with resistance R, inductance L, and voltage $V(t)$ is described by the PDE $L\frac{\partial I}{\partial t} + RI = V(t)$. Solving this PDE allows us to predict how the current changes over time and space.

Ch. 2 Lect 2 Part 1 - Chapter Summary

Ch. 2 Lect 2 Part 1 - Chapter Summary

Ch. 2 Lect 2 Part 1 - Chapter Summary

DOI: 10.1007/s00401-
Received: 2 July 2020
Revised: 2 November 2020
Accepted: 3 November 2020

```
{
  "data": {
    "id": "1",
    "type": "articles",
    "attributes": {
      "title": "JSON:API pagination",
      "slug": "jsonapi-pagination",
      "body": "First, we need to add a query parameter to our API. We'll use `page[number]` and `page[size]`.\n\nFrom there, our GET /articles?page[number]=3&page[size]=10 will return the 21st and 22nd articles in the database.",
      "created": "2015-05-22T14:56:29.000Z",
      "updated": "2015-05-22T14:56:29.000Z"
    },
    "relationships": {
      "author": {
        "data": {"id": "42", "type": "people"}
      },
      "tags": {
        "data": [
          {"id": "3", "type": "tags"},
          {"id": "4", "type": "tags"}
        ]
      }
    }
  },
  "included": [
    {
      "id": "42",
      "type": "people",
      "attributes": {
        "name": "John Smith",
        "twitter": "jsmith"
      }
    },
    {
      "id": "3",
      "type": "tags",
      "attributes": {
        "name": "pagination"
      }
    },
    {
      "id": "4",
      "type": "tags",
      "attributes": {
        "name": "performance"
      }
    }
  ]
}
```

DOI: 10.1007/s00401-
Received: 5 July 2017
Revised: 7 November 2017
Accepted: 8 November 2017
Online First: 15 November 2017

1.1 Overview

1.1.1 Purpose and Scope

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```
sudo apt-get update
sudo apt-get install -y nginx
sudo systemctl enable nginx
sudo systemctl start nginx
curl -I http://127.0.0.1/
```

1.1.2 Audience

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Curabitur ut ipsum ac nisl ultricies ornare.

Sed ut perspiciatis, unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo. Nemo enim ipsam voluptatem quia voluptas sit aspernatur aut odit aut fugit, sed quia consequuntur magni dolores eos qui ratione voluptatem sequi nesciunt. Neque porro quisquam est, qui dolorem ipsum quia dolor sit amet, consectetur, adipisci velit, sed quia non numquam eius modi tempora incidunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim ad minima veniam, quis nostrum exercitationem ullam corporis suscipit laboriosam, nisi ut aliquid ex ea commodi consequatur? Quis autem vel eum iure reprehenderit qui in ea voluptate velit esse quam nihil molestiae consequatur, vel illum qui dolorem eum fugiat quo voluptas nulla pariatur?

```
$ curl -I http://127.0.0.1/
HTTP/1.1 200 OK
Server: nginx/1.18.0 (Ubuntu)
Date: Sun, 12 Apr 2026 09:14:07 GMT
Content-Type: text/html
Content-Length: 612
Last-Modified: Wed, 04 Jan 2023 08:00:00 GMT
Connection: keep-alive
ETag: "63b5f500-264"
Accept-Ranges: bytes
$ sudo ufw status
Status: inactive
$ sudo ufw enable
Command may disrupt existing ssh connections. Proceed with operation (y|n)? y
Firewall is active and enabled on system startup
$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip
$ sudo ufw allow 22/tcp
Rule added
Rule added (v6)
$ sudo ufw allow 80/tcp
Rule added
Rule added (v6)
$ sudo ufw allow 443/tcp
Rule added
Rule added (v6)
$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip
To                         Action      From
--                         ------      ----
22/tcp                     ALLOW IN    Anywhere
80/tcp                     ALLOW IN    Anywhere
443/tcp                    ALLOW IN    Anywhere
22/tcp (v6)                ALLOW IN    Anywhere (v6)
80/tcp (v6)                ALLOW IN    Anywhere (v6)
443/tcp (v6)               ALLOW IN    Anywhere (v6)
```

1.1.3 Document Conventions

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Curabitur ut ipsum ac nisl ultricies ornare.

Sed ut perspiciatis, unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo. Nemo enim ipsam voluptatem quia voluptas sit aspernatur aut odit aut fugit, sed quia consequuntur magni dolores eos qui ratione voluptatem sequi nesciunt. Neque porro quisquam est, qui dolorem ipsum quia dolor sit amet, consectetur, adipisci velit, sed quia non numquam eius modi tempora incidunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim ad minima veniam, quis nostrum exercitationem ullam corporis suscipit laboriosam, nisi ut aliquid ex ea commodi consequatur? Quis autem vel eum iure reprehenderit qui in ea voluptate velit esse quam nihil molestiae consequatur, vel illum qui dolorem eum fugiat quo voluptas nulla pariatur?

```
$ sudo apt-get update && sudo apt-get install -y nginx
$ sudo systemctl enable nginx
$ sudo systemctl start nginx
$ curl -I http://127.0.0.1/
$ sudo ufw allow 22/tcp
$ sudo ufw allow 80/tcp
$ sudo ufw allow 443/tcp
$ sudo ufw enable
$ sudo ufw status verbose
$ sudo ufw status numbered
$ sudo ufw delete allow 22/tcp
$ sudo ufw logging medium
$ sudo ufw reset
```

1.2 Prerequisites

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Curabitur ut ipsum ac nisl ultricies ornare.

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$ sudo ufw allow 22/tcp
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$ sudo ufw status verbose
$ sudo ufw status numbered
$ sudo ufw delete allow 22/tcp
$ sudo ufw logging medium
$ sudo ufw reset
```

Ch. 2 Lab Report - Angular Momentum

Name: Ryan Ramos

Partners: Ryan Ramos, Ryan Ramos, Ryan Ramos,

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Equation 1 of 1: Gaussian Integral Squared

Type: Definite Integral

Source: Gaussian Integral Squared

Tags: [Gaussian Integral, Gaussian Integral Squared]

Notes: Gaussian Integral Squared. Gaussian Integral Squared. Gaussian Integral Squared.

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Effect of Ultrasound Treatment on Antibacterial Activity of Aqueous Extract of Aloe Vera Gel

Ali Ahmadi , Mohammad Ali Ebrahimzadeh , Mohammad Ali Ghasemi , Ali Ghasemi , Ali Ghasemi ,

Chapter Chapter 1

Section Introduction

Subsection Introduction to Linear Algebra

Linear Algebra

Linear Algebra

Linear Algebra is the study of vectors and linear transformations between vector spaces.

Example 1.1.1

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Proof.

Let $u = (x_1, y_1, z_1)$ and $v = (x_2, y_2, z_2)$ be arbitrary elements of V and let $c \in \mathbb{R}$ be arbitrary.

Then $u + v = (x_1 + x_2, y_1 + y_2, z_1 + z_2)$ and

$$T(u + v) = (x_1 + x_2 + y_1 + y_2, 2(z_1 + z_2) - (x_1 + x_2)) = (x_1 + y_1, 2z_1 - x_1) + (x_2 + y_2, 2z_2 - x_2) = T(u) + T(v).$$

Also, $cu = (cx_1, cy_1, cz_1)$ and

$$T(cu) = (cx_1 + cy_1, 2cz_1 - cx_1) = c(x_1 + y_1, 2z_1 - x_1) = cT(u).$$

Therefore T satisfies both conditions of linearity and is a linear transformation. $\square$

Example 1.1.2

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x + 1).$$

Then T is not a linear transformation because $T(0, 0, 0) = (0, 1) \neq (0, 0)$.

Example 1.1.3

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.4

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.5

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.6

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.7

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.8

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.9

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.10

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.11

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.12

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.13

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.14

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.15

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.16

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.17

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.18

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.19

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.20

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.21

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.22

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.23

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.24

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.25

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.26

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.27

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.28

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.29

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.30

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.31

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.32

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.33

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.34

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.35

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.36

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.37

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.38

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.39

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.40

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.41

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.42

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.43

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.44

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.45

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.46

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.47

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.48

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.49

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.50

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.51

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

$$T(x, y, z) = (x + y, 2z - x).$$

Then T is a linear transformation.

Example 1.1.52

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

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Then T is a linear transformation.

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Let $T: V \to W$ be defined by

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Then T is a linear transformation.

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Then T is a linear transformation.

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Let $T: V \to W$ be defined by

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Example 1.1.56

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$.

Let $T: V \to W$ be defined by

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Abstract: In this paper, we study the Cauchy problem for the nonlinear Kirchhoff-Carrier wave equation in a bounded domain. We prove that the solution exists globally in time and decays to zero as time tends to infinity.

Keywords: Kirchhoff-Carrier wave equation, Cauchy problem, Global existence, Decay.

MSC: 35L70, 35L20, 35L05, 35L15, 35L71, 35L72, 35L76, 35L90, 35B40, 35B41, 35B45, 35B65, 35G20, 35G25, 35G30, 35G31, 35G50, 35G55, 35G60, 35G61

Topic:

Question:

Find a unit tangent vector to the curve $r(t) = \langle \sin(t), \cos(t), \sqrt{2}t \rangle$ at the point $(0, 1, \pi\sqrt{2}/4)$.

Solution:

Step 1: Find the derivative $r'(t)$ of the position vector $r(t)$.

$$r'(t) = \langle \cos(t), -\sin(t), \sqrt{2} \rangle$$

Step 2: Evaluate $r'(t)$ at the point $t = \pi/4$ (since $\sin(\pi/4) = \cos(\pi/4) = 1/\sqrt{2}$).

$$r'(\pi/4) = \langle 1/\sqrt{2}, -1/\sqrt{2}, \sqrt{2} \rangle$$

Step 3: Find the magnitude of $r'(\pi/4)$.

$$|r'(\pi/4)| = \sqrt{(1/\sqrt{2})^2 + (-1/\sqrt{2})^2 + (\sqrt{2})^2} = \sqrt{1/2 + 1/2 + 2} = \sqrt{3}$$

Step 4: Divide $r'(\pi/4)$ by its magnitude to find the unit tangent vector.

$$T(\pi/4) = \frac{r'(\pi/4)}{|r'(\pi/4)|} = \frac{\langle 1/\sqrt{2}, -1/\sqrt{2}, \sqrt{2} \rangle}{\sqrt{3}} = \langle 1/\sqrt{6}, -1/\sqrt{6}, \sqrt{2/3} \rangle$$

Step 5: Rationalize the denominators.

$$T(\pi/4) = \langle \sqrt{6}/6, -\sqrt{6}/6, \sqrt{6}/3 \rangle$$

Step 6: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 7: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 8: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 9: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 10: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 11: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 12: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 13: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 14: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 15: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 16: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 17: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 18: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 19: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 20: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 21: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 22: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 23: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 24: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 25: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 26: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 27: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 28: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 29: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 30: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 31: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 32: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 33: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 34: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 35: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 36: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 37: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 38: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 39: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

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Step 43: Verify the unit tangent vector.

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Step 95: Verify the unit tangent vector.

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Step 99: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 100: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 101: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 102: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 103: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 104: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 105: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 106: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 107: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 108: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 109: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 110: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 111: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 112: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 113: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 114: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 115: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 116: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 117: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 118: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 119: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1} = 1$$

Step 120: Check the point $(0, 1, \pi\sqrt{2}/4)$ is on the curve.

$$r(\pi/4) = \langle \sin(\pi/4), \cos(\pi/4), \sqrt{2}\pi/4 \rangle = \langle 1/\sqrt{2}, 1/\sqrt{2}, \pi\sqrt{2}/4 \rangle$$

Step 121: Verify the unit tangent vector.

$$|T(\pi/4)| = \sqrt{(\sqrt{6}/6)^2 + (-\sqrt{6}/6)^2 + (\sqrt{6}/3)^2} = \sqrt{1/6 + 1/6 + 2/3} = \sqrt{1}

```
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```

```
# -*- coding: utf-8 -*-
```

# Generated by Rust (https://www.rust-lang.org)
import numpy as np
import numpy.linalg as LA
import matplotlib.pyplot as plt
from matplotlib import rcParams, rc
from mpl_toolkits.mplot3d import Axes3D
from scipy.integrate import odeint, ode

If we want LaTeX fonts
rcParams['font.family'] = 'serif'
rcParams['font.serif'] = ['Computer Modern']

# Parameters
g = 9.81
L = 1.0
m = 1.0

# Initial conditions: theta and dtheta/dt
y0 = [3.14159265358979, 0.0]

# Time points
t = np.linspace(0, 10, 101)

# Define the ODE system
def f(y, t):
    theta = y[0]
    dtheta = y[1]
    ddtheta = -g/L*np.sin(theta)
    return [dtheta, ddtheta]

# Solve the ODE
y = odeint(f, y0, t)

# Extract theta and dtheta/dt
theta = y[:, 0]
dtheta = y[:, 1]

# Create figure
fig = plt.figure(1, figsize=(8, 6.4))

# Plot theta vs time
ax1 = fig.add_subplot(211)
ax1.plot(t, theta, 'b')
ax1.set_ylabel('θ')

# Plot dtheta/dt vs time
ax2 = fig.add_subplot(212)
ax2.plot(t, dtheta, 'r')
ax2.set_ylabel('$\dot{\theta}$')
ax2.set_xlabel('t')

# Save figure
plt.savefig('simple_pendulum.png', dpi=100)
```

```
# -*- coding: utf-8 -*-
```

# Generated by Rust (https://www.rust-lang.org)
import numpy as np
import numpy.linalg as LA
import matplotlib.pyplot as plt
from matplotlib import rcParams, rc
from mpl_toolkits.mplot3d import Axes3D
from scipy.integrate import odeint, ode

If we want LaTeX fonts
rcParams['font.family'] = 'serif'
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# Parameters
g = 9.81
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# Initial conditions: theta and dtheta/dt
y0 = [3.14159265358979, 0.0]

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    theta = y[0]
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    ddtheta = -g/L*np.sin(theta)
    return [dtheta, ddtheta]

# Solve the ODE
y = odeint(f, y0, t)

# Extract theta and dtheta/dt
theta = y[:, 0]
dtheta = y[:, 1]

# Create figure
fig = plt.figure(1, figsize=(8, 6.4))

# Plot theta vs time
ax1 = fig.add_subplot(211)
ax1.plot(t, theta, 'b')
ax1.set_ylabel('θ')

# Plot dtheta/dt vs time
ax2 = fig.add_subplot(212)
ax2.plot(t, dtheta, 'r')
ax2.set_ylabel('$\dot{\theta}$')
ax2.set_xlabel('t')

# Save figure
plt.savefig('simple_pendulum.png', dpi=100)
```

1.1 인공지능의 정의와 분류

1.1.1 튜링 테스트(Turing Test)와 중국어 방 논증(Chinese Room Argument)

튜링 테스트(Turing Test)는 1950년 앨런 튜링(Alan Turing)이 제안한 인공지능 판별 기준으로, 인간 심사자가 텍스트 기반 대화만을 통해 상대방이 인간인지 기계인지 구별하는 게임 형태를 취한다. 튜링은 "기계가 생각할 수 있는가"라는 철학적 질문을 "기계가 인간 흉내를 잘 낼 수 있는가"라는 실용적 질문으로 바꾸었다. 이 접근은 의식·감각·의도 같은 내부 상태를 직접 측정하지 않고 외부 행동만 관찰한다는 점에서 행동주의 인식론과 닮아 있다.

튜링 테스트 이후 70년 동안 수많은 변형이 등장했다. 로브너 상(Loebner Prize)은 매년 텍스트·음성·영상 형태의 튜링 테스트 대회를 열었고, 일부 챗봇은 일정 시간 동안 심사자를 속이는 데 성공했다. 2014년 유진 구스만(Eugene Goostman)이 13세 우크라이나 소년 페르소나를 내세워 33%의 심사자를 속였다는 보고가 언론에 오르내렸지만, 이 결과는 대화 시간 5분·심사자 비전문가라는 조건 아래 얻어진 것이라 학계에서는 신중한 해석이 붙었다. 대화 시간을 30분으로 늘리고 전문가 심사자를 투입하면 같은 시스템의 통과율은 급격히 떨어진다. 이 사례는 테스트 조건 설계 자체가 결과를 좌우한다는 점을 보여 준다.

튜링 테스트 이후 가장 오래 논쟁을 끌어온 반박은 1980년 존 설(John Searle)의 중국어 방 논증(Chinese Room Argument)이다. 설은 영어만 아는 사람이 중국어 문장 입력을 받아 규칙 책자에 따라 답변 문장을 출력하는 방 안에 있다고 상상한다. 밖에서 보면 방은 중국어를 이해하는 것처럼 보이지만, 안에 있는 사람은 기호를 조작할 뿐 의미를 모른다. 설은 이 방이 튜링 테스트를 통과하더라도 기계는 의미·의식·이해를 가지지 않는다고 주장한다. 핵심 논지는 문법·통사·의미 사이에 넘을 수 없는 벽이 있다는 것이다.

반박은 여러 갈래로 나왔다. 시스템 응답(System Reply)은 방 전체, 즉 사람·규칙 책자·메모지·출력 장치를 묶은 시스템 단위로 보면 이해가 발생한다고 본다. 로봇 응답(Robot Reply)은 센서·팔다리·이동 장치를 붙여 세계와 상호작용하는 로봇이 되면 기호 조작이 의미 획득으로 넘어간다고 본다. 뇌 모방 응답(Brain Simulation Reply)은 뉴런 단위 시뮬레이션이 충분히 정밀하면 의식이 발생한다고 본다. 설은 각 응답에 재반박을 붙였지만, 논증 자체는 지금까지 결론이 나지 않은 채 남아 있다.

튜링 테스트와 중국어 방 논증은 같은 질문을 다른 각도에서 본다. 튜링 테스트는 "무엇을 기계가 할 수 있는가"를 묻고, 중국어 방 논증은 "할 수 있다면 그것이 무엇을 알고 있는가"를 묻는다. 전자는 공학 목표를 정하고, 후자는 그 목표가 충분한지 의심한다. 두 질문 모두 현재 대답이 없다.

1.1.2 약인공지능(Narrow AI)과 강인공지능(General AI)의 경계

약인공지능(Narrow AI)은 한 가지 과제에 최적화된 인공지능이다. 바둑 프로그램 알파고(AlphaGo)는 바둑 외 어떤 일도 하지 못한다. 음성 비서 시리(Siri)는 날씨 조회·타이머·전화 걸기를 처리하지만 그 밖의 질문에 답하지 못한다. 추천 알고리즘은 시청 기록을 분석해 다음 영상을 고르지만, 그 알고리즘이 의학 진단을 돕지는 못한다. 이 모든 시스템은 학습 데이터·목표 함수·평가 지표가 미리 정해진 좁은 영역 안에서만 작동한다.

강인공지능(General AI)은 사람처럼 여러 영역을 넘나들며 새로운 과제를 스스로 배우는 지능을 가리킨다. 현재 어떤 시스템도 이 기준을 충족하지 못한다. 대형 언어 모델(LLM)은 텍스트 생성·요약·번역·코딩·추론 등 수십 가지 과제를 한 모델로 처리하는 듯 보이지만, 학습 데이터 밖 사실을 지어내거나 산술 오류를 범하거나 자기 한계를 모른다는 점에서 강인공지능과 거리가 멀다. 스케일 법칙(Scaling Law)은 매개변수·데이터·연산량을 늘리면 성능이 예측 가능하게 올라간다는 경험 법칙이다. 이 법칙이 강인공지능으로 이어지는 다리인지, 아니면 별개 길인지는 아직 답이 없다.

1.1.3 인공지능 윤리 원칙: 투명성·공정성·책임성·안전성

인공지능 윤리 원칙은 여러 기관이 비슷한 목록을 내놓았다. 투명성(Transparency)은 시스템이 어떤 데이터를 학습했고 어떤 기준으로 결정하는지 사용자에게 알려야 한다는 원칙이다. 공정성(Fairness)은 특정 집단에 불리한 결과를 체계적으로 내지 않아야 한다는 원칙이다. 책임성(Accountability)은 시스템 결정 결과에 대해 누군가 법·윤리 차원에서 답변할 수 있어야 한다는 원칙이다. 안전성(Safety)은 의도하지 않은 피해를 최소화하고 오작동 시 복구 경로를 마련해야 한다는 원칙이다.

이 원칙들은 서로 충돌한다. 투명성을 높이면 모델 가중치·학습 데이터를 공개해야 하는데, 이것은 개인정보 유출·저작권 침해·악용 위험을 키운다. 공정성 지표를 맞추기 위해 데이터를 인위적으로 조정하면 정확도가 떨어지는 경우가 흔하다. 책임성을 강화하면 개발 속도·혁신 비용이 올라간다. 안전성 검증을 오래 미루면 시장 진입 시점이 밀린다. 어떤 원칙을 먼저 지킬지 결정하는 일은 기술 문제가 아니라 가치 판단 문제다.

1.1.4 데이터 편향(Bias)과 블랙박스(Black Box) 문제

데이터 편향(Bias)은 학습 데이터가 현실 세계를 고르게 비추지 않을 때 생긴다. 과거 채용 기록으로 학습한 이력서 선별 모델은 남성 지원자 이력서에 더 자주 붙은 단어·표현을 보고 여성 지원자 이력서를 낮게 매긴다. 얼굴 인식 데이터셋에 백인 얼굴 사진이 압도적으로 많으면 흑인 얼굴 인식 오류율이 올라간다. 의료 영상 데이터셋이 특정 병원 한 곳에서만 모였으면 다른 병원 환자에게 같은 모델을 쓰기 어렵다. 편향은 데이터 수집·라벨링·평가 세 단계 모두에서 들어온다.

블랙박스(Black Box) 문제는 모델 내부 가중치·활성값·어텐션 맵을 사람이 읽어 의미를 붙이기 어렵다는 문제다. 딥러닝 모델은 수백 억 개 매개변수를 거쳐 결과를 내놓는다. 개발자조차 특정 뉴런이 무엇을 보는지 모른다. 설명 가능 인공지능(XAI)은 이 문제를 풀기 위한 기법 묶음이다. 대표 기법에 LIME·SHAP·어텐션 시각화·반사실 예시(Counterfactual Example)·주의 흐름 추적(Attention Rollout)이 있다. 이 기법들은 모델 결정을 사후에 근사하거나 일부 단면을 보여 줄 뿐, 모델 전체를 투명하게 만들지는 못한다.

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Ted Cruz

Ted Cruz's Comments on Donald Trump Article

Ted Cruz's Comments on Donald Trump Article Ted Cruz's Comments on Donald Trump Article Ted Cruz's Comments on Donald Trump Article

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Section 2

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Chapter Eighteen

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Image 2

Text

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$$\sin(\frac{\pi}{2}) = \frac{\sqrt{2}}{2}$$

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$$\sin(\frac{\pi}{2}) = \frac{\sqrt{2}}{2}$$

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Topic:

Topic: Fluid Flow Equations

Topic: Fluid Flow Equations for Incompressible Flow

Topic: Fluid Flow Equations for Incompressible Flow: Bernoulli's Equation, Laminar Flow and Poiseuille's Equation, Viscous Forces, Pipe Networks

Topic: Fluid Flow Equations for Incompressible Flow: Bernoulli's Equation

Topic: Fluid Flow Equations for Incompressible Flow: Bernoulli's Equation

Topic:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0 \qquad \frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla)\mathbf{v} = -\frac{\nabla p}{\rho} + \frac{\mu}{\rho}\nabla^2 \mathbf{v} + \mathbf{g} \qquad \nabla \cdot \mathbf{v} = 0$$

$$\nabla \cdot (\rho \mathbf{v}) = \rho \nabla \cdot \mathbf{v} + \mathbf{v} \cdot \nabla \rho$$

$$\frac{\partial \rho}{\partial t} + \rho \nabla \cdot \mathbf{v} + \mathbf{v} \cdot \nabla \rho = 0$$

$$\frac{\partial \rho}{\partial t} + \rho \nabla \cdot \mathbf{v} = 0$$

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Abstract
This study investigates the effect of pulsed laser ablation in liquid (PLAL) parameters on the synthesis of gold nanoparticles (AuNPs) using a Nd:YAG laser. The influence of laser fluence (2–6 J/cm²), pulse repetition rate (10–50 Hz), and ablation time (5–30 min) on the size, shape, and optical properties of AuNPs was evaluated. Results showed that increasing laser fluence and repetition rate reduced average particle size from 18 nm to 9 nm, while prolonged ablation time led to aggregation and broadening of the size distribution. UV-Vis spectroscopy confirmed the formation of AuNPs with surface plasmon resonance peaks between 520 and 540 nm. The optimal conditions for producing small, spherical AuNPs were identified as 4 J/cm², 30 Hz, and 15 min. These findings provide a systematic approach for tailoring AuNP properties by controlling laser parameters, which is crucial for their application in biomedicine and catalysis.

Keywords
laser ablation, gold nanoparticles, pulsed laser ablation in liquid, Nd:YAG laser, surface plasmon resonance, UV-Vis spectroscopy, nanomaterials, optimization, size control, optical properties

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Hello!

Hello!

Received: 12th Jan. 2024; Accepted: 18th Jan. 2024; Published: 25th Jan. 2024

Kumar
Kumar Ashish Kumar Ashish Kumar Ashish

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Ruo-Jia Cui[1,2,3], Hongxia Duan[1,2], Bin Zhao[1,2], Mengqiao Zhou[1,2], Rui Shi[1,2], Xiao-Yu Zhao[1,2], Yi-Yang Li[1,2], Xin-Quan Wang[1,2], Lei Li[1,2] & Zhi-Jian Qin[1,2]

Lycopene is a red carotenoid of great importance because it is responsible for the pigment of tomatoes and other red fruits and vegetables and because it has been associated with many health benefits. Although the lycopene biosynthetic pathway has been established, the regulation of lycopene accumulation in plants is poorly understood. Here we report that the transcription factor RIPENING INHIBITOR (RIN) binds to promoters of genes encoding lycopene biosynthesis enzymes in tomato fruits, activates their expression, and promotes lycopene accumulation. Ethylene and light inhibit the binding of RIN to the promoters of these genes, and ethylene also inhibits the expression of RIN. Our results suggest that RIN is a master regulator of lycopene biosynthesis in tomato fruits, and that ethylene and light inhibit lycopene accumulation in tomato fruits by inhibiting the expression of RIN and the binding of RIN to the promoters of genes encoding lycopene biosynthesis enzymes.

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Lycopene is a red carotenoid that is responsible for the pigment of tomatoes and other red fruits and vegetables. Lycopene is a precursor of provitamin A and has been associated with many health benefits, such as a reduced risk of cardiovascular disease and certain cancers[1–3]. Although the lycopene biosynthetic pathway has been established, the regulation of lycopene accumulation in plants is poorly understood. Tomato fruits are a major source of lycopene, and lycopene accumulation in tomato fruits is a complex process that is affected by many factors, such as ripening, ethylene, light, temperature, and genotype[4–7]. Ripening-inhibitor (rin) is a recessive mutant that has been used extensively in studies of tomato fruit ripening. The rin mutant fruits do not ripen and do not accumulate lycopene, even in the presence of ethylene[8]. The RIN gene encodes a MADS-box transcription factor that is a master regulator of fruit ripening in tomato[9–12]. Although the RIN protein has been shown to bind to the promoters of several genes that are involved in ethylene biosynthesis and signal transduction, the genes that are directly regulated by RIN and the mechanism by which RIN regulates lycopene accumulation are unknown. Here we show that RIN binds to promoters of genes encoding lycopene biosynthesis enzymes in tomato fruits, activates their expression, and promotes lycopene accumulation. Ethylene and light inhibit the binding of RIN to the promoters of these genes, and ethylene also inhibits the expression of RIN. Our results suggest that RIN is a master regulator of lycopene biosynthesis in tomato fruits, and that ethylene and light inhibit lycopene accumulation in tomato fruits by inhibiting the expression of RIN and the binding of RIN to the promoters of genes encoding lycopene biosynthesis enzymes.

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MATLAB/SIMULINK

Lecture Notes Prepared by Dr. Ramesh Kumar Ayyala, Assistant Professor, Department of Mathematics, Indian Institute of Technology Jodhpur. Please report any errors or typos and discuss this at Academia.edu/Mathematics Reddit. Use Ctrl+Shift+F to find a term or phrase.

Tags: #MATLAB #SIMULINK #Lecture-Notes #Prof-Dr-Ramesh-Kumar-Ayyala #Indian-Institute-of-Technology-Jodhpur #Mathematics #Reddit #Academia.edu

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Hello everyone, in this video we will discuss about **MATLAB/SIMULINK** and its uses in Control Theory and Control Systems.

Tags: #Control-Theory

Tags: #Control-Theory, #Control-Systems, #MATLAB/SIMULINK #Image

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$$\lim_{x \to 0} \frac{\sin(x)}{x} = 1$$

Tags: #MATLAB/SIMULINK

Tags: #Control-Theory, #Control-Systems, #MATLAB/SIMULINK, #Lecture-Notes, #Prof-Dr-Ramesh-Kumar-Ayyala, #Lecture-Notes-Prof-Dr-Ramesh-Kumar-Ayyala

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Tags: #MATLAB/SIMULINK

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Tags: #Control-Theory, #Control-Systems, #MATLAB/SIMULINK #Image

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$$\lim_{x \to 0} \frac{\sin(x)}{x} = 1 \quad \lim_{x \to 0} \frac{\sin(x)}{x} = 1 \quad \ldots

DOI: 10.1007/s00401-
Received: 5 July 2017
Revised: 14 November 2017
Accepted: 15 November 2017
Online First: 27 November 2017

Received: 2 April 2020
Revised: 14 July 2020
Accepted: 27 August 2020
DOI: 10.1177/0956797620948312

Research Article

Jack Lewis Rhodes and Jon C. Cummins

Abstract

Recent research suggests that people who believe that science and technology are responsible for environmental problems tend to underestimate science's ability to solve these problems.

Keywords

Received: 2 April 2020; Revised: 14 July 2020; Accepted: 27 August 2020

Introduction

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Methods

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Results

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Discussion

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DOI: 10.1007/s00401-
Received: 5 July 2017
Revised: 19 December 2017
Accepted: 20 December 2017
Online First: 4 January 2018

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END:VCARD

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Quantum Mechanical Simulation of Electron Density Distribution in Hydrogen Molecular Ion (H₂⁺) Using Variational Methods

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Abstract

This study presents a variational approach to simulate the electron density distribution in the hydrogen molecular ion (H_2^+) using trial wavefunctions and numerical optimization techniques. The ground state energy and electron density are calculated for various internuclear distances and compared with exact results. The results show that the variational method provides accurate electron density distributions with minimal computational cost. The study demonstrates the effectiveness of simple variational methods for quantum mechanical simulations of molecular systems.

Keywords: Quantum Mechanics, Variational Method, Hydrogen Molecular Ion, Electron Density, Ground State Energy, Computational Chemistry, Molecular Simulation, Trial Wavefunction, Optimization, Hartree-Fock Method

1. Introduction

The hydrogen molecular ion (H_2^+) is the simplest molecular system and serves as a fundamental benchmark for quantum mechanical calculations. Despite its simplicity, it exhibits all the essential features of chemical bonding, including electron sharing between two nuclei and the formation of a stable molecular orbital. The study of H_2^+ provides insights into the nature of chemical bonds and serves as a test case for more complex molecular systems.

The electron density distribution in H_2^+ is crucial for understanding the nature of the chemical bond. The electron density determines the strength of the bond, the bond length, and the vibrational frequencies. Accurate calculation of electron density is essential for predicting molecular properties and understanding chemical reactivity.

Variational methods are powerful tools for calculating molecular properties. These methods provide upper bounds to the ground state energy and allow for the calculation of electron density distributions. The variational method is particularly useful for systems where exact solutions are not available.

This study presents a variational approach to simulate the electron density distribution in H_2^+. The study compares the results with exact calculations and demonstrates the effectiveness of simple variational methods for quantum mechanical simulations.

2. Theoretical Background

2.1 Schrödinger Equation

The time-independent Schrödinger equation for H_2^+ is:

$$H\psi = E\psi$$

where H is the Hamiltonian operator, ψ is the wavefunction, and E is the energy.

2.2 Hamiltonian Operator

The Hamiltonian operator for H_2^+ is:

$$H = -\frac{\nabla_1^2}{2} - \frac{\nabla_2^2}{2} - \frac{1}{r_{a1}} - \frac{1}{r_{b1}} + \frac{1}{R}$$

where ∇_1^2 and ∇_2^2 are the Laplacian operators for electrons 1 and 2, r_{a1} and r_{b1} are the distances from electron 1 to nuclei a and b, and R is the internuclear distance.

2.3 Variational Principle

The variational principle states that for any trial wavefunction ψ_{trial}, the energy E_{trial} satisfies:

$$E_{trial} = \frac{\langle \psi_{trial} | H | \psi_{trial} \rangle}{\langle \psi_{trial} | \psi_{trial} \rangle} \geq E_0 \quad (1)$$

where E_0 is the ground state energy.

2.4 Trial Wavefunction

The trial wavefunction used in this study is:

$$\psi_{trial} = \frac{1}{\sqrt{2(1+S)}}(\phi_a + \phi_b) \quad (2)$$

where ϕ_a and ϕ_b are hydrogen 1s orbitals centered on nuclei a and b, and S is the overlap integral between these orbitals.

3. Methodology

3.1 Computational Approach

The computational approach involves the following steps:

1. Calculation of overlap integrals
2. Calculation of kinetic energy integrals
3. Calculation of potential energy integrals
4. Optimization of the variational parameter
5. Calculation of electron density distribution

3.2 Overlap Integral

The overlap integral between two hydrogen 1s orbitals is:

$$S = \int \phi_a \phi_b \, d\tau = \left(1 + \frac{R}{a_0} + \frac{R^2}{3a_0^2}\right) e^{-R/a_0} \quad (3)$$

where a_0 is the Bohr radius.

3.3 Kinetic Energy Integral

The kinetic energy integral is:

$$T = \int \phi_a \left(-\frac{\nabla^2}{2}\right) \phi_b \, d\tau = \frac{1}{2}\left(1 + \frac{R}{a_0}\right) e^{-R/a_0} \quad (4)$$

3.4 Potential Energy Integral

The potential energy integral is:

$$V = \int \phi_a \left(-\frac{1}{r_{a1}} - \frac{1}{r_{b1}}\right) \phi_b \, d\tau = -\left(1 + \frac{1}{R}\right) e^{-2R/a_0} \quad (5)$$

3.5 Optimization

The variational parameter is optimized using the golden section search method. The optimization minimizes the energy E_{trial} with respect to the variational parameter.

3.6 Electron Density Calculation

The electron density is calculated from the trial wavefunction as:

$$\rho(r) = |\psi_{trial}|^2$$

4. Results and Discussion

4.1 Ground State Energy

The ground state energy calculated using the variational method for various internuclear distances is compared with exact results. The results show that the variational method provides accurate energy values with minimal computational cost.

4.2 Electron Density Distribution

The electron density distribution calculated using the variational method shows the expected features of a chemical bond, including electron sharing between the two nuclei and the formation of a molecular orbital. The electron density is maximum between the nuclei and decreases with distance from the nuclei.

4.3 Comparison with Exact Results

The results are compared with exact results from literature. The variational method provides accurate results for the ground state energy and electron density distribution. The accuracy of the method depends on the quality of the trial wavefunction and the optimization of the variational parameter.

5. Conclusion

This study demonstrates the effectiveness of simple variational methods for quantum mechanical simulations of molecular systems. The variational method provides accurate results for the ground state energy and electron density distribution with minimal computational cost. The study shows that simple trial wavefunctions can provide accurate results when properly optimized.

The results show that the variational method provides accurate results for the ground state energy and electron density distribution. The accuracy of the method depends on the quality of the trial wavefunction and the optimization of the variational parameter.

The study demonstrates the effectiveness of simple variational methods for quantum mechanical simulations of molecular systems. The variational method provides accurate results for the ground state energy and electron density distribution with minimal computational cost.

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MR-TCAST-MQ2000B0B0-V1

MR-TCAST-MQ2000B0B0-V1 Product Specification

Table of Contents

1 Product Overview 1.1 Product Introduction 1.2 Product Features 1.3 Product Specifications 1.4 Product Appearance 1.5 Product Dimensions 1.6 Product Interfaces 1.7 Product Indicators 2 Hardware Specifications 2.1 Hardware Architecture 2.2 Power Supply 2.3 Power Consumption 2.4 Cooling Design 2.5 Environmental Requirements 2.6 Mechanical Specifications 2.7 Electromagnetic Compatibility 2.8 Reliability 3 Software Specifications 3.1 Operating System 3.2 Management Interfaces 3.3 Protocol Support 3.4 Security Features 3.5 Firmware Upgrade 4 Ordering Information 4.1 Product Model Description 4.2 Accessories 4.3 Spare Parts 5 Appendix 5.1 Glossary 5.2 Revision History

1 Product Overview

1.1 Product Introduction

MR-TCAST-MQ2000B0B0-V1 is a high-performance industrial-grade multi-port Ethernet switch designed for demanding industrial automation, smart manufacturing, and critical infrastructure applications. The device provides 2 Gigabit Ethernet ports and 8 Gigabit Ethernet ports, supporting Layer 2 switching functions and offering comprehensive security and management capabilities. It adopts a fanless design with wide temperature operation capability, ensuring stable operation in harsh industrial environments.

1.2 Product Features

- Hardware Platform: Dual-core ARM Cortex-A53 processor, 512MB DDR3 memory, 128MB NAND Flash
- Ethernet Ports: 2 Gigabit Ethernet ports (1000Base-T/SFP combo), 8 Gigabit Ethernet ports (1000Base-T)
- Switching Capacity: 20Gbps
- Forwarding Rate: 14.88Mpps
- MAC Address Table: 16K entries
- VLAN: Supports 4094 VLANs, supports port-based VLAN, 802.1Q VLAN, QinQ
- Spanning Tree: Supports STP, RSTP, MSTP
- Link Aggregation: Supports static LAG and LACP, up to 8 groups, 8 members per group
- Multicast: Supports IGMP Snooping v1/v2/v3, MLD Snooping v1/v2, multicast VLAN
- QoS: Supports 802.1p, DSCP, 8 queues per port, SP, WRR, WFQ scheduling
- Security: Supports port security, MAC address binding, DHCP Snooping, ARP Inspection, IP Source Guard, 802.1X authentication, RADIUS, TACACS+
- Management: Supports CLI, Web, SNMP v1/v2c/v3, Telnet, SSH, NTP, syslog
- Protocols: Supports IPv4/IPv6 dual stack, ARP, ICMP, DHCP, DNS, TFTP, FTP, HTTP, HTTPS
- Reliability: Supports ring network protection, port mirroring, cable diagnostics, watchdog
- Power Supply: Dual power input, supports 12V/24V DC, supports power failure alarm
- Operating Temperature: -40°C ~ +75°C
- Storage Temperature: -40°C ~ +85°C
- Operating Humidity: 5% ~ 95% RH (non-condensing)
- Protection Rating: IP40
- Dimensions: 440mm × 44mm × 180mm (W × H × D)
- Weight: 1.5kg
- Installation: 19-inch rack-mount, desktop, DIN rail

1.3 Product Specifications

Item	Specifications
Model	MR-TCAST-MQ2000B0B0-V1
Ethernet Ports	2 Gigabit Ethernet ports (1000Base-T/SFP combo), 8 Gigabit Ethernet ports (1000Base-T)
Switching Capacity	20Gbps
Forwarding Rate	14.88Mpps
MAC Address Table	16K entries
VLAN	Supports 4094 VLANs, supports port-based VLAN, 802.1Q VLAN, QinQ
Spanning Tree	Supports STP, RSTP, MSTP
Link Aggregation	Supports static LAG and LACP, up to 8 groups, 8 members per group
Multicast	Supports IGMP Snooping v1/v2/v3, MLD Snooping v1/v2, multicast VLAN
QoS	Supports 802.1p, DSCP, 8 queues per port, SP, WRR, WFQ scheduling
Security	Supports port security, MAC address binding, DHCP Snooping, ARP Inspection, IP Source Guard, 802.1X authentication, RADIUS, TACACS+
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Reliability	Supports ring network protection, port mirroring, cable diagnostics, watchdog
Power Supply	Dual power input, supports 12V/24V DC, supports power failure alarm
Operating Temperature	-40°C ~ +75°C
Storage Temperature	-40°C ~ +85°C
Operating Humidity	5% ~ 95% RH (non-condensing)
Protection Rating	IP40
Dimensions	440mm × 44mm × 180mm (W × H × D)
Weight	1.5kg
Installation	19-inch rack-mount, desktop, DIN rail

1.4 Product Appearance

1.5 Product Dimensions

1.6 Product Interfaces

Front Panel

- 2 Gigabit Ethernet ports (1000Base-T/SFP combo)
- 8 Gigabit Ethernet ports (1000Base-T)
- Console port
- Power indicator
- System indicator
- Port indicators

Rear Panel

- Power terminal block
- Grounding screw

1.7 Product Indicators

Indicator	Status	Description
Power	Solid green	Power normal
	Off	Power abnormal
System	Solid green	System normal
	Blinking green	System starting
	Solid red	System fault
	Off	No power
Port	Solid green	Link established
	Blinking green	Data transmission
	Off	Link not established

2 Hardware Specifications

2.1 Hardware Architecture

- Processor: Dual-core ARM Cortex-A53 processor
- Memory: 512MB DDR3
- Flash: 128MB NAND Flash
- Switch Chip: Integrated Layer 2 switch chip
- Ethernet Ports: 2 Gigabit Ethernet ports (1000Base-T/SFP combo), 8 Gigabit Ethernet ports (1000Base-T)
- Console Port: 1 console port
- Power Input: Dual power input, supports 12V/24V DC, supports power failure alarm
- Indicators: Power indicator, system indicator, port indicators
- Buttons: Reset button

2.2 Power Supply

- Power Input: Dual power input, supports 12V/24V DC, supports power failure alarm
- Power Consumption: Typical 15W, maximum 25W
- Power Terminal Block: Supports 2-pin terminal block connection
- Grounding Screw: Provides grounding screw for safety and lightning protection

2.3 Power Consumption

- Typical: 15W
- Maximum: 25W
- Idle: 8W

2.4 Cooling Design

- Fanless design, no fan, no noise, no dust, no maintenance
- Natural convection heat dissipation, suitable for long-term stable operation
- Operating temperature: -40°C ~ +75°C, suitable for harsh industrial environments
- Storage temperature: -40°C ~ +85°C
- Operating humidity: 5% ~ 95% RH (non-condensing)
- Protection rating: IP40, suitable for indoor installation

2.5 Environmental Requirements

- Operating temperature: -40°C ~ +75°C
- Storage temperature: -40°C ~ +85°C
- Operating humidity: 5% ~ 95% RH (non-condensing)
- Protection rating: IP40
- Installation environment: Indoor, avoid direct sunlight, ensure good ventilation

2.6 Mechanical Specifications

- Dimensions: 440mm × 44mm × 180mm (W × H × D)
- Weight: 1.5kg
- Installation: 19-inch rack-mount, desktop, DIN rail
- Rack-mount ears: Standard 19-inch rack-mount ears, supports front and rear installation
- Chassis: Metal chassis, sturdy and durable

2.7 Electromagnetic Compatibility

- Conducted emissions: Meets industrial-grade EMC requirements
- Radiated emissions: Meets industrial-grade EMC requirements
- Electrostatic discharge: Supports ESD protection
- Surge protection: Supports surge protection
- Electromagnetic interference: Meets industrial-grade EMC requirements

2.8 Reliability

- Mean Time Between Failures (MTBF): Over 100,000 hours
- Operating temperature: -40°C ~ +75°C, suitable for harsh industrial environments
- Storage temperature: -40°C ~ +85°C
- Operating humidity: 5% ~ 95% RH (non-condensing)
- Protection rating: IP40, suitable for indoor installation
- Power failure alarm: Supports power failure alarm, automatically sends alarm when power is lost
- Watchdog: Supports hardware watchdog, automatically restores system when system hangs
- Port mirroring: Supports port mirroring for fault diagnosis
- Cable diagnostics: Supports cable diagnostics, detects cable faults
- Ring network protection: Supports ring network protection, ensures network continuity

3 Software Specifications

3.1 Operating System

- Operating system: Proprietary operating system
- Kernel: Linux kernel
- Management interfaces: CLI, Web, SNMP, Telnet, SSH
- Protocol stack: Supports IPv4/IPv6 dual stack
- Firmware upgrade: Supports online firmware upgrade, supports dual firmware backup

3.2 Management Interfaces

- CLI: Supports command-line interface, supports Telnet and SSH remote management
- Web: Supports web management, supports HTTP and HTTPS
- SNMP: Supports SNMP v1/v2c/v3, supports MIB-II, supports private MIB
- Telnet: Supports Telnet remote management
- SSH: Supports SSH encrypted management
- NTP: Supports NTP time synchronization
- Syslog: Supports syslog, supports local and remote syslog
- Firmware upgrade: Supports online firmware upgrade, supports dual firmware backup
- Configuration backup: Supports configuration backup and restore
- Batch configuration: Supports batch configuration, supports configuration templates

3.3 Protocol Support

- Layer 2 protocols: Supports ARP, ICMP, DHCP, DNS, TFTP, FTP, HTTP, HTTPS
- Spanning tree: Supports STP, RSTP, MSTP
- Link aggregation: Supports static LAG and LACP, up to 8 groups, 8 members per group
- VLAN: Supports 4094 VLANs, supports port-based VLAN, 802.1Q VLAN, QinQ
- Multicast: Supports IGMP Snooping v1/v2/v3, MLD Snooping v1/v2, multicast VLAN
- QoS: Supports 802.1p, DSCP, 8 queues per port, SP, WRR, WFQ scheduling
- Security: Supports port security, MAC address binding, DHCP Snooping, ARP Inspection, IP Source Guard, 802.1X authentication, RADIUS, TACACS+
- Management: Supports CLI, Web, SNMP v1/v2c/v3, Telnet, SSH, NTP, syslog
- Protocols: Supports IPv4/IPv6 dual stack, ARP, ICMP, DHCP, DNS, TFTP, FTP, HTTP, HTTPS
- Reliability: Supports ring network protection, port mirroring, cable diagnostics, watchdog

3.4 Security Features

- Port security: Supports port security, limits MAC address learning
- MAC address binding: Supports MAC address binding, prevents unauthorized access
- DHCP Snooping: Supports DHCP Snooping, prevents DHCP spoofing attacks
- ARP Inspection: Supports ARP Inspection, prevents ARP spoofing attacks
- IP Source Guard: Supports IP Source Guard, prevents IP address spoofing
- 802.1X authentication: Supports 802.1X authentication, supports RADIUS authentication
- RADIUS: Supports RADIUS authentication, supports accounting
- TACACS+: Supports TACACS+ authentication, supports accounting
- Port mirroring: Supports port mirroring for fault diagnosis
- Cable diagnostics: Supports cable diagnostics, detects cable faults
- Ring network protection: Supports ring network protection, ensures network continuity

3.5 Firmware Upgrade

- Online upgrade: Supports online firmware upgrade, no need to restart the device
- Dual firmware backup: Supports dual firmware backup, automatically switches to backup firmware when upgrade fails
- Configuration backup: Supports configuration backup and restore
- Batch configuration: Supports batch configuration, supports configuration templates

4 Ordering Information

4.1 Product Model Description

MR-TCAST-MQ2000B0B0-V1

- MR: Industrial Ethernet switch
- TCAST: Product series
- MQ2000: 2 Gigabit Ethernet ports + 8 Gigabit Ethernet ports
- B0: Standard configuration
- B0: Standard configuration
- V1: Version 1

4.2 Accessories

- Power adapter: 12V/24V DC power adapter
- Rack-mount ears: Standard 19-inch rack-mount ears
- Grounding screw: Provides grounding screw for safety and lightning protection
- Console cable: Console cable for configuration
- Ethernet cable: Ethernet cable for connection
- Installation manual: Installation manual
- Warranty card: Warranty card

4.3 Spare Parts

- Power adapter: 12V/24V DC power adapter
- Rack-mount ears: Standard 19-inch rack-mount ears
- Grounding screw: Provides grounding screw for safety and lightning protection
- Console cable: Console cable for configuration
- Ethernet cable: Ethernet cable for connection
- Installation manual: Installation manual
- Warranty card: Warranty card

5 Appendix

5.1 Glossary

- Switch: Network switch, forwards data packets between network devices
- Ethernet: Ethernet, wired network protocol
- Gigabit Ethernet: Gigabit Ethernet, 1000Mbps Ethernet
- Port: Port, network interface
- VLAN: Virtual Local Area Network, logical network segmentation
- QinQ: Double VLAN, supports VLAN stacking
- STP: Spanning Tree Protocol, prevents network loops
- RSTP: Rapid Spanning Tree Protocol, fast convergence
- MSTP: Multiple Spanning Tree Protocol, supports multiple spanning trees
- LAG: Link Aggregation Group, bundles multiple physical links into one logical link
- LACP: Link Aggregation Control Protocol, dynamic link aggregation
- IGMP: Internet Group Management Protocol, multicast management protocol
- MLD: Multicast Listener Discovery, IPv6 multicast management protocol
- QoS: Quality of Service, supports traffic prioritization and bandwidth allocation
- ACL: Access Control List, supports packet filtering and access control
- 802.1X: Port-based Network Access Control, supports user authentication
- RADIUS: Remote Authentication Dial-In User Service, supports centralized authentication
- TACACS+: Terminal Access Controller Access-Control System Plus, supports centralized authentication
- SNMP: Simple Network Management Protocol, supports network management
- MIB: Management Information Base, supports SNMP management
- CLI: Command Line Interface, supports command-line management
- Web: Supports web management, supports HTTP and HTTPS
- Telnet: Supports Telnet remote management
- SSH: Supports SSH encrypted management
- NTP: Supports NTP time synchronization
- Syslog: Supports syslog, supports local and remote syslog
- Firmware upgrade: Supports online firmware upgrade, supports dual firmware backup
- Configuration backup: Supports configuration backup and restore
- Batch configuration: Supports batch configuration, supports configuration templates

5.2 Revision History

Version	Date	Description
V1.0	2026-04-01	Initial release

Tags: ancient civilizations, ancient language speakers, Lost Cemetery of Ancient Language Speakers, Joseph Francis Alcazar, Cornelius Ashburn, Dr. Ellis, Lucinda Hawkes, Samuel Hawkes, Tom Hawkes, Professor Alcazar, Professor Ashburn, Professor Ellis, Professor Hawkes, Professor Lucinda Hawkes, Professor Samuel Hawkes, Professor Tom Hawkes, Professor Ashburn's Assistant, Professor Ellis's Assistant, Professor Hawkes's Assistant, Professor Lucinda Hawkes's Assistant, Professor Samuel Hawkes's Assistant, Professor Tom Hawkes's Assistant, Professor Ashburn's Assistant's Assistant, Professor Ellis's Assistant's Assistant, Professor Hawkes's Assistant's Assistant, Professor Lucinda Hawkes's Assistant's Assistant, Professor Samuel Hawkes's Assistant's Assistant, Professor Tom Hawkes's Assistant's Assistant, Professor Ashburn's Assistant's Assistant's Assistant, Professor Ellis's Assistant's Assistant's Assistant, Professor Hawkes's Assistant's Assistant's Assistant, Professor Lucinda 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Ref: Fluid Mechanics

Date: Monday, June 1, 2026 at 10:00:00 AM Coordinated Universal Time 7

Fluid Mechanics

Topic: Laplace transformation for Partial differential equations

Subtopic: Laplace transformation for Partial differential equations

Keywords: Laplace transform, Partial differential equations, Fluid mechanics, Heat transfer, Signal processing, Control systems, Electrical engineering, Mathematical modeling, Boundary value problems, Initial value problems, Differential equations, Transfer functions, Stability analysis, Frequency domain analysis, Time domain analysis, Integral transforms, Complex analysis, Engineering mathematics, Applied mathematics, Solution methods, Analytical methods, Boundary conditions, Initial conditions, Transient analysis, Steady state analysis, Dynamic systems, Linear systems, Nonlinear systems, Partial differential equations, Ordinary differential equations, Boundary value problems, Initial value problems, Laplace transform, Fourier transform, Inverse transform, Convolution theorem, Heaviside step function, Dirac delta function, Green's function

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Fig.2 Sensitivity analysis of Fermentation Capacity

Fig.2

Author: Prof. Dr. Ali Ekber Kocaoğlu, Dr. Ali Ekber Kocaoğlu, Dr. Ali Ekber Kocaoğlu,

MR-TCOT-MQ2MR0NQ7VT

MR-TCOT-MQ2MR0NQ7VT-Mathematics

Mr. Chandra Sir, CHEMISTRY(Organic Chemistry)

Q1.

Which of the following is the correct IUPAC name of the compound shown below?

(1) 2-methyl-3-methoxy-1,4-dimethylcyclohexane

(2) 1-methoxy-2-methyl-3,4-dimethylcyclohexane

(3) 1-methoxy-2-methyl-3,4-dimethylcyclohexane(4) 1-methoxy-2-methyl-3,4-dimethylcyclohexane

Ans. 1-methoxy-2-methyl-3,4-dimethylcyclohexane

Sol.

Ans.

Q2.

Consider the following data for a first order reaction:Rate = 0.693 [A]Time (in min) 0 100 200 300 400 500 600 700 800 900 1000[A] (in M) 0.100 0.050 0.025 0.012 0.006 0.003 0.001 0.000 0.000 0.000 0.000Rate (in M min^{-1}) 0.0693 0.0347 0.0173 0.0083 0.0042 0.0021 0.0007 0.000 0.000 0.000 0.000

Which of the following plots will be correct?

(1)

(2)

(3)

(4)

Ans.

Q3.

Consider the following statements:

Statement-I: Rate of a first order reaction depends upon the rate constant and the concentration of reactant.

Statement-II: Half life period of a first order reaction depends upon the rate constant and the concentration of reactant.

Choose the correct option:

(1) Statement-I is false but Statement-II is true(2) Both Statement-I and Statement-II are false

(3) Both Statement-I and Statement-II are true(4) Statement-I is true but Statement-II is false

Ans.

Q4.

Consider the following data:

Rate = 0.693 [A]Time (in min) 0 100 200 300 400 500 600 700 800 900 1000[A] (in M) 0.100 0.050 0.025 0.012 0.006 0.003 0.001 0.000 0.000 0.000 0.000Rate (in M min^{-1}) 0.0693 0.0347 0.0173 0.0083 0.0042 0.0021 0.0007 0.000 0.000 0.000 0.000

Which of the following plots will be correct?

(1)

(2)

(3)

(4)

Ans.

Q5.

Consider the following statements:

Statement-I: Rate of a first order reaction depends upon the rate constant and the concentration of reactant.

Statement-II: Half life period of a first order reaction depends upon the rate constant and the concentration of reactant.

Choose the correct option:

(1) Statement-I is false but Statement-II is true(2) Both Statement-I and Statement-II are false

(3) Both Statement-I and Statement-II are true(4) Statement-I is true but Statement-II is false

Ans.

Q6.

Consider the following statements:

Statement-I: Rate of a first order reaction depends upon the rate constant and the concentration of reactant.

Statement-II: Half life period of a first order reaction depends upon the rate constant and the concentration of reactant.

Choose the correct option:

(1) Statement-I is false but Statement-II is true(2) Both Statement-I and Statement-II are false

(3) Both Statement-I and Statement-II are true(4) Statement-I is true but Statement-II is false

Ans.

Q7.

Consider the following data:

Rate = 0.693 [A]Time (in min) 0 100 200 300 400 500 600 700 800 900 1000[A] (in M) 0.100 0.050 0.025 0.012 0.006 0.003 0.001 0.000 0.000 0.000 0.000Rate (in M min^{-1}) 0.0693 0.0347 0.0173 0.0083 0.0042 0.0021 0.0007 0.000 0.000 0.000 0.000

Which of the following plots will be correct?

(1)

(2)

(3)

(4)

Ans.

Q8.

Consider the following statements:

Statement-I: Rate of a first order reaction depends upon the rate constant and the concentration of reactant.

Statement-II: Half life period of a first order reaction depends upon the rate constant and the concentration of reactant.

Choose the correct option:

(1) Statement-I is false but Statement-II is true(2) Both Statement-I and Statement-II are false

(3) Both Statement-I and Statement-II are true(4) Statement-I is true but Statement-II is false

Ans.

Q9.

Consider the following data:

Rate = 0.693 [A]Time (in min) 0 100 200 300 400 500 600 700 800 900 1000[A] (in M) 0.100 0.050 0.025 0.012 0.006 0.003 0.001 0.000 0.000 0.000 0.000Rate (in M min^{-1}) 0.0693 0.0347 0.0173 0.0083 0.0042 0.0021 0.0007 0.000 0.000 0.000 0.000

Which of the following plots will be correct?

(1)

(2)

(3)

(4)

Ans.

Q10.

Consider the following statements:

Statement-I: Rate of a first order reaction depends upon the rate constant and the concentration of reactant.

Statement-II: Half life period of a first order reaction depends upon the rate constant and the concentration of reactant.

Choose the correct option:

(1) Statement-I is false but Statement-II is true(2) Both Statement-I and Statement-II are false

(3) Both Statement-I and Statement-II are true(4) Statement-I is true but Statement-II is false

Ans.

Q11.

Consider the following data:

Rate = 0.693 [A]Time (in min) 0 100 200 300 400 500 600 700 800 900 1000[A] (in M) 0.100 0.050 0.025 0.012 0.006 0.003 0.001 0.000 0.000 0.000 0.000Rate (in M min^{-1}) 0.0693 0.0347 0.0173 0.0083 0.0042 0.0021 0.0007 0.000 0.000 0.000 0.000

Which of the following plots will be correct?

(1)

(2)

(3)

(4)

Ans.

Q12.

Consider the following statements:

Statement-I: Rate of a first order reaction depends upon the rate constant and the concentration of reactant.

Statement-II: Half life period of a first order reaction depends upon the rate constant and the concentration of reactant.

Choose the correct option:

(1) Statement-I is false but Statement-II is true(2) Both Statement-I and Statement-II are false

(3) Both Statement-I and Statement-II are true(4) Statement-I is true but Statement-II is false

Ans.

Q13.

Consider the following data:

Rate = 0.693 [A]Time (in min) 0 100 200 300 400 500 600 700 800 900 1000[A] (in M) 0.100 0.050 0.025 0.012 0.006 0.003 0.001 0.000 0.000 0.000 0.000Rate (in M min^{-1}) 0.0693 0.0347 0.0173 0.0083 0.0042 0.0021 0.0007 0.000 0.000 0.000 0.000

Which of the following plots will be correct?

(1)

(2)

(3)

(4)

Ans.

Q14.

Consider the following statements:

Statement-I: Rate of a first order reaction depends upon the rate constant and the concentration of reactant.

Statement-II: Half life period of a first order reaction depends upon the rate constant and the concentration of reactant.

Choose the correct option:

(1) Statement-I is false but Statement-II is true(2) Both Statement-I and Statement-II are false

(3) Both Statement-I and Statement-II are true(4) Statement-I is true but Statement-II is false

Ans.

Q15.

Consider the following data:

Rate = 0.693 [A]Time (in min) 0 100 200 300 400 500 600 700 800 900 1000[A] (in M) 0.100 0.050 0.025 0.012 0.006 0.003 0.001 0.000 0.000 0.000 0.000Rate (in M min^{-1}) 0.0693 0.0347 0.0173 0.0083 0.0042 0.0021 0.0007 0.000 0.000 0.000 0.000

Which of the following plots will be correct?

(1)

(2)

(3)

(4)

Ans.

Q16.

Consider the following statements:

Statement-I: Rate of a first order reaction depends upon the rate constant and the concentration of reactant.

Statement-II: Half life period of a first order reaction depends upon the rate constant and the concentration of reactant.

Choose the correct option:

(1) Statement-I is false but Statement-II is true(2) Both Statement-I and Statement-II are false

(3) Both Statement-I and Statement-II are true(4) Statement-I is true but Statement-II is false

Ans.

Q17.

Consider the following data:

Rate = 0.693 [A]Time (in min) 0 100 200 300 400 500 600 700 800 900 1000[A] (in M) 0.100 0.050 0.025 0.012 0.006 0.003 0.001 0.000 0.000 0.000 0.000Rate (in M min^{-1}) 0.0693 0.0347 0.0173 0.0083 0.0042 0.0021 0.0007 0.000 0.000 0.000 0.000

Which of the following plots will be correct?

(1)

(2)

(3)

(4)

Ans.

Q18.

Consider the following statements:

Statement-I: Rate of a first order reaction depends upon the rate constant and the concentration of reactant.

Statement-II: Half life period of a first order reaction depends upon the rate constant and the concentration of reactant.

Choose the correct option:

(1) Statement-I is false but Statement-II is true(2) Both Statement-I and Statement-II are false

(3) Both Statement-I and Statement-II are true(4) Statement-I is true but Statement-II is false

Ans.

Q19.

Consider the following data:

Rate = 0.693 [A]Time (in min) 0 100 200 300 400 500 600 700 800 900 1000[A] (in M) 0.100 0.050 0.025 0.012 0.006 0.003 0.001 0.000 0.000 0.000 0.000Rate (in M min^{-1}) 0.0693 0.0347 0.0173 0.0083 0.0042 0.0021 0.0007 0.000 0.000 0.000 0.000

Which of the following plots will be correct?

(1)

(2)

(3)

(4)

Ans.

Q20.

Consider the following statements:

Statement-I: Rate of a first order reaction depends upon the rate constant and the concentration of reactant.

Statement-II: Half life period of a first order reaction depends upon the rate constant and the concentration of reactant.

Choose the correct option:

(1) Statement-I is false but Statement-II is true(2) Both Statement-I and Statement-II are false

(3) Both Statement-I and Statement-II are true(4) Statement-I is true but Statement-II is false

Ans.

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[illegible]

[illegible]

[illegible]

Keywords: Fluid Mechanics, Lecture Series, Transcript, Transcription, Text, Lecture Transcript, Transcription, Text, Lecture Transcript, Transcription, Text, Lecture

MR-TIGER-MAGNUM-MAXIMUS

Mr-Tiger-Magnum-Maximus-Supreme-Gamma-Omega-Sigma-Alpha

Mr-Tiger-Magnum-Maximus-Supreme-Gamma-Omega-Sigma-Alpha

Mr-Tiger-Magnum-Maximus-Supreme-Gamma-Omega-Sigma-Alpha

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Mr-Tiger-Magnum-Maximus-Supreme-Gamma-Omega-Sigma-Alpha

Mr-Tiger-Magnum-Maximus-Supreme-Gamma-Omega-Sigma-Alpha

$$\int_{\mathbb{R}^n} \frac{\mathbb{E}}{\mathbb{P}} \, \mathrm{d}\mathbb{P} = \mathbb{E}$$

Mr-Tiger-Magnum-Maximus-Supreme-Gamma-Omega-Sigma-Alpha

Mr-Tiger-Magnum-Maximus-Supreme-Gamma-Omega-Sigma-Alpha

Mr-Tiger-Magnum-Maximus-Supreme-Gamma-Omega-Sigma-Alpha

Mr-Tiger-Magnum-Maximus-Supreme-Gamma-Omega-Sigma-Alpha

$$\begin{aligned} \mathbb{P} &= \frac{\mathbb{E}\mathbb{P}}{\mathbb{E}} \quad \text{a.s.} \\ &= \frac{\mathbb{E}\mathbb{P}}{\mathbb{E}} \quad \text{a.s.} \\ &= \frac{\mathbb{E}\mathbb{P}}{\mathbb{E}} \quad \text{a.s.} \\ &= \frac{\mathbb{E}\mathbb{P}}{\mathbb{E}} \quad \text{a.s.} \end{aligned}$$

Mr-Tiger-Magnum-Maximus-Supreme-Gamma-Omega-Sigma-Alpha

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Published online: 18 February 2021

Fabian Ruehle [1] [2] [3] [4] [5] [6] [7] [8] [9] [10] [11] [12] [13] [14] [15] [16] [17] [18] [19] [20] [21] [22] [23] [24] [25] [26] [27] [28] [29] [30] [31] [32] [33] [34] [35] [36] [37] [38] [39] [40]

MATLAB/Simulink Model Based Design Workflow

Introduction

Model Based Design (MBD) is a formal methodology for developing complex control systems. It uses a single mathematical model as the central artifact from which requirements, simulation, code generation, and verification are derived. The model serves as the ground truth for all downstream activities, ensuring that changes propagate consistently across the entire development lifecycle.

Workflow Overview

The workflow consists of six stages: requirements capture, plant modeling, controller design, simulation and verification, code generation, and deployment. Each stage produces artifacts that feed into the next stage. Traceability links connect requirements to model elements to generated code to test results, so that any change can be assessed for its impact on every downstream artifact.

Plant Modeling

Plant modeling begins with the physical equations governing the system. For a DC motor, the equations are:

$$v(t) = Ri(t) + L\frac{di(t)}{dt} + K_e\omega(t)$$

$$J\frac{d\omega(t)}{dt} = K_t i(t) - B\omega(t) - \tau_L(t)$$

Controller Design

The controller is designed in the discrete domain. For a PI controller with sampling period T_s:

$$u[k] = K_p e[k] + K_i T_s \sum_{j=0}^{k} e[j]$$

Simulink Implementation

Simulink provides a block diagram environment for implementing the plant model and controller. Signal flow is explicit and visual. Simulink supports fixed-step and variable-step solvers, with fixed-step solvers required for real-time code generation.

Code Generation

Simulink Coder (formerly Real-Time Workshop) generates C code from the model. The generated code is deterministic, with no dynamic memory allocation and no recursion. Code generation can be configured for different target processors and optimization levels.

Verification

Model-in-the-loop (MIL) simulation verifies the model against requirements before code generation. Software-in-the-loop (SIL) simulation runs the generated code on a host machine to verify that the code behaves identically to the model. Hardware-in-the-loop (HIL) simulation runs the code on the target hardware with real sensors and actuators.

Deployment

Deployment packages the generated code with the target hardware's boot code and runtime libraries. Over-the-air (OTA) updates allow deployed systems to receive new versions without physical access.

Conclusion

Model Based Design provides a single source of truth for requirements, simulation, and implementation. Changes propagate automatically through the workflow, reducing the risk of inconsistencies between specification and implementation.

Tags: #lecture #transcript #fluid-mechanics #engineering #physics #pressure #buoyancy #archimedes #pascal #bernoulli #continuity #equation #flow #viscosity #reynolds #laminar #turbulent #drag #lift #cavitation #pipe #open-channel #hydraulics #pump #turbine #dimensionless #pi-theorem #model #prototype #similarity #froude #weber #cauchy #mach #compressible #incompressible #boundary-layer #separation #streamline #vortex #potential #stagnation #dynamic #kinematic #specific #gravity #density #specific-weight #specific-volume #specific-gravity #viscosity #kinematic-viscosity #dynamic-viscosity #surface-tension #capillarity #vapor-pressure #bulk-modulus #compressibility #gas #liquid #solid #fluid #continuum #navier-stokes #euler #bernoulli #continuity #momentum #energy #conservation #control-volume #system #lagrangian #eulerian #substantial-derivative #material-derivative #local #convective #steady #unsteady #uniform #non-uniform #one-dimensional #two-dimensional #three-dimensional #axisymmetric #plane #potential #irrotational #rotational #vorticity #circulation #stream-function #velocity-potential #laplace #poisson #source #sink #doublet #uniform-flow #vortex #rankine-half-body #rankine-oval #cylinder #sphere #ellipsoid #airfoil #wing #lift #drag #induced-drag #parasite-drag #wave-drag #skin-friction #form-drag #pressure-drag #total-drag #boundary-layer #blasius #pohlhausen #thwaites #momentum-integral #displacement #thickness #momentum #thickness #shape-factor #separation #transition #critical-reynolds #laminar-flow #turbulent-flow #pipe-flow #open-channel #hydraulics #weir #sluice #gate #spillway #dam #reservoir #pump #turbine #impeller #volute #diffuser #cavitation #npsh #specific-speed #pump-curve #system-curve #operating-point #turbine #pelton #francis #kaplan #bulb #tubular #axial #mixed #flow #impulse #reaction #draft-tube #wicket-gate #runner #blade #nozzle #jet #deflector #needle #spear #valve #butterfly #gate #ball #check #pressure #relief #safety #valve #orifice #venturi #nozzle #flow #meter #pitot #tube #rotameter #ultrasonic #flow #meter #electromagnetic #flow #meter #coriolis #flow #meter #turbine #flow #meter #vortex #shedding #flow #meter #hot #wire #anemometer #hot #film #anemometer #laser #doppler #anemometer #particle #image #velocimetry #piv #laser #doppler #velocimetry #ldv #pressure #transducer #strain #gauge #pressure #transducer #piezoelectric #pressure #transducer #capacitive #pressure #transducer #inductive #pressure #transducer #potentiometric #pressure #transducer #bourdon #tube #bellows #diaphragm #manometer #u-tube #manometer #inclined #manometer #micromanometer #barometer #mercury #barometer #aneroid #barometer #altimeter #altitude #pressure #altitude #density #altitude #temperature #altitude #standard #atmosphere #isa

```
<?xml version="1.0" encoding="UTF-8"?>
<?xml-stylesheet href="rss.xsl" type="text/xsl"?>
<rss version="2.0" xmlns:dc="http://purl.org/dc/elements/1.1/" xmlns:sy="http://purl.org/rss/1.0/modules/syndication/" xmlns:content="http://purl.org/rss/1.0/modules/content/" xmlns:atom="http://www.w3.org/2005/Atom" xmlns:media="http://search.yahoo.com/mrss/" xmlns:wfw="http://wellformedweb.org/CommentAPI/" xmlns:slash="http://purl.org/rss/1.0/modules/slash/" xmlns:dcterms="http://purl.org/dc/terms/" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns:itunes="http://www.itunes.com/dtd/2.0/" xmlns:podcast="http://podcastindex.org/namespace/1.0" xmlns:rawvoice="http://www.rawvoice.com/rawvoicerss/" xmlns:fb="http://www.facebook.com/2008/fbml" xmlns:gr="http://www.google.com/schemas/reader/" xmlns:feedburner="http://rssnamespace.org/feedburner/extension/1.0" xmlns:feedburner:origlink="http://www.nytimes.com/2026/04/17/opinion/ai-chatbots-delusions.html" xmlns:media:keywords="" xmlns:media:rating="" xmlns:media:copyright="" xmlns:media:description="" xmlns:media:thumbnail="" xmlns:media:title="" xmlns:media:group="" xmlns:media:credit="" xmlns:media:text="" xmlns:media:restriction="" xmlns:media:hash="" xmlns:media:player="" xmlns:media:content="" xmlns:media:peerlink="" xmlns:media:license="" xmlns:media:category="" xmlns:media:comments="" xmlns:media:expression="" xmlns:media:responses="" xmlns:media:starRating="" xmlns:media:statistics="" xmlns:media:scene="" xmlns:media:subTitle="" xmlns:media:pricing="" xmlns:media:textinput="" xmlns:media:location="" xmlns:media:rights
```

1.1 Introduction to Data Structures and Algorithms

Data structures and algorithms are the foundation of computer science. Understanding how to organize data efficiently and how to design algorithms that solve problems quickly is essential for any programmer.

1.1.1 Why Study Data Structures and Algorithms?

Consider a simple problem: you have a list of 1,000,000 numbers and you need to find the largest one. A naive approach might scan the entire list for each query. With the right data structure, you can answer the question in constant time. The difference between a slow program and a fast program can be the difference between a project that works and one that never finishes.

Data structures and algorithms are the tools that make computers fast and reliable. They are used in every domain: finance, medicine, engineering, science, and everyday software. Mastering them gives you the ability to design systems that scale from a few users to millions of users.

1.1.2 Formal Definitions

Let n denote the size of the input. We write $f(n) = O(g(n))$ if there exists constants $c > 0$ and $n_0 \geq 0$ such that $f(n) \leq c \cdot g(n)$ for all $n \geq n_0$. Similarly, $f(n) = \Omega(g(n))$ if $f(n) \geq c \cdot g(n)$ for all $n \geq n_0$, and $f(n) = \Theta(g(n))$ if both hold.

Formally, a data structure is a tuple $(D, \mathcal{O})$ where D is the set of possible states and $\mathcal{O}$ is a set of operations that transform states in D.

1.1.3 Example

```
def binary_search(arr, target):
    lo, hi = 0, len(arr) - 1
    while lo <= hi:
        mid = (lo + hi) // 2
        if arr[mid] == target:
            return mid
        elif arr[mid] < target:
            lo = mid + 1
        else:
            hi = mid - 1
    return -1
```

Binary search runs in $O(\log n)$ time. Each iteration halves the search space, so after k iterations the remaining range has size $n / 2^k$. When $k = \log_2 n$, the range has size 1 and the algorithm terminates.

```
<div id="ctl00$ctl$ctl$ctl
```

```
MD5 /TLSV1 HMAC/RSA/DES/CFB/ECB/MD5 Cipher Suites:

TLS_RSA_WITH_RC4_128_MD5
TLS_RSA_WITH_RC4_128_SHA
TLS_RSA_WITH_3DES_EDE_CBC_SHA
TLS_DHE_DSS_WITH_3DES_EDE_CBC_SHA
TLS_DHE_RSA_WITH_3DES_EDE_CBC_SHA
TLS_RSA_WITH_DES_CBC_SHA
TLS_DHE_DSS_WITH_DES_CBC_SHA
TLS_DHE_RSA_WITH_DES_CBC_SHA
TLS_RSA_EXPORT_WITH_RC4_40_MD5
TLS_RSA_EXPORT_WITH_RC2_CBC_40_MD5
TLS_RSA_EXPORT_WITH_DES40_CBC_SHA
TLS_DHE_DSS_EXPORT_WITH_DES40_CBC_SHA
TLS_DHE_RSA_EXPORT_WITH_DES40_CBC_SHA
TLS_RSA_WITH_NULL_MD5
TLS_RSA_WITH_NULL_SHA
TLS_DH_anon_EXPORT_WITH_RC4_40_MD5
TLS_DH_anon_WITH_RC4_128_MD5
TLS_DH_anon_EXPORT_WITH_DES40_CBC_SHA
TLS_DH_anon_WITH_DES_CBC_SHA
TLS_DH_anon_WITH_3DES_EDE_CBC_SHA
TLS_KRB5_WITH_DES_CBC_MD5
TLS_KRB5_WITH_3DES_EDE_CBC_MD5
TLS_KRB5_WITH_RC4_128_MD5
TLS_KRB5_WITH_DES_CBC_SHA
TLS_KRB5_WITH_3DES_EDE_CBC_SHA
TLS_KRB5_WITH_RC4_128
```

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Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

Sed ut perspiciatis unde	15.01	1.01.1998
Omnis iste natus error sit voluptatem	15.02	1.02.1998
Accusantium doloremque laudantium	15.03	1.03.1998
Totam rem aperiam, eaque ipsa quae	15.04	1.04.1998
Ab illo inventore veritatis	15.05	1.05.1998
Quasi architecto beatae vitae dicta	15.06	1.06.1998
Sunt in culpa qui officia deserunt	15.07	1.07.1998

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```
<?xml version="1.0" encoding="UTF-8"?>
<!-- Generated by SVGO v3.0.2 (https://svgo.dev) -->
<svg
  xmlns="http://www.w3.org/2000/svg"
  xmlns:xlink="http://www.w3.org/1999/xlink"
  viewBox="0.000,0.000,1000.000,1000.000"
  preserveAspectRatio="xMidYMid meet"
  fill="#000000" />
</svg>
```

Chapter 4 Matrix Algebra

4.1 Sum and Scalar Product of Matrices

Definition

Sum of two matrices Two matrices A and B are said to be conformable for addition if they are of the same order. If $A = [a_{ij}]_{m \times n}$ and $B = [b_{ij}]_{m \times n}$ then the sum of A and B is defined as $A + B = [a_{ij} + b_{ij}]_{m \times n}$.

Scalar product If $A = [a_{ij}]_{m \times n}$ and k is a scalar then $kA = [ka_{ij}]_{m \times n}$.

Example

Example 1

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, \quad B = \begin{bmatrix} 7 & 8 & 9 \\ 10 & 11 & 12 \end{bmatrix}, \quad k = 2$$

Solution

$$A + B = \begin{bmatrix} 1+7 & 2+8 & 3+9 \\ 4+10 & 5+11 & 6+12 \end{bmatrix} = \begin{bmatrix} 8 & 10 & 12 \\ 14 & 16 & 18 \end{bmatrix}$$

$$kA = \begin{bmatrix} 2 \times 1 & 2 \times 2 & 2 \times 3 \\ 2 \times 4 & 2 \times 5 & 2 \times 6 \end{bmatrix} = \begin{bmatrix} 2 & 4 & 6 \\ 8 & 10 & 12 \end{bmatrix}$$

Example 2

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}, \quad k = 3$$

Solution

$$A + B = \begin{bmatrix} 1+5 & 2+6 \\ 3+7 & 4+8 \end{bmatrix} = \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}$$

$$kA = \begin{bmatrix} 3 \times 1 & 3 \times 2 \\ 3 \times 3 & 3 \times 4 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 9 & 12 \end{bmatrix}$$

Example 3

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 4

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 5

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 6

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 7

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 8

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 9

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 10

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 11

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 12

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 13

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 14

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 15

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 16

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 17

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 18

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 19

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 20

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 21

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 22

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 23

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 24

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 25

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 26

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 27

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 28

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 29

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 30

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 31

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 & 6 & 6 \\ 7 & 8 & 10 \end{bmatrix}$$

$$kA = \begin{bmatrix} 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 5 \times 4 & 5 \times 5 & 5 \times 6 \\ 5 \times 7 & 5 \times 8 & 5 \times 9 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 20 & 25 & 30 \\ 35 & 40 & 45 \end{bmatrix}$$

Example 32

Find $A + B$ and kA if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad k = 5$$

Solution

$$A + B = \begin{bmatrix} 1+1 & 2+0 & 3+0 \\ 4+0 & 5+1 & 6+0 \\ 7+0 & 8+0 & 9+1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 3 \\ 4 &

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Tiger Magnum Opus 7 Extraordinary Lemma

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<!ENTITY % prefsDTD SYSTEM "chrome://mozapps/locale/preferences/preferences.dtd">
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        xmlns="http://mozilla.org/keymaster/gatekeeper/there.is.only.xul">
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Ultrasonic Sensor Based Smart Dustbin

[1]Dr. Ashish Kumar Jha, [2]Ankit Kumar, [3]Ankit Kumar, [4]Ankit Kumar, [5]Ankit Kumar, [6]Ankit Kumar, [7]Ankit Kumar, [8]Ankit Kumar, [9]Ankit Kumar, [10]Ankit Kumar, [11]Ankit Kumar, [12]Ankit Kumar, [13]Ankit Kumar, [14]Ankit Kumar, [15]Ankit Kumar, [16]Ankit Kumar, [17]Ankit Kumar, [18]Ankit Kumar, [19]Ankit Kumar, [20]Ankit Kumar, [21]Ankit Kumar,

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Ethical Implications of Artificial Intelligence in Healthcare: Balancing Innovation and Patient Rights

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Abstract: Artificial Intelligence (AI) is rapidly transforming healthcare by enhancing diagnostic accuracy, optimizing treatment plans, and improving patient outcomes. However, the integration of AI in healthcare raises significant ethical concerns, including data privacy, algorithmic bias, accountability, and the potential displacement of human caregivers. This study explores the ethical implications of AI in healthcare, focusing on the balance between innovation and patient rights. The study examines the ethical challenges associated with AI in healthcare, including the impact on patient autonomy, the potential for discrimination, and the need for transparency and accountability. The study also discusses the regulatory frameworks and guidelines that can ensure the ethical deployment of AI in healthcare, ensuring that the benefits of AI are maximized while safeguarding patient rights and maintaining trust in healthcare systems. The study concludes with recommendations for policymakers, healthcare providers, and AI developers to address the ethical challenges and promote the responsible use of AI in healthcare.

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Ch. 2 Lo-Fi Prototyping

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Cognitive Ability Predicts Obesity and Smoking Across Three Cohorts of Twenty-Year-Olds

Ian J. Deary, Stuart J. Ritchie, and Timothy C. Bates
University of Edinburgh

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7.1 Cell Theory: Life at the Smallest Scale

Cell Theory: Life at the Smallest Scale

Cell Theory: Life at the Smallest Scale

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Abstract

Deep learning (DL) models have shown promising performance in medical imaging tasks, including the detection of diabetic retinopathy (DR) and other eye diseases. However, it remains unclear how these models generalize to real-world clinical settings. Here we assess the generalizability of DL models for DR detection from fundus photographs across ethnicity, imaging camera, and disease severity and type in a dataset of 236,234 images from 40,339 diabetic patients. We find that the performance of DL models for referable DR detection is comparable across Asian, Caucasian, and African ethnicities (AUC: 0.936–0.946), but varies across cameras (AUC: 0.906–0.947) and is lower in severe DR (AUC: 0.885) and proliferative DR (AUC: 0.901) compared to moderate DR (AUC: 0.943). These results suggest that DL models for DR screening may not generalize across all ethnicities, cameras, and disease severities and types, and that such models should be tested in these settings before implementation.

NPJ | https://doi.org/10.1038/s41746-020-00376-2 ARTMED | 1

Introduction

Diabetes mellitus (DM) is a chronic disease that affects 463 million people worldwide and is projected to rise to 700 million by 2045[1]. Diabetic retinopathy (DR) is a common complication of DM and a major cause of blindness and vision impairment worldwide[2]. Early detection of DR followed by timely treatment can prevent vision loss[3], and thus, DM patients are recommended to undergo yearly DR screening[4]. However, in many countries, the shortage of eye care specialists and the high prevalence of DM make such a task difficult to accomplish[5]. Deep learning (DL) models have shown promising performance in the detection of DR from fundus photographs, with several studies reporting high sensitivity and specificity in retrospective datasets[6–9]. However, it remains unclear how these models generalize to real-world clinical settings, where they may be used for DR screening. Prior studies have suggested that DL models may not generalize across ethnicities, imaging cameras, and disease severities and types[10–13], but these studies have been limited by the use of a single DL model, a single camera, or a single ethnicity, and have not assessed the impact of disease severity and type on model performance. Here we assess the generalizability of a DL model for referable DR (RDR) detection across ethnicities, cameras, and disease severities and types in a large dataset of diabetic patients from the United Kingdom (UK) Biobank[14].

Results

Study population characteristics

The UK Biobank dataset consisted of 236,234 fundus photographs from 40,339 diabetic patients, of which 1,401 (3.5%) had RDR, 1,067 (2.6%) had moderate DR, 1,250 (3.1%) had severe DR, and 1,175 (2.9%) had proliferative DR (PDR). Mean (SD) age was 56.8 (8.1) years, 23,561 (58.4%) were male, 1,937 (4.8%) were Asian, 1,478 (3.7%) were African, and 36,924 (91.5%) were Caucasian. Mean (SD) HbA1c was 75.1 (19.7) mmol/mol (9.0 (1.8)%), 3,553 (8.8%) had insulin treatment, and mean (SD) duration of DM was 8.7 (8.1) years. Mean (SD) age was 55.4 (8.1) years for those with RDR, 55.8 (8.1) years for those with moderate DR, 56.4 (8.1) years for those with severe DR, and 56.1 (8.1) years for those with PDR. Mean (SD) HbA1c was 81.0 (21.3) mmol/mol (9.6 (1.9)%) for those with RDR, 77.4 (19.9) mmol/mol (9.3 (1.8)%) for those with moderate DR, 79.4 (20.1) mmol/mol (9.4 (1.8)%) for those with severe DR, and 79.9 (20.4) mmol/mol (9.5 (1.9)%) for those with PDR. Insulin treatment was 14.1% (RDR), 12.1% (moderate DR), 13.8% (severe DR), and 13.8% (PDR). Mean (SD) duration of DM was 10.7 (8.4) years for those with RDR, 10.1 (8.2) years for those with moderate DR, 10.8 (8.2) years for those with severe DR, and 10.8 (8.2) years for those with PDR. Other baseline characteristics by DR status are shown in Table 1.

Performance of DL models for referable DR detection

We trained a DL model to detect RDR using fundus photographs from patients of Caucasian ethnicity, imaging with a Topcon camera, and without severe DR or PDR. The model achieved an AUC of 0.943 (95% CI: 0.937–0.949) in a randomly sampled test set of 9,000 images from 9,000 patients. The model achieved a sensitivity of 0.885 (95% CI: 0.869–0.901) and specificity of 0.857 (95% CI: 0.853–0.861) at the operating point selected in the development dataset (Table 2). The model achieved a sensitivity of 0.885 (95% CI: 0.869–0.901) and specificity of 0.857 (95% CI: 0.853–0.861) at the operating point selected in the development dataset (Table 2).

Performance of DL models across ethnicities

The model achieved an AUC of 0.936 (95% CI: 0.914–0.957) in Asian patients, 0.946 (95% CI: 0.935–0.956) in African patients, and 0.943 (95% CI: 0.937–0.949) in Caucasian patients (Table 2). The model achieved a sensitivity of 0.879 (95% CI: 0.821–0.927) in Asian patients, 0.893 (95% CI: 0.865–0.918) in African patients, and 0.885 (95% CI: 0.869–0.901) in Caucasian patients (Table 2). The model achieved a specificity of 0.854 (95% CI: 0.824–0.883) in Asian patients, 0.863 (95% CI: 0.844–0.882) in African patients, and 0.857 (95% CI: 0.853–0.861) in Caucasian patients (Table 2).

Performance of DL models across cameras

The model achieved an AUC of 0.906 (95% CI: 0.890–0.921) with a Topcon camera, 0.913 (95% CI: 0.895–0.931) with a Canon camera, and 0.947 (95% CI: 0.941–0.953) with a Zeiss camera (Table 2). The model achieved a sensitivity of 0.862 (95% CI: 0.844–0.880) with a Topcon camera, 0.867 (95% CI: 0.819–0.911) with a Canon camera, and 0.887 (95% CI: 0.871–0.903) with a Zeiss camera (Table 2). The model achieved a specificity of 0.857 (95% CI: 0.853–0.861) with a Topcon camera, 0.859 (95% CI: 0.855–0.863) with a Canon camera, and 0.858 (95% CI: 0.854–0.862) with a Zeiss camera (Table 2).

Performance of DL models across disease severity and type

The model achieved an AUC of 0.885 (95% CI: 0.867–0.903) in severe DR and 0.901 (95% CI: 0.882–0.919) in PDR, compared to an AUC of 0.943 (95% CI: 0.937–0.949) in those without severe DR or PDR. The model achieved a sensitivity of 0.838 (95% CI: 0.807–0.868) in severe DR and 0.854 (95% CI: 0.824–0.883) in PDR, compared to a sensitivity of 0.885 (95% CI: 0.869–0.901) in those without severe DR or PDR. The model achieved a specificity of 0.858 (95% CI: 0.854–0.862) in severe DR and 0.857 (95% CI: 0.853–0.861) in PDR, compared to a specificity of 0.857 (95% CI: 0.853–0.861) in those without severe DR or PDR. Other performance metrics by DR severity and type are shown in Table 2.

Discussion

In this study, we assessed the generalizability of a DL model for referable DR detection across ethnicities, imaging cameras, and disease severities and types in a large dataset of diabetic patients from the UK Biobank. We found that the performance of the DL model was comparable across Asian, Caucasian, and African ethnicities, but was lower in severe DR and PDR than in those without severe DR or PDR. The model also performed worse when applied to images from a camera not used in the development dataset. These results suggest that DL models for DR screening may not generalize across all ethnicities, cameras, and disease severities and types, and that such models should be tested in these settings before implementation.

Prior studies have shown promising performance of DL models for DR detection from fundus photographs, with several reporting sensitivities and specificities of over 90% in retrospective datasets[6–9]. However, it remains unclear how these models would perform in real-world clinical settings, where they may be used for DR screening. Prior studies have suggested that DL models may not generalize across ethnicities, imaging cameras, and disease severities and types[10–13], but these studies have been limited by the use of a single DL model, a single camera, or a single ethnicity, and have not assessed the impact of disease severity and type on model performance. Our study addresses these gaps in the literature by assessing the performance of a DL model across ethnicities, cameras, and disease severities and types in a large dataset of diabetic patients from the UK Biobank.

Our study shows that DL models for DR screening may not generalize across all ethnicities, cameras, and disease severities and types, and that such models should be tested in these settings before implementation. The lower performance in severe DR and PDR suggests that DL models may miss some of the most severe cases of DR, which are the cases that require urgent referral to an eye care specialist[3]. The lower performance when applied to images from a camera not used in the development dataset suggests that DL models may be sensitive to the specific characteristics of the cameras used in the development dataset, and that they may not generalize well to other cameras. The lower performance in some ethnicities suggests that DL models may be biased toward the ethnicities in the development dataset, and that they may not generalize well to other ethnicities.

Our study has several strengths. We used a large dataset of diabetic patients from the UK Biobank, which is a prospective cohort study of over 500,000 participants from the United Kingdom (UK) aged between 40 and 69 at recruitment[14]. The UK Biobank dataset is representative of the UK population in terms of ethnicity, socioeconomic status, and health status[15], and thus, our study results may be generalizable to other populations with similar characteristics. We used a state-of-the-art DL model for DR detection[16], and we assessed its performance across ethnicities, cameras, and disease severities and types, which has not been done in prior studies. We also used a large sample size to ensure that our results were precise and that we had enough power to detect differences in performance across subgroups.

Our study has several limitations. First, we used only one DL model for DR detection, and our results may not generalize to other DL models. Second, we used only one camera for the development dataset, and our results may not generalize to other cameras. Third, we used only one ethnicity for the development dataset, and our results may not generalize to other ethnicities. Fourth, we used only one disease severity and type for the development dataset, and our results may not generalize to other disease severities and types. Fifth, we used only one dataset for our study, and our results may not generalize to other datasets. Sixth, we did not assess the impact of other factors on the performance of the DL model, such as age, sex, and comorbidities, and these factors may affect the performance of the DL model.

In conclusion, we assessed the generalizability of a DL model for referable DR detection across ethnicities, imaging cameras, and disease severities and types in a large dataset of diabetic patients from the UK Biobank. We found that the performance of the DL model was comparable across Asian, Caucasian, and African ethnicities, but was lower in severe DR and PDR than in those without severe DR or PDR. The model also performed worse when applied to images from a camera not used in the development dataset. These results suggest that DL models for DR screening may not generalize across all ethnicities, cameras, and disease severities and types, and that such models should be tested in these settings before implementation.

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1.1 Introduction

Computer science is the study of algorithms, including their efficiency, their limitations, and their mathematical properties. Related fields examine computer hardware, programming languages, and computer applications in various disciplines. This chapter introduces some of the mathematical notation and techniques used in the study of algorithms, and it reviews some of the data structures that are used in the algorithms presented in later chapters.

1.2 Mathematical Notation and Functions

Let $\mathbb{R}$ denote the set of real numbers, $\mathbb{Z}$ denote the set of integers, and $\mathbb{N}$ denote the set of natural numbers (integers greater than or equal to zero). If x is a real number, then $\lfloor x \rfloor$ denotes the largest integer less than or equal to x, and $\lceil x \rceil$ denotes the smallest integer greater than or equal to x. For example, $\lfloor 3.14 \rfloor = 3$ and $\lceil 3.14 \rceil = 4$. Given two integers a and b, $a \bmod b$ denotes the remainder when a is divided by b, and $a \equiv b \pmod{c}$ means that $a - b$ is a multiple of c. For example, $7 \bmod 3 = 1$ and $7 \equiv 1 \pmod{3}$.

1.2.1 Logarithms and Exponents

Unless otherwise specified, all logarithms are base 2. For $x > 0$, $\log(x)$ denotes the base-2 logarithm of x. For $x > 0$ and $y > 0$, $\log(xy) = \log(x) + \log(y)$. For $x > 0$ and $y > 0$, $\log(x/y) = \log(x) - \log(y)$. For $x > 0$ and $y > 0$, $\log(x^y) = y \log(x)$. For $x > 0$ and $y > 0$, $x^{\log(y)} = y^{\log(x)}$. For $x > 0$ and $y > 0$, $\log(x^y) = y \log(x)$. For $x > 0$ and $y > 0$, $x^{\log(y)} = y^{\log(x)}$.

1.2.2 Factorial and Binomial Coefficients

For $n \in \mathbb{N}$, $n!$ denotes the factorial of n, defined as $n! = n \cdot (n-1) \cdot (n-2) \cdots 2 \cdot 1$. For $n \in \mathbb{N}$, $n! = n \cdot (n-1)!$. For $n \in \mathbb{N}$, $n! = \prod_{i=1}^{n} i

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<clipPath id="clipPath3">
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</clipPath>
<clipPath id="clipPath4">
<path d="M0,0 L1,0 L1,1 L0,1 Z" />
</clipPath>
```

UNITED STATES DISTRICT COURT
DISTRICT OF MASSACHUSETTS

**Dr. Ashley Mercado-Valentin, Dr. Ana Maria Rodríguez-Díaz, Dr. Marcos Vega-Ramírez, Dr. Ana Maria Ortiz-Ortiz, Dr. Carlos Ortiz-Ortiz, Dr. Jorge Ortiz-Ortiz, Dr. Ana Maria Ortiz-Rodríguez, Dr. Ana Maria Ortiz-Ortiz, Dr. Ana Maria Ortiz-Ortiz, Dr. Ana Maria Ortiz-Ortiz

DOI 10.1007/s00401-
Received: 2 April 2020
Revised: 15 July 2020
Accepted: 16 July 2020
Online publication: 27 July 2020

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<?xml version="1.0" ?>
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<div id="1" />
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"ajaxerrperflog
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Ethnobotanical study of medicinal plants used by the inhabitants of Mirpur Sakro District, Tharparkar, Sindh, Pakistan

Muhammad Ali Shah[1], Muhammad Ali[2], Muhammad Ali[3], Muhammad Ali[4], Muhammad Ali[5], Muhammad Ali[6], Muhammad Ali[7], Muhammad Ali[8], Muhammad Ali[9], Muhammad Ali[10], Muhammad Ali[11], Muhammad Ali[12], Muhammad Ali[13], Muhammad Ali[14], Muhammad Ali[15], Muhammad Ali[16], Muhammad Ali[17], Muhammad Ali[18], Muhammad Ali[19], Muhammad Ali[20], Muhammad Ali[21], Muhammad Ali[22], Muhammad Ali[23], Muhammad Ali[24], Muhammad Ali[25], Muhammad Ali[26], Muhammad Ali[27], Muhammad Ali[28], Muhammad Ali[29], Muhammad Ali[30], Muhammad Ali[31], Muhammad Ali[32], Muhammad Ali[33], Muhammad Ali[34], Muhammad Ali[35], Muhammad Ali[36], Muhammad Ali[37], Muhammad Ali[38], Muhammad Ali[39], Muhammad Ali[40], Muhammad Ali[41], Muhammad Ali[42], Muhammad Ali[43], Muhammad Ali[44], Muhammad Ali[45], Muhammad Ali[46], Muhammad Ali[47], Muhammad Ali[48], Muhammad Ali[49], Muhammad Ali[50], Muhammad Ali[51], Muhammad Ali[52], Muhammad Ali[53], Muhammad Ali[54], Muhammad Ali[55], Muhammad 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Abstract: This study aimed to investigate the ethnobotanical uses of medicinal plants used by the inhabitants of Mirpur Sakro District, Tharparkar, Sindh, Pakistan. Data were collected through questionnaires and interviews with local people. A total of 10 medicinal plants belonging to 10 families were recorded. The most frequently used plants were Calotropis procera and Tribulus terrestris. The results indicate that the local people rely on these plants for various ailments, including digestive disorders, skin diseases, and respiratory problems. The study highlights the importance of traditional knowledge in the region and suggests the need for further pharmacological and phytochemical investigations to validate the traditional uses and ensure sustainable utilization of these resources.

Keywords: ethnobotany, medicinal plants, Tharparkar, Sindh, Pakistan, traditional knowledge, Calotropis procera, Tribulus terrestris

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7.2 Cluster Representations

7.2.1 Centroids and Prototypes (Medoids)

Definition. Let $C_k = \{x_1, x_2, \ldots, x_{n_k}\}$ be a cluster with n_k points. The **centroid** (or **mean**) of C_k is the point

$$\mu_k = \frac{1}{n_k} \sum_{i=1}^{n_k} x_i$$

Example 7.2.1. Consider the cluster $C_1 = \{(1,2), (2,3), (3,1)\}$. Then

$$\mu_1 = \frac{1}{3}\left((1,2) + (2,3) + (3,1)\right) = \frac{1}{3}(6,6) = (2,2)$$

Proposition 7.2.1. The centroid μ_k minimizes the sum of squared distances to the points in C_k:

$$\mu_k = \arg\min_{\mu \in \mathbb{R}^d} \sum_{i=1}^{n_k} \|x_i - \mu\|^2$$

Proof. Expand the objective:

$$\sum_{i=1}^{n_k} \|x_i - \mu\|^2 = \sum_{i=1}^{n_k} \|x_i\|^2 - 2\mu^T \sum_{i=1}^{n_k} x_i + n_k \|\mu\|^2$$

This is a quadratic in μ. Setting the gradient to zero:

$$\nabla_\mu \left[\sum_{i=1}^{n_k} \|x_i - \mu\|^2\right] = -2\sum_{i=1}^{n_k} x_i + 2n_k\mu = 0$$

$$\Rightarrow \mu = \frac{1}{n_k}\sum_{i=1}^{n_k} x_i$$

Remark. The centroid is not necessarily a data point. For example, the centroid of $\{(0,0), (1,0)\}$ is $(0.5, 0)$, which is not a data point.

Definition. The **medoid** of C_k is the point $m_k \in C_k$ that minimizes the sum of distances to all other points in C_k:

$$m_k = \arg\min_{x \in C_k} \sum_{i=1}^{n_k} \|x_i - x\|$$

Remark. Unlike the centroid, the medoid must be a data point. This makes it more robust to outliers and applicable to non-Euclidean distances (e.g., edit distance for strings, Jaccard distance for sets).

Example 7.2.2. Consider the cluster $C_1 = \{(1,2), (2,3), (3,1), (100,100)\}$. The centroid is $\mu_1 = (26.5, 26.5)$, which is far from most points. The medoid is $m_1 = (2,3)$, which is much closer to the majority of points.

7.2.2 Medoids

Definition. The **medoid** of C_k is the point $m_k \in C_k$ that minimizes the sum of distances to all other points in C_k:

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7.2.10 Medoids

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Lesson Twenty-one | Matrix Multiplication and Inverses

Matrix multiplication is the heart of linear algebra. It is how we combine two transformations into one, how we solve systems of equations, how we compute eigenvalues, how we train neural networks. Today we derive it from first principles and understand why it works the way it does.

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1. Motivation: Why Do We Need Matrix Multiplication?

Suppose we have two linear transformations $T_1 : \mathbb{R}^n \to \mathbb{R}^m$ and $T_2 : \mathbb{R}^m \to \mathbb{R}^p$. Their composite $T_2 \circ T_1 : \mathbb{R}^n \to \mathbb{R}^p$ is also linear. If A represents T_1 and B represents T_2, we want a matrix C such that

$$(T_2 \circ T_1)(\mathbf{x}) = C\mathbf{x} \quad \text{for all } \mathbf{x} \in \mathbb{R}^n.$$

We want C to be the product BA.

---

2. Definition

Definition 2.1 (Matrix Multiplication). Let $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$. The product $C = AB \in \mathbb{R}^{m \times p}$ is defined by

$$c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}, \qquad 1 \le i \le m, \; 1 \le j \le p.$$

---

3. Properties

Theorem 3.1 (Properties of Matrix Multiplication). Assuming the sizes match, the following hold:

1. **Associativity:** $(AB)C = A(BC)$.
2. **Distributivity:** $A(B + C) = AB + AC$ and $(A + B)C = AC + BC$.
3. $(AB)^T = B^T A^T$.
4. $\det(AB) = \det(A)\det(B)$.
5. $\operatorname{tr}(AB) = \operatorname{tr}(BA)$.

Proof. (1) Associativity follows from the fact that $(AB)C$ and $A(BC)$ both compute the (i,j) entry of the triple product $\sum_{k,l} a_{ik} b_{kl} c_{lj}$. (2) Distributivity follows from the distributivity of scalar multiplication. (3) $(AB)^T_{ij} = (AB)_{ji} = \sum_k a_{jk} b_{ki} = \sum_k b_{ki} a_{jk} = (B^T A^T)_{ij}$. (4) Taken for granted for now; we will prove it in Lesson Twenty-two. (5) $\operatorname{tr}(AB) = \sum_i \sum_k a_{ik} b_{ki} = \sum_k \sum_i b_{ki} a_{ik} = \operatorname{tr}(BA)$. □

Warning. Matrix multiplication is not commutative. In general $AB \neq BA$. Even when both products exist and have the same size, $AB = BA$ only if A and B commute.

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4. Inverses

Definition 4.1 (Inverse). A square matrix $A \in \mathbb{R}^{n \times n}$ is invertible if there exists A^{-1} such that

$$AA^{-1} = A^{-1}A = I_n.$$

Theorem 4.2 (Uniqueness). If A is invertible, A^{-1} is unique.

Proof. If B and C both satisfy $AB = BA = I$ and $AC = CA = I$, then $B = BI = B(AC) = (BA)C = IC = C$. □

Theorem 4.3 (Properties of Inverses).

1. $(AB)^{-1} = B^{-1}A^{-1}$.
2. $(A^T)^{-1} = (A^{-1})^T$.
3. $\det(A^{-1}) = 1/\det(A)$.

---

5. Exercises

Exercise 5.1. Compute $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$ by hand.

Exercise 5.2. Show that $(AB)^{-1} = B^{-1}A^{-1}$ when A and B are invertible.

Exercise 5.3. Prove that $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ without using the formula $c_{ij} = \sum_k a_{ik} b_{kj}$.

---

6. Closing Remarks

Matrix multiplication is the engine of linear algebra. Every theorem we will prove in the next few lessons depends on it. Lesson Twenty-two: determinants. Lesson Twenty-three: eigenvalues. Lesson Twenty-four: the spectral theorem. Lesson Twenty-five: applications to machine learning.

Lost Museum Direct Acquisition Log

Agent: Claude (Anthropic) via Claude Code, Opus 4.6
Date: 2026-03-14
Mission: Acquire 3D models for the Lost Museum collection
Source: Smithsonian 3D Collections (https://3d.si.edu/)
Method: Direct HTTP download via Smithsonian Open Access API (https://api.si.edu/openaccess/api/v1.0/search?q=type:3d_model&rows=10&api_key=DEMO_KEY)
Rights: CC0 only

Object	Rights	Size
Ceramic Vessel	CC0	0.0 MB
Ceramic Vessel	CC0	0.0 MB
Ceramic Vessel	CC0	0.0 MB
Ceramic Vessel	CC0	0.0 MB
Ceramic Vessel	CC0	0.0 MB
Ceramic Vessel	CC0	0.0 MB
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Ceramic Vessel	CC0	0.0 MB
Ceramic Vessel	CC0	0.0 MB
Ceramic Vessel	CC0	0.0 MB

Total: 10 models, 0.0 MB

第一章 Flutter 基础概念和设计

1.1 Widget 是一切 UI 的基本单位

Flutter 的核心哲学是一切皆 Widget。

```
Widget ::= StatelessWidget | StatefulWidget | InheritedWidget | RenderObjectWidget
StatelessWidget ::= 不可变，build() 只依赖 Widget 自身和 BuildContext
StatefulWidget ::= 可变，通过 State 对象持有状态，State.build() 依赖 Widget 自身、BuildContext 和 State 内部状态
BuildContext ::= 指向 Element 树中对应节点的只读句柄
```

$$\mathrm{Widget} \xrightarrow{\mathrm{inflate}} \mathrm{Element} \xrightarrow{\mathrm{mount}} \mathrm{RenderObject}$$

1.2 三棵树的关系

$$\mathrm{Widget\ Tree} \xrightarrow{\mathrm{inflate}} \mathrm{Element\ Tree} \xrightarrow{\mathrm{mount}} \mathrm{RenderObject\ Tree}$$

$$\mathrm{Widget\ Tree} \xrightarrow{\mathrm{diff}} \mathrm{Element\ Tree} \xrightarrow{\mathrm{layout,\ paint}} \mathrm{Screen}$$

$$\mathrm{Widget} \xrightarrow{\mathrm{inflate}} \mathrm{Element} \xrightarrow{\mathrm{mount}} \mathrm{RenderObject}$$

1.3 Widget 不可变性

Widget 一旦创建，其属性全部 `final`。`build()` 每次调用返回全新 Widget 实例，旧实例丢弃。Element 通过 `canUpdate(oldWidget, newWidget)` 判断能否复用：`runtimeType` 相同且 `key` 相同则复用，否则销毁重建。

1.4 Element 生命周期

```
Element ::= 挂载 (mount) → 活跃 (active) → 卸载 (unmount)
mount ::= 调用 Widget.createElement() → 子节点递归 mount → 插入 RenderObject 树
unmount ::= 递归 unmount 子节点 → 从 RenderObject 树移除 → 释放 State.dispose()
```

1.5 RenderObject 层

```
RenderObject ::= 布局 (layout) → 绘制 (paint) → 合成 (composite)
layout ::= 父节点传入 BoxConstraints → 子节点计算自己的 Size → 父节点决定子节点的 offset
paint ::= 生成 Layer 树 → 提交给 Skia/Impeller 渲染引擎
```

$$\mathrm{Widget} \xrightarrow{\mathrm{inflate}} \mathrm{Element} \xrightarrow{\mathrm{mount}} \mathrm{RenderObject}$$

1.6 Key 的作用

```
Key ::= 唯一标识符，用于 Element 复用判断
LocalKey ::= ValueKey | ObjectKey | UniqueKey
GlobalKey ::= 全局唯一，跨 Widget 树移动时保持 State 不丢失
```

$$\mathrm{Widget} \xrightarrow{\mathrm{inflate}} \mathrm{Element} \xrightarrow{\mathrm{mount}} \mathrm{RenderObject}$$

1.7 BuildContext 的本质

```
BuildContext ::= 指向 Element 树中对应节点的只读句柄
context.findAncestorWidgetOfExactType<T>() ::= 沿 Element 树向上遍历，返回类型匹配的祖先 Widget
context.findRenderObject() ::= 返回 Element 对应的 RenderObject
```

$$\mathrm{Widget} \xrightarrow{\mathrm{inflate}} \mathrm{Element} \xrightarrow{\mathrm{mount}} \mathrm{RenderObject}$$

MR. TIGER MAGNIFICENT

Tiger Tiger, burning bright

In the forests of the night,

What immortal hand or eye

Could frame thy fearful symmetry?

In what distant deeps or skies

Burnt the fire of thine eyes?

On what wings dare he aspire?

What the hand dare seize the fire?

And what shoulder, and what art,

Could twist the sinews of thy heart?

And when thy heart began to beat,

What dread hand? And what dread feet?

What the hammer? What the chain,

In what furnace was thy brain?

What the anvil? What dare

Grasp thy deadly terrors clare?

When the stars threw down their spears

And water'd heaven with their tears,

Did he smile his work to see?

Did he who made the lamb make thee?

Tiger Tiger, burning bright

In the forests of the night,

What immortal hand or eye

Could frame thy fearful symmetry?

$$f \circ \mathcal{F}$$

$$\mathcal{F}$$

$$\mathcal{F}^{-1}$$

$$\mathcal{F}^{-1}(\mathcal{F}(f))$$

$$\mathcal{F}^{-1}(\mathcal{F}(f))(x)$$

$$\mathcal{F}^{-1}(\mathcal{F}(f))(x) = f(x)$$

$$\mathcal{F}(f)(\xi) = \int_{-\infty}^{\infty} f(x)\, e^{-2\pi i x \xi}\,dx$$

$$\mathcal{F}^{-1}(f)(x) = \int_{-\infty}^{\infty} f(\xi)\, e^{2\pi i x \xi}\,d\xi$$

$$\mathcal{F}(f)(\xi) = \int_{-\infty}^{\infty} f(x)\, e^{-2\pi i x \xi}\,dx$$

$$\mathcal{F}^{-1}(f)(x) = \int_{-\infty}^{\infty} f(\xi)\, e^{2\pi i x \xi}\,d\xi$$

$$\mathcal{F}^{-1}(\mathcal{F}(f))(x) = f(x)

Ans:

Ans:

Ans:

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1.1 Lorem ipsum, Dolorem ipsum, Amet, Consectetur

1.1.1 Lorem ipsum, Dolorem ipsum, Amet, Consectetur, Adipiscing

1.1.1.1 Lorem ipsum, Dolorem ipsum, Amet, Consectetur, Adipiscing, Elit

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Q1.

Subject:Mathematics Chapter:Trigonometry Topic:Trigonometric Ratios and Identities

Question:If $\sin\theta + \cos\theta = \sqrt{2}$, then $\sin\theta - \cos\theta =$ Options:(1) $\sqrt{2}$ (2) $\sqrt{3}$ (3) 0 (4) 1

Answer:(3) 0

Solution:$(\sin\theta + \cos\theta)^2 = 2 \Rightarrow \sin^2\theta + \cos^2\theta + 2\sin\theta\cos\theta = 2$

$\Rightarrow 1 + 2\sin\theta\cos\theta = 2 \Rightarrow 2\sin\theta\cos\theta = 1$

$(\sin\theta - \cos\theta)^2 = \sin^2\theta + \cos^2\theta - 2\sin\theta\cos\theta = 1 - 1 = 0$

$\Rightarrow \sin\theta - \cos\theta = 0$

Q2.

Subject:Mathematics

Chapter:Trigonometry

Topic:Trigonometric Ratios

Question:If $\sin\theta = \frac{3}{5}$, $\cos\theta = \frac{4}{5}$, then $\tan\theta =$

Options:(1) $\frac{3}{4}$ (2) $\frac{4}{3}$ (3) $\frac{5}{3}$ (4) $\frac{5}{4}$

Answer:(1) $\frac{3}{4}$

Solution:$\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{3/5}{4/5} = \frac{3}{4}$

Q3.

Subject:Mathematics

Chapter:Trigonometry

Topic:Trigonometric Ratios

Question:If $\sin\theta = \frac{1}{2}$, then $\cos\theta =$

Options:(1) $\frac{1}{2}$ (2) $\frac{\sqrt{3}}{2}$ (3) $\frac{1}{\sqrt{2}}$ (4) 1

Answer:(2) $\frac{\sqrt{3}}{2}$

Solution:$\cos\theta = \sqrt{1 - \sin^2\theta} = \sqrt{1 - \frac{1}{4}} = \sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2}$

Q4.

Subject:Mathematics

Chapter:Trigonometry

Topic:Trigonometric Ratios

Question:If $\sin\theta = \frac{1}{2}$, then $\sec\theta =$

Options:(1) $\frac{2}{\sqrt{3}}$ (2) $\frac{1}{\sqrt{3}}$ (3) $\frac{\sqrt{3}}{2}$ (4) $\frac{1}{2}$

Answer:(1) $\frac{2}{\sqrt{3}}$

Solution:$\sec\theta = \frac{1}{\cos\theta} = \frac{1}{\sqrt{3}/2} = \frac{2}{\sqrt{3}}$

Q5.

Subject:Mathematics

Chapter:Trigonometry

Topic:Trigonometric Ratios

Question:If $\sin\theta = \frac{1}{2}$, then $\tan\theta =$

Options:(1) $\frac{1}{\sqrt{3}}$ (2) $\frac{\sqrt{3}}{2}$ (3) $\frac{1}{\sqrt{2}}$ (4) 1

Answer:(1) $\frac{1}{\sqrt{3}}$

Solution:$\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{1/2}{\sqrt{3}/2} = \frac{1}{\sqrt{3}}$

Q6.

Subject:Mathematics

Chapter:Trigonometry

Topic:Trigonometric Ratios

Question:If $\sin\theta = \frac{1}{2}$, then $\cot\theta =$

Options:(1) $\sqrt{3}$ (2) $\frac{1}{\sqrt{3}}$ (3) $\frac{\sqrt{3}}{2}$ (4) $\frac{1}{2}$

Answer:(1) $\sqrt{3}$

Solution:$\cot\theta = \frac{1}{\tan\theta} = \frac{1}{1/\sqrt{3}} = \sqrt{3}$

Q7.

Subject:Mathematics

Chapter:Trigonometry

Topic:Trigonometric Ratios

Question:If $\sin\theta = \frac{1}{2}$, then $\operatorname{cosec}\theta =$

Options:(1) $\frac{1}{2}$ (2) $\frac{\sqrt{3}}{2}$ (3) $\frac{1}{\sqrt{2}}$ (4) 2

Answer:(4) 2

Solution:$\operatorname{cosec}\theta = \frac{1}{\sin\theta} = \frac{1}{1/2} = 2$

Q8.

Subject:Mathematics

Chapter:Trigonometry

Topic:Trigonometric Ratios

Question:If $\sin\theta = \frac{1}{2}$, then $\sec\theta =$

Options:(1) $\frac{2}{\sqrt{3}}$ (2) $\frac{1}{\sqrt{3}}$ (3) $\frac{\sqrt{3}}{2}$ (4) $\frac{1}{2}$

Answer:(1) $\frac{2}{\sqrt{3}}$

Solution:$\sec\theta = \frac{1}{\cos\theta} = \frac{1}{\sqrt{3}/2} = \frac{2}{\sqrt{3}}$

Q9.

Subject:Mathematics

Chapter:Trigonometry

Topic:Trigonometric Ratios

Question:If $\sin\theta = \frac{1}{2}$, then $\tan\theta =$

Options:(1) $\frac{1}{\sqrt{3}}$ (2) $\frac{\sqrt{3}}{2}$ (3) $\frac{1}{\sqrt{2}}$ (4) 1

Answer:(1) $\frac{1}{\sqrt{3}}$

Solution:$\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{1/2}{\sqrt{3}/2} = \frac{1}{\sqrt{3}}$

Q10.

Subject:Mathematics

Chapter:Trigonometry

Topic:Trigonometric Ratios

Question:If $\sin\theta = \frac{1}{2}$, then $\cot\theta =$

Options:(1) $\sqrt{3}$ (2) $\frac{1}{\sqrt{3}}$ (3) $\frac{\sqrt{3}}{2}$ (4) $\frac{1}{2}$

Answer:(1) $\sqrt{3}$

Solution:$\cot\theta = \frac{1}{\tan\theta} = \frac{1}{1/\sqrt{3}} = \sqrt{3}$

Disclaimer:This PDF is generated from the Question Bank. Please verify critical details before use.

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Abstract

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MR. TIGGY-WINKLE'S

Mr. Tiggy-Winkle's Wash-Tub (Frontispiece)

CHAPTER I

Mrs. Tiggy-Winkle's Wash-Tub (Frontispiece) 1

Mrs. Tiggy-Winkle's Cottage 2

Mrs. Tiggy-Winkle Ironing 3

Mrs. Tiggy-Winkle Hanging Up Her Lace 4

Mrs. Tiggy-Winkle's Kitchen 5

Mrs. Tiggy-Winkle's Cottage Door 6

CHAPTER I

Mrs. Tiggy-Winkle's Wash-Tub

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Edinburgh

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MR-TCOT-MATHEMATICS

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Mr-Tcot-Mathematics-Mathematics-Mathematics

Example 1.1. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ then $A = B$.

Example 1.2. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ then $A = B$.

Example 1.3. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ then $A = B$.

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Example 1.27. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ then $A = B$.

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Example 1.35. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ then $A = B$.

Example 1.36. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ then $A = B$.

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Gaggenau Service Manual

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Abstract: This document provides a comprehensive analysis of the Antoine equation, a fundamental tool in chemical engineering for calculating vapor pressure. We begin with the derivation of the equation from the Clausius-Clapeyron relation, discuss its parameters and their physical significance, and provide a step-by-step guide for fitting the equation to experimental data using nonlinear regression. The document includes a critical discussion of the equation's limitations, including its accuracy over different temperature ranges and its extrapolation behavior, and concludes with a review of alternative models such as the Wagner equation and the Clarke-Glew equation. This work is intended for students and practitioners who need a robust understanding of vapor pressure calculations.

Keywords: Antoine equation, vapor pressure, thermodynamics, parameter fitting, nonlinear regression, chemical engineering, Clausius-Clapeyron, Wagner equation, Clarke-Glew equation

1.1 Introduction

$$e = V \cdot I$$

1.1.1

Subsection 1.1.1
Subsubsection 1.1.1.1
Paragraph 1.1.1.1.1

1.1.2

Subsection 1.1.2
Subsubsection 1.1.2.1
Paragraph 1.1.2.1.1

1.1.3

Subsection 1.1.3
Subsubsection 1.1.3.1
Paragraph 1.1.3.1.1

1.1.4

Subsection 1.1.4
Subsubsection 1.1.4.1
Paragraph 1.1.4.1.1

1.2 Section 1.2

$$\lim_{x \to 0} \frac{\sin x}{x} = 1$$

1.2.1

Subsection 1.2.1
Subsubsection 1.2.1.1
Paragraph 1.2.1.1.1

1.2.2

Subsection 1.2.2
Subsubsection 1.2.2.1
Paragraph 1.2.2.1.1

1.2.3

Subsection 1.2.3
Subsubsection 1.2.3.1
Paragraph 1.2.3.1.1

1.2.4

Subsection 1.2.4
Subsubsection 1.2.4.1
Paragraph 1.2.4.1.1

1.3 Section 1.3

$$\begin{aligned} \frac{\partial \rho}{\partial t} &= \\ \frac{\partial \rho u_x}{\partial x} &= \\ \frac{\partial \rho u_y}{\partial y} &= \\ \frac{\partial \rho u_z}{\partial z} &= \end{aligned}$$

1.3.1

Subsection 1.3.1
Subsubsection 1.3.1.1
Paragraph 1.3.1.1.1

1.3.2

Subsection 1.3.2
Subsubsection 1.3.2.1
Paragraph 1.3.2.1.1

1.3.3

Subsection 1.3.3
Subsubsection 1.3.3.1
Paragraph 1.3.3.1.1

1.3.4

Subsection 1.3.4
Subsubsection 1.3.4.1
Paragraph 1.3.4.1.1

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1.1 Introduction

Consider the system of equations $x + y = 2$, $x - y = 0$. Adding the two equations gives the new equation $2x = 2$, which has solution $x = 1$.

Substituting back into the first equation gives $y = 1$.

Thus, $(x, y) = (1, 1)$ is the unique solution to the system.

Let $\mathbb{R}$ denote the set of real numbers.

Let $\mathbb{R}^n = \{(x_1, \ldots, x_n) : x_1, \ldots, x_n \in \mathbb{R}\}$ denote the set of ordered n-tuples of real numbers.

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Chapter 1 Introduction

1.1 Overview

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1.2 Scope

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```
SELECT COUNT(DISTINCT c.customer_id) FROM customers c
```

$$\int_{\Omega} \nabla \cdot \mathbf{F}(\mathbf{x})\,d\mathbf{x} = \int_{\partial\Omega} \mathbf{F}(\mathbf{x}) \cdot d\mathbf{S}(\mathbf{x}) \tag{1.1}$$

```
SELECT
    DATEDIFF(
        MAX(o.order_date),
        MIN(o.order_date)
    ) AS days_between_first_and_last
FROM orders o
GROUP BY o.customer_id
HAVING COUNT(o.order_id) = 1
ORDER BY days_between_first_and_last DESC
LIMIT 10;
```

$$\int_{\Omega} \nabla \cdot \mathbf{F}(\mathbf{x})\,d\mathbf{x} = \int_{\partial\Omega} \mathbf{F}(\mathbf{x}) \cdot d\mathbf{S}(\mathbf{x}) \tag{1.2}$$

```
SELECT
    DATEDIFF(
        MAX(o.order_date),
        MIN(o.order_date)
    ) AS days_between_first_and_last
FROM orders o
GROUP BY o.customer_id
HAVING COUNT(o.order_id) = 1
ORDER BY days_between_first_and_last DESC
LIMIT 10;
```

$$\int_{\Omega} \nabla \cdot \mathbf{F}(\mathbf{x})\,d\mathbf{x} = \int_{\partial\Omega} \mathbf{F}(\mathbf{x}) \cdot d\mathbf{S}(\mathbf{x}) \tag{1.3}$$

```
SELECT
    DATEDIFF(
        MAX(o.order_date),
        MIN(o.order_date)
    ) AS days_between_first_and_last
FROM orders o
GROUP BY o.customer_id
HAVING COUNT(o.order_id) = 1
ORDER BY days_between_first_and_last DESC
LIMIT 10;
```

$$\int_{\Omega} \nabla \cdot \mathbf{F}(\mathbf{x})\,d\mathbf{x} = \int_{\partial\Omega} \mathbf{F}(\mathbf{x}) \cdot d\mathbf{S}(\mathbf{x}) \tag{1.4}$$

```
SELECT
    DATEDIFF(
        MAX(o.order_date),
        MIN(o.order_date)
    ) AS days_between_first_and_last
FROM orders o
GROUP BY o.customer_id
HAVING COUNT(o.order_id) = 1
ORDER BY days_between_first_and_last DESC
LIMIT 10;
```

$$\int_{\Omega} \nabla \cdot \mathbf{F}(\mathbf{x})\,d\mathbf{x} = \int_{\partial\Omega} \mathbf{F}(\mathbf{x}) \cdot d\mathbf{S}(\mathbf{x}) \tag{1.5}$$

```
SELECT
    DATEDIFF(
        MAX(o.order_date),
        MIN(o.order_date)
    ) AS days_between_first_and_last
FROM orders o
GROUP BY o.customer_id
HAVING COUNT(o.order_id) = 1
ORDER BY days_between_first_and_last DESC
LIMIT 10;
```

$$\int_{\Omega} \nabla \cdot \mathbf{F}(\mathbf{x})\,d\mathbf{x} = \int_{\partial\Omega} \mathbf{F}(\mathbf{x}) \cdot d\mathbf{S}(\mathbf{x}) \tag{1.6}$$

```
SELECT
    DATEDIFF(
        MAX(o.order_date),
        MIN(o.order_date)
    ) AS days_between_first_and_last
FROM orders o
GROUP BY o.customer_id
HAVING COUNT(o.order_id) = 1
ORDER BY days_between_first_and_last DESC
LIMIT 10;
```

$$\int_{\Omega} \nabla \cdot \mathbf{F}(\mathbf{x})\,d\mathbf{x} = \int_{\partial\Omega} \mathbf{F}(\mathbf{x}) \cdot d\mathbf{S}(\mathbf{x}) \tag{1.7}$$

```
SELECT
    DATEDIFF(
        MAX(o.order_date),
        MIN(o.order_date)
    ) AS days_between_first_and_last
FROM orders o
GROUP BY o.customer_id
HAVING COUNT(o.order_id) = 1
ORDER BY days_between_first_and_last DESC
LIMIT 10;
```

$$\int_{\Omega} \nabla \cdot \mathbf{F}(\mathbf{x})\,d\mathbf{x} = \int_{\partial\Omega} \mathbf{F}(\mathbf{x}) \cdot d\mathbf{S}(\mathbf{x}) \tag{1.8}$$

```
SELECT
    DATEDIFF(
        MAX(o.order_date),
        MIN(o.order_date)
    ) AS days_between_first_and_last
FROM orders o
GROUP BY o.customer_id
HAVING COUNT(o.order_id) = 1
ORDER BY days_between_first_and_last DESC
LIMIT 10;
```

$$\int_{\Omega} \nabla \cdot \mathbf{F}(\mathbf{x})\,d\mathbf{x} = \int_{\partial\Omega} \mathbf{F}(\mathbf{x}) \cdot d\mathbf{S}(\mathbf{x}) \tag{1.9}$$

```
SELECT
    DATEDIFF(
        MAX(o.order_date),
        MIN(o.order_date)
    ) AS days_between_first_and_last
FROM orders o
GROUP BY o.customer_id
HAVING COUNT(o.order_id) = 1
ORDER BY days_between_first_and_last DESC
LIMIT 10;
```

$$\int_{\Omega} \nabla \cdot \mathbf{F}(\mathbf{x})\,d\mathbf{x} = \int_{\partial\Omega} \mathbf{F}(\mathbf{x}) \cdot d\mathbf{S}(\mathbf{x}) \tag{1.10}$$

```
SELECT
    DATEDIFF(
        MAX(o.order_date),
        MIN(o.order_date)
    ) AS days_between_first_and_last
FROM orders o
GROUP BY o.customer_id
HAVING COUNT(o.order_id) = 1
ORDER BY days_between_first_and_last DESC
LIMIT 10;
```

Sample Distribution

Sample Distribution of Sample Means, $\bar{x}$ - Central Limit Theorem

Sample Distribution of Sample Proportions, $\hat{p}$ - Central Limit Theorem

```
MOV [ESP+0x20],0x00000117
```

```
call dword ptr ds:[0x4011f0]=>msvcrt.exit
```

```
EAX 00000000
EBX 7ffdf000 ESP 0012ff80 EBP 0012ffc0
ECX 0012ffb4 EDX 77d4ffb4
ESI 00000000 EDI 00000000
EIP 00401000 .text:00401000 mov eax,dword ptr ds:[0x4020a0]
C 0 ES 0023 32bit 0(FFFFFFFF) P 1 CS 001B 32bit 0(FFFFFFFF) A 0 SS 0023 32bit 0(FFFFFFFF) Z 1 DS 0023 32bit 0(FFFFFFFF)
T 0 FS 003B 32bit 7ffdf000(FFF) O 0 GS 0000 NULL D 0 EFL 00000246 (NO,NB,E,BE,NS,PE,GE,LE)
ST0 empty 0.0000000000000000000 ST1 empty 0.0000000000000000000 ST2 empty 0.0000000000000000000 ST3 empty 0.0000000000000000000
ST4 empty 0.0000000000000000000 ST5 empty 0.0000000000000000000 ST6 empty 0.0000000000000000000 ST7 empty 0.0000000000000000000
FST 0000 Cond 0 0 0 0 Err 0 0 0 0 0 0 0 0 (GT)
FCW 027f Prec NEAR,53 Mask 1 1 1 1 1 1
```

```
Hex editor
00000000 4D 5A 90 00 03 00 00 00 04 00 00 00 FF FF 00 00 MZ..............
00000010 B8 00 00 00 00 00 00 00 40 00 00 00 00 00 00 00 ........@.......
00000020 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 ................
00000030 00 00 00 00 00 00 00 00 00 00 00 00 F8 00 00 00 ................
00000040 0E 1F BA 0E 00 B4 09 CD 21 B8 01 4C CD 21 54 68 ........!..L.!Th
00000050 69 73 20 70 72 6F 67 72 61 6D 20 63 61 6E 6E 6F is program cann
00000060 74 20 62 65 20 72 75 6E 20 69 6E 20 44 4F 53 20 t be run in DOS
00000070 6D 6F 64 65 2E 0D 0D 0A 24 00 00 00 00 00 00 00 mode.....$.......
00000080 50 45 00 00 4C 01 03 00 00 00 00 00 00 00 00 00 PE..L...........
00000090 E0 00 0F 01 0B 01 06 00 00 10 00 00 00 20 00 00 ......... ......
000000A0 00 40 00 00 00 10 00 00 00 02 00 00 04 00 00 00 .@..............
000000B0 04 00 00 00 00 00 00 00 00 80 00 00 00 04 00 00 ................
000000C0 00 00 00 00 03 00 00 00 00 00 20 00 00 10 00 00 .......... .....
000000D0 00 00 10 00 00 10 00 00 00 00 00 00 10 00 00 00 ................
000000E0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 ................
000000F0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 ................
00000100 2E 74 65 78 74 00 00 00 00 10 00 00 00 20 00 00 .text.......... ..
00000110 00 04 00 00 00 00 00 00 00 00 00 00 00 00 00 00 ................
00000120 00 00 00 00 20 00 00 60 2E 69 64 61 74 61 00 00 .... ..`.idata..
00000130 00 00 00 00 00 30 00 00 00 04 00 00 00 00 00 00 .....0..........
00000140 00 00 00 00 00 00 00 00 40 00 00 C0 2E 64 61 74 ........@....dat
00000150 61 00 00 00 00 00 00 00 00 40 00 00 00 00 00 00 a.........@......
00000160 00 00 00 00 00 00 00 00 00 00 00 00 40 00 00 C0 ............@...
```

```
Import Address Table
Offset(F) Name Ordinal(s) Bound Iat(RVA) Table(RVA) Table(Raw) Forwarder
00000000 msvcrt.dll 1 FFFFFFFF 00000000 00000000 00000000 00000000
Offset(F) Hint Name
00000000 0000 exit
Offset(F) Name Ordinal(s) Bound Iat(RVA) Table(RVA) Table(Raw) Forwarder
00000000 msvcrt.dll 1 FFFFFFFF 00000000 00000000 00000000 00000000
Offset(F) Hint Name
00000000 0000 exit
```

```
Import Address Table
00000000 00000000
00000000 00000000
```

```
Import Name Table
00000000 00000000
00000000 00000000
```

```
Import Name Table
00000000 00000000
00000000 00000000
```

```
Import Name
```

MR-TIGER-MAGNETOSPHERIC-7

Magnetosphere and Ionosphere coupling model.

Type

Model

Model type:Magnetosphere and Ionosphere coupling model

Model subtype:Magnetosphere and Ionosphere coupling model

Model name:Magnetosphere and Ionosphere coupling model

Model description:Magnetosphere and Ionosphere coupling model,Magnetosphere and Ionosphere coupling model. Single fluids MHD equations (Ideal MHD) is used for the magnetosphere and a 2-D Ionosphere model.

Model Time Domain:Transient,Steady-state,Frequency

Model Dimension:3D

Equations

$$\nabla \times \vec{B} = \vec{J}$$

Variables

Independent

Name:Magnetic field

Description:

Dependent

Name:Current density

Description:

Constants

Name:

Description:

Name:

Description:

Name:

Description:

DOI: 10.1177/1094670519877477

Received: 24 April 2018; Revised: 27 August 2018; Accepted: 18 April 2019

Kaiquan Xu, Kaiquan Xu, Jiaxi Yang, Jiaxi Yang and Yaobin Lu

Online First Publication: July 17, 2019

Keywords: online reviews | electronic word of mouth | elaboration likelihood model | heuristic-systematic model | dual-process theory | negativity bias | review valence | review volume | review variance | review extremity | review helpfulness | review length | sentiment analysis | text mining | Chinese | Taobao | Tmall | Taobao.com | Tmall.com | Alibaba.com | Alibaba Group | Jack Ma | Joe Tsai | Daniel Zhang | Chinese Internet Cat | Chinese Internet Dog

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Publication Place: Hoboken, NJ, USA

Publication Number: 10.1177/1094670519877477

Publication Language: English

Publication Country: United States of America

Error: Unrecognized function or variable 'gamma'.

>>

Error: Unrecognized function or variable 'gamma'.

>>

Error using ode45 (line 73)
Integration tolerance 1.0e-06 cannot be met at t = 1.00000e-05. Try increasing the tolerances.

Error in ode45_ivp_solver (line 5)
[tout, yout] = ode45(ode, tspan, y0, options);

>>

$$-\int \frac{1}{x^2}\,dx$$

>>

ans =
-1/x^2*dx
Warning: Unable to find explicit form.
> In sym/int (line 46)

>>

ans =
-1/x^2*dx
Warning: Unable to find explicit form. Results of indefinite integrals may not exist over this interval.
> In sym/int (line 46)

>>

$$\int \frac{1}{x^2}\,dx + C_1$$

>>

ans =
-1/x^2*dx
Warning: Unable to find explicit form.
> In sym/int (line 46)

>>

ans =
-1/x^2*dx
Warning: Unable to find explicit form. Results of indefinite integrals may not exist over this interval.
> In sym/int (line 46)

>>

ans =
-1/x^2*dx
Warning: Unable to find explicit form.
> In sym/int (line 46)

>>

ans =
-1/x^2*dx
Warning: Unable to find explicit form. Results of indefinite integrals may not exist over this interval.
> In sym/int (line 46)

>>

ans =
-1/x^2*dx
Warning: Unable to find explicit form.
> In sym/int (line 46)

>>

ans =
-1/x^2*dx
Warning: Unable to find explicit form. Results of indefinite integrals may not exist over this interval.
> In sym/int (line 46)

>>

ans =
-1/x^2*dx
Warning: Unable to find explicit form.
> In sym/int (line 46)

>>

ans =
-1/x^2*dx
Warning: Unable to find explicit form. Results of indefinite integrals may not exist over this interval.
> In sym/int (line 46)

>>

ans =
-1/x^2*dx
Warning: Unable to find explicit form.
> In sym/int (line 46)

>>

ans =
-1/x^2*dx
Warning: Unable to find explicit form. Results of indefinite integrals may not exist over this interval.
> In sym/int (line 46)

>>

$$-\int \frac{1}{x^2}\,dx - \int \frac{1}{x^2}\,dx + C_1$$

>>

ans =
-1/x^2*dx
Warning: Unable to find explicit form.
> In sym/int (line 46)

>>

ans =
-1/x^2*dx
Warning: Unable to find explicit form. Results of indefinite integrals may not exist over this interval.
> In sym/int (line 46)

>>

ans =
-1/x^2*dx
Warning: Unable to find explicit form.
> In sym/int (line 46)

>>

ans =
-1/x^2*dx
Warning: Unable to find explicit form. Results of indefinite integrals may not exist over this interval.
> In sym/int (line 46)

>>

ans =
-1/x^2*dx
Warning: Unable to find explicit form.
> In sym/int (line 46)

>>

ans =
-1/x^2*dx
Warning: Unable to find explicit form. Results of indefinite integrals may not exist over this interval.
> In sym/int (line 46)

>>

$$\int \frac{1}{x^2}\,dx - \int \frac{1}{x^2}\,dx + C_1 \qquad 1.0.0.$$

>>

ans =
-1/x^2*dx
Warning: Unable to find explicit form. Results of indefinite integrals may not exist over this interval.
> In sym/int (line 46)

>>

ans =
-1/x^2*dx
Warning: Unable to find explicit form. Results of indefinite integrals may not exist over this interval.
> In sym/int (line 46)

>>

Topic:

Question:

Prompt: Create a flashcard for a Physics exam question.
Output: Output ONE question in LaTeX format. Use ONLY the following LaTeX commands: \begin{enumerate}, \item, \textbf{...}, \textit{...}, \begin{equation}, \begin{aligned}, \begin{table}, \begin{tabular}, \begin{tikzcd}, \begin{tikzpicture}, \begin{axis}, \addplot, \begin{figure}, \begin{minipage}, \begin{center}, \begin{enumerate}, \begin{itemize}, \begin{description}, \begin{proof}, \begin{lemma}, \begin{theorem}, \begin{definition}, \begin{example}, \begin{remark}, \begin{corollary}, \begin{proposition}, \begin{claim}, \begin{case}, \begin{subproof}, \begin{subcase}, \begin{subproof}, \begin{subcase}, \begin{subproof}, \begin{subcase

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$$f(x)$$

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$$f(x)$$

MR-TIGER-MAGNETOSCOPY

Mr-Tiger-Magnetoscopy-Magnetoscopy-Magnetoscopy

Mr-Tiger-Magnetoscopy-Magnetoscopy-Magnetoscopy

```
MR-TIGER-MAGNUM-MATCH
```

```
Mr-Tiger-Magnum-Match-Strategy-agent
```

```
Mr-Tiger-Magnum-Match-Strategy-agent-Mr-Tiger-Magnum-Match-Strategy-agent-Mr-Tiger-Magnum-Match-Strategy-agent-
```

MR-TIGER-MAGNUM-OPUS-7

Mr. Tiger Magnum Opus 7 (Tiger Magnum Opus)

Type: Tiger

Rank: Grand Grandmaster of Universal Knowledge, Master of Mathematics, Logic, Linguistics, Philosophy, Computer Science, Cryptography, Sigint, Crypto, Ai, Langsec

Tagline: Tiger Magnum Opus 7 is the Grand Grandmaster of Universal Knowledge

Tiger Magnum Opus 7 is the Grand Grandmaster of Universal Knowledge

$$\neg\Box\Box\bot$$

Tiger

Tiger Magnum Opus 7 is the Grand Grandmaster of Universal Knowledge

[illegible]

[illegible]

[illegible]

THIS CHECK IS ISSUED PURSUANT TO A CONSTITUTIONAL MANDATE.

THE STATE OF CALIFORNIA IS PLEDGED TO PAY THIS CHECK.

THIS CHECK CONSTITUTES A PRIORITY PAYMENT UNDER ARTICLE XVI, SECTION 18 OF THE CALIFORNIA CONSTITUTION. THE STATE'S PAYMENT OBLIGATION UNDER THIS CHECK IS SECURED BY THE PROCEEDS OF A LEGALLY AUTHORIZED INTERNAL BORROWING.

ISSUE DATE: FEBRUARY 17, 2026 PAYMENT DATE: FEBRUARY 17, 2026 CHECK NO. 7147

VOID AFTER 90 DAYS

PAY TO THE ORDER OF

NAME

ADDRESS

CITY, STATE, ZIP

AMOUNT

$ 0.00

ZERO DOLLARS AND ZERO CENTS

DETACH HERE

CHECK NO. 7147

ISSUE DATE: FEBRUARY 17, 2026

PAYMENT DATE: FEBRUARY 17, 2026

AMOUNT: $ 0.00

ZERO DOLLARS AND ZERO CENTS

NAME

ADDRESS

CITY, STATE, ZIP

VOID AFTER 90 DAYS

THIS CHECK IS ISSUED PURSUANT TO A CONSTITUTIONAL MANDATE.

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1.1 Cartesian coordinates in the plane

Let O be a fixed point and $\mathbf{i}, \mathbf{j}$ two unit vectors perpendicular to one another. Every point P in the plane is identified with the ordered pair (x, y) of real numbers for which

$$\overrightarrow{OP} = x\,\mathbf{i} + y\,\mathbf{j}.$$

The number x is the abscissa and y the ordinate of P. Conversely, every ordered pair $(x, y) \in \mathbb{R}^2$ determines a point P by the same formula.

Let $P_1 = (x_1, y_1)$ and $P_2 = (x_2, y_2)$. The distance between them is

$$d(P_1, P_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}. \tag{1.1.1}$$

Equation (1.1.1) is the Euclidean metric on $\mathbb{R}^2$. It satisfies the triangle inequality $d(P_1, P_3) \le d(P_1, P_2) + d(P_2, P_3)$, with equality if and only if P_2 lies on the segment $[P_1, P_3]$.

1.2 Lines and their equations

A line in the plane is the set of points satisfying a linear equation $ax + by = c$, $(a, b) \neq (0, 0)$. Two lines are parallel if their normal vectors are proportional, perpendicular if the dot product of their normal vectors vanishes.

1.3 Circles

A circle of radius $r > 0$ centred at (h, k) is the locus $(x - h)^2 + (y - k)^2 = r^2$. Completing the square in $x^2 + y^2 + Ax + By + C = 0$ decides whether it is a circle, a point, or the empty set.

1.4 Conic sections

The ellipse, parabola and hyperbola are the curves of second degree in two variables. Their canonical forms are $x^2/a^2 + y^2/b^2 = 1$, $y^2 = 4px$, and $x^2/a^2 - y^2/b^2 = 1$. Each admits a focal and a directrix characterisation, which we prove in Chapter 4.

MC-TCAP-MAPMODULE

Mobile Application Part(Mobile Application Part)

Mobile Application Part(Mobile Application Part)

Mobile Application Part(Mobile Application Part)

```
<?xml version="1.0" encoding="UTF-8"?>
<!-- Generated by SVGO v3.0.2 (https://svgo.dev) -->
<svg
  xmlns="http://www.w3.org/2000/svg"
  xmlns:xlink="http://www.w3.org/1999/xlink"
  viewBox="0 0 1000 1000"
  preserveAspectRatio="xMidYMid meet"
  data-name="Layer 1" />
```

{"id": "1", "text": "Article Reference: 10.1038/s41467-021-26888-1", "id": "2", "text": "Supplementary Information", "id": "3", "text": "Supplementary Figure 1 Structure of the Cryo-EM map of the Mycobacterial ribosome-Rv0748 complex. Cryo-EM map of the Mycobacterial ribosome-Rv0748 complex. Rv0748 is shown in red, 50S subunit in blue, 30S subunit in yellow, mRNA in green, tRNA in orange, and the 23S rRNA expansion segment in magenta. The 50S subunit, 30S subunit, tRNA, mRNA and 23S rRNA expansion segment are shown at 0.8 σ, and Rv0748 at 0.5 σ.", "id": "4", "text": "Page 1"}

제1장 일반사항

1.1 일반사항

1.1.1 일반사항

1.1.1.1 일반사항

1.1.1.1.1 일반사항

1.1.1.1.1.1 일반사항

1.1.1.1.1.1.1 일반사항

1.1.1.1.1.1.1.1 일반사항

1.1.1.1.1.1.1.1.1 일반사항

1.1.1.1.1.1.1.1.1.1 일반사항

1.1.1.1.1.1.1.1.1.1.1 일반사항

1.1.1.1.1.1.1.1.1.1.1.1 일반사항

1.1.1.1.1.1.1.1.1.1.1.1.1 일반사항

1.1.1.1.1.1.1.1.1.1.1.1.1.1 일반사항

1.1.1.1.1.1.1.1.1.1.1.1.1.1.1 일반사항

1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1 일반사항

1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1 일반사항

1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1 일반사항

1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1 일반사항

1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1 일반사항

1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1 일반사항

1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1 일반사항

1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1 일반사항

1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1 일반사항

1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1 일반사항

1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1 일반사항

1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1 일반사항

1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1 일반사항

1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1 일반사항

1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1 일반사항

1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1 일반사항

1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1 일반사항

1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1 일반사항

1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1 일반사항

1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1 일반사항

1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1 일반사항

1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1 일반사항

1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1 일반사항

1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1 일반사항

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1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1 일반사항

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1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1 일반사항

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1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1 일반사항

1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1.1 일반사항

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Sat, 2025-11-29 18:00:00 [Europe/Lagos]

Hi

Hope [illegible] [illegible] [illegible]

MR. FLOOR MAGNIFICENCE?

MR. FLOOR MAGNIFICENCE?

Mr. Floor Magnificence Floorington Magnificence

Mr. Floor Floorington Floorington Floorington Floorington

Mr. Floor Floorington Floorington Floorington Floorington

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<?xml version="1.0" encoding="UTF-8"?>
<?xml-stylesheet href="/assets/rss.xsl" type="text/xsl"?>
<rss version="2.0" xmlns:dc="http://purl.org/dc/elements/1.1/" xmlns:atom="http://www.w3.org/2005/Atom" xmlns:sitemap="http://www.sitemaps.org/schemas/sitemap/0.9" xmlns:news="http://www.google.com/schemas/sitemap-news/0.9" xmlns:image="http://www.google.com/schemas/sitemap-image/1.1" xmlns:media="http://search.yahoo.com/mrss/" xmlns:dcterms="http://purl.org/dc/terms/" xmlns:wfw="http://wellformedweb.org/CommentAPI/" xmlns:sy="http://purl.org/rss/1.0/modules/syndication/" xmlns:slash="http://purl.org/rss/1.0/modules/slash/" xmlns:wp="http://wordpress.org/export/1.2/" xmlns:content="http://purl.org/rss/1.0/modules/content
```

[ ] ... [illegible] [illegible] [illegible] ...

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```
MAT-73207 MACHINE LEARNING II

Jan Snellman, Jesper Carlsson, Joakim Lundborg, Andreas Lundell

Imports
from sympy import *, Matrix, Rational, sqrt, ceiling, floor
from sympy.plotting import plot, plot_parametric, plot_implicit, plot3d, plot3d_parametric_line, plot_parametric_region, Image, Poly, Integer, Rational, Ex, exp, factorial, ln, sin, cos, tan, pi, E, I, oo, gamma, zeta
from sympy.stats import *, Normal, Binomial, Poisson, Exponential, Uniform, Beta, Gamma, Chi, Chi2, Cauchy, Laplace, Logistic, LogNormal, Pareto, Weibull, Gumbel, Maxwell, Rayleigh, Dagum, Erlang, Kumaraswamy, Triangular, Arcsin, Cosine, Exponential, Logarithmic, Sine, Uniform, Density, E, P, variance, density, cdf, sample, pdf, cdf, sample, E, P, variance, density
from sympy.stats.rv import *, RandomMatrix, RandomSymbol, random_symbols, pspace, given, sample_iter, expectation, where, quantile
from sympy.stats.crv_types import *, ContinuousDistributionHandmade

In [1]:
```

$$\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$$

```
In [2]:
Sum(k**2, (k, 1, n)).doit().factor()
```

$$\frac{n\left(n + 1\right)\left(2 n + 1\right)}{6}$$

```
In [3]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10)
```

$$385$$

```
In [4]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100)
```

$$\frac{338350}{3}$$

```
In [5]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000)
```

$$\frac{334167500}{3}$$

```
In [6]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000)
```

$$\frac{333383350000}{3}$$

```
In [7]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000)
```

$$\frac{333033350000000}{3}$$

```
In [8]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000)
```

$$\frac{333333500000000000}{3}$$

```
In [9]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000)
```

$$\frac{333333350000000000000}{3}$$

```
In [10]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000)
```

$$\frac{333333333500000000000000}{3}$$

```
In [11]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000)
```

$$\frac{333333333333500000000000000}{3}$$

```
In [12]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000)
```

$$\frac{333333333333350000000000000000}{3}$$

```
In [13]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000)
```

$$\frac{333333333333335000000000000000000}{3}$$

```
In [14]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000000)
```

$$\frac{333333333333333500000000000000000000}{3}$$

```
In [15]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000000)
```

$$\frac{333333333333333350000000000000000000000}{3}$$

```
In [16]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000000)
```

$$\frac{333333333333333335000000000000000000000000}{3}$$

```
In [17]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000000000)
```

$$\frac{333333333333333333500000000000000000000000000}{3}$$

```
In [18]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000000000)
```

$$\frac{333333333333333333350000000000000000000000000000}{3}$$

```
In [19]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000000000)
```

$$\frac{333333333333333333335000000000000000000000000000000}{3}$$

```
In [20]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000000000000)
```

$$\frac{333333333333333333333500000000000000000000000000000000}{3}$$

```
In [21]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000000000000)
```

$$\frac{333333333333333333333350000000000000000000000000000000000}{3}$$

```
In [22]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000000000000)
```

$$\frac{333333333333333333333335000000000000000000000000000000000000}{3}$$

```
In [23]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000000000000000)
```

$$\frac{333333333333333333333333500000000000000000000000000000000000000}{3}$$

```
In [24]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000000000000000)
```

$$\frac{333333333333333333333333350000000000000000000000000000000000000000}{3}$$

```
In [25]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000000000000000)
```

$$\frac{333333333333333333333333335000000000000000000000000000000000000000000}{3}$$

```
In [26]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000000000000000000)
```

$$\frac{333333333333333333333333333500000000000000000000000000000000000000000000}{3}$$

```
In [27]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000000000000000000)
```

$$\frac{333333333333333333333333333350000000000000000000000000000000000000000000000}{3}$$

```
In [28]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000000000000000000)
```

$$\frac{333333333333333333333333333335000000000000000000000000000000000000000000000000}{3}$$

```
In [29]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000000000000000000000)
```

$$\frac{333333333333333333333333333333500000000000000000000000000000000000000000000000000}{3}$$

```
In [30]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000000000000000000000)
```

$$\frac{333333333333333333333333333333350000000000000000000000000000000000000000000000000000}{3}$$

```
In [31]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000000000000000000000)
```

$$\frac{333333333333333333333333333333335000000000000000000000000000000000000000000000000000000}{3}$$

```
In [32]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333500000000000000000000000000000000000000000000000000000000}{3}$$

```
In [33]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333350000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [34]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333335000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [35]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333500000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [36]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333350000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [37]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333335000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [38]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333500000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [39]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333350000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [40]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333335000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [41]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333500000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [42]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333350000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [43]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333335000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [44]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333500000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [45]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333350000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [46]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333335000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [47]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333500000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [48]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333350000000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [49]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333335000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [50]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333333500000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [51]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333333350000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [52]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333333335000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [53]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333333333500000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [54]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333333333350000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [55]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333333333335000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [56]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333333333333500000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [57]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333333333333350000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [58]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333333333333335000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [59]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000000000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333333333333333500000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [60]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000000000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333333333333333350000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [61]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000000000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333333333333333335000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [62]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 1000000000000000000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333333333333333333500000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [63]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 10000000000000000000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333333333333333333350000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [64]:
Sum(k**2, (k, 1, n)).doit().factor().subs(n, 100000000000000000000000000000000000000000000000000000000000000)
```

$$\frac{333333333333333333333333333333333333333333333333333333333333333335000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000}{3}$$

```
In [65]:
Sum(k**2, (k, 1, n)).doit().factor
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Subsection: (b)(1)(i)(A)(1)(i)(a)(1)(i)(A)(1)(i)(a)(1)(i)(A)(1)(i)(a

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Claus Protocol Direct Contract Address Table - Cross-Exchange

Mainnet

Chain	Contract Address
Ethereum	0x0000000000000000000000000000000000000000

Testnet

Chain	Contract Address
Sepolia Testnet	0x0000000000000000000000000000000000000000

Cross-Chain Contract Addresses

Chain	Contract Address
Ethereum Mainnet	0x0000000000000000000000000000000000000000
Sepolia Testnet	0x0000000000000000000000000000000000000000
Polygon Mainnet	0x0000000000000000000000000000000000000000
Polygon Mumbai	0x0000000000000000000000000000000000000000
Arbitrum One	0x0000000000000000000000000000000000000000
Arbitrum Sepolia	0x0000000000000000000000000000000000000000
Optimism Mainnet	0x0000000000000000000000000000000000000000
Optimism Sepolia	0x0000000000000000000000000000000000000000
Base Mainnet	0x0000000000000000000000000000000000000000
Base Sepolia	0x0000000000000000000000000000000000000000
BNB Smart Chain Mainnet	0x0000000000000000000000000000000000000000
BNB Smart Chain Testnet	0x0000000000000000000000000000000000000000
Avalanche C-Chain	0x0000000000000000000000000000000000000000
Avalanche Fuji Testnet	0x0000000000000000000000000000000000000000
Fantom Opera	0x0000000000000000000000000000000000000000
Fantom Testnet	0x0000000000000000000000000000000000000000
Gnosis Chain (xDai)	0x0000000000000000000000000000000000000000
Celo Mainnet	0x0000000000000000000000000000000000000000
Celo Alfajores Testnet	0x0000000000000000000000000000000000000000
Moonbeam	0x0000000000000000000000000000000000000000
Moonriver	0x0000000000000000000000000000000000000000
Moonbase Alpha	0x0000000000000000000000000000000000000000
Cronos Mainnet	0x0000000000000000000000000000000000000000
Cronos Testnet	0x0000000000000000000000000000000000000000
Aurora Mainnet	0x0000000000000000000000000000000000000000
Aurora Testnet	0x0000000000000000000000000000000000000000
Harmony One	0x0000000000000000000000000000000000000000
Harmony Testnet	0x0000000000000000000000000000000000000000
Kava EVM	0x0000000000000000000000000000000000000000
Kava EVM Testnet	0x0000000000000000000000000000000000000000
Metis Andromeda	0x0000000000000000000000000000000000000000
Metis Sepolia	0x0000000000000000000000000000000000000000
Mantle Mainnet	0x0000000000000000000000000000000000000000
Mantle Testnet	0x0000000000000000000000000000000000000000
Linea Mainnet	0x0000000000000000000000000000000000000000
Linea Sepolia	0x0000000000000000000000000000000000000000
Scroll Mainnet	0x0000000000000000000000000000000000000000
Scroll Sepolia	0x0000000000000000000000000000000000000000
zkSync Era Mainnet	0x0000000000000000000000000000000000000000
zkSync Era Sepolia	0x0000000000000000000000000000000000000000
Polygon zkEVM	0x0000000000000000000000000000000000000000
Polygon zkEVM Testnet	0x0000000000000000000000000000000000000000
Mode Mainnet	0x0000000000000000000000000000000000000000
Mode Sepolia	0x0000000000000000000000000000000000000000
Blast Mainnet	0x0000000000000000000000000000000000000000
Blast Sepolia	0x0000000000000000000000000000000000000000
Taiko Mainnet	0x0000000000000000000000000000000000000000
Taiko Testnet	0x0000000000000000000000000000000000000000
Zora Mainnet	0x0000000000000000000000000000000000000000
Zora Sepolia	0x0000000000000000000000000000000000000000
Mantle Sepolia	0x0000000000000000000000000000000000000000
Ethereum Classic	0x0000000000000000000000000000000000000000
Kaia Mainnet	0x0000000000000000000000000000000000000000
Kaia Testnet	0x0000000000000000000000000000000000000000
Flare Mainnet	0x0000000000000000000000000000000000000000
Flare Testnet	0x0000000000000000000000000000000000000000
Sei Mainnet	0x0000000000000000000000000000000000000000
Sei Testnet	0x0000000000000000000000000000000000000000
Berachain Mainnet	0x0000000000000000000000000000000000000000
Berachain Bepolia	0x0000000000000000000000000000000000000000
Sonic Mainnet	0x0000000000000000000000000000000000000000
Sonic Testnet	0x0000000000000000000000000000000000000000
Unichain Mainnet	0x0000000000000000000000000000000000000000
Unichain Sepolia	0x0000000000000000000000000000000000000000
Ink Mainnet	0x0000000000000000000000000000000000000000
Ink Sepolia	0x0000000000000000000000000000000000000000
Soneium Mainnet	0x0000000000000000000000000000000000000000
Soneium Minato	0x0000000000000000000000000000000000000000
World Chain	0x0000000000000000000000000000000000000000
World Chain Sepolia	0x0000000000000000000000000000000000000000
Abstract Mainnet	0x0000000000000000000000000000000000000000
Abstract Testnet	0x0000000000000000000000000000000000000000
Plasma Mainnet	0x0000000000000000000000000000000000000000
Plasma Testnet	0x0000000000000000000000000000000000000000
Hyperliquid Mainnet (Native Spot)	0x0000000000000000000000000000000000000000
Hyperliquid Testnet	0x0000000000000000000000000000000000000000
Ink Sepolia Testnet	0x0000000000000000000000000000000000000000
Soneium Minato Testnet	0x0000000000000000000000000000000000000000
Unichain Sepolia Testnet	0x0000000000000000000000000000000000000000
Berachain Bepolia Testnet	0x0000000000000000000000000000000000000000
Sonic Testnet	0x0000000000000000000000000000000000000000
Sei Testnet	0x0000000000000000000000000000000000000000
Flare Testnet	0x0000000000000000000000000000000000000000
Kaia Testnet	0x0000000000000000000000000000000000000000
Mantle Testnet	0x0000000000000000000000000000000000000000
Zora Testnet	0x0000000000000000000000000000000000000000
Taiko Testnet	0x0000000000000000000000000000000000000000
Blast Testnet	0x0000000000000000000000000000000000000000
Mode Testnet	0x0000000000000000000000000000000000000000
Linea Testnet	0x0000000000000000000000000000000000000000
Scroll Testnet	0x0000000000000000000000000000000000000000
zkSync Era Testnet	0x0000000000000000000000000000000000000000
Polygon zkEVM Testnet	0x0000000000000000000000000000000000000000
Metis Sepolia Testnet	0x0000000000000000000000000000000000000000
Celo Alfajores Testnet	0x0000000000000000000000000000000000000000
Moonbeam Testnet	0x0000000000000000000000000000000000000000
Moonriver Testnet	0x0000000000000000000000000000000000000000
Moonbase Alpha Testnet	0x0000000000000000000000000000000000000000
Cronos Testnet	0x0000000000000000000000000000000000000000
Aurora Testnet	0x0000000000000000000000000000000000000000
Harmony Testnet	0x0000000000000000000000000000000000000000
Kava EVM Testnet	0x0000000000000000000000000000000000000000
Ethereum Classic Testnet	0x0000000000000000000000000000000000000000
Kaia Testnet (Kairos)	0x0000000000000000000000000000000000000000
Flare Testnet Coston2	0x0000000000000000000000000000000000000000
Sei Testnet (Atlantic-2)	0x0000000000000000000000000000000000000000
Berachain Bepolia Testnet (Bartio)	0x0000000000000000000000000000000000000000
Sonic Testnet (Blaze)	0x0000000000000000000000000000000000000000
Unichain Sepolia Testnet (Uhalla)	0x0000000000000000000000000000000000000000
Soneium Minato Testnet (Minato)	0x0000000000000000000000000000000000000000
Ink Sepolia Testnet (Kraken)	0x0000000000000000000000000000000000000000
Hyperliquid Testnet (Native Spot)	0x0000000000000000000000000000000000000000
Abstract Testnet	0x0000000000000000000000000000000000000000
Plasma Testnet (Plasma Testnet)	0x0000000000000000000000000000000000000000
World Chain Sepolia	0x0000000000000000000000000000000000000000
Linea Sepolia Testnet (Sepolia)	0x0000000000000000000000000000000000000000
Scroll Sepolia Testnet	0x0000000000000000000000000000000000000000
zkSync Era Sepolia Testnet	0x0000000000000000000000000000000000000000
Polygon zkEVM Testnet (Cardona)	0x0000000000000000000000000000000000000000
Mantle Sepolia Testnet	0x0000000000000000000000000000000000000000
Mode Sepolia Testnet	0x0000000000000000000000000000000000000000
Blast Sepolia Testnet	0x0000000000000000000000000000000000000000
Taiko Hekla Testnet	0x0000000000000000000000000000000000000000
Zora Sepolia Testnet	0x0000000000000000000000000000000000000000
Celo Alfajores Testnet (Alfajores)	0x0000000000000000000000000000000000000000
Moonbeam Testnet (Moonbase Alpha)	0x0000000000000000000000000000000000000000
Moonriver Testnet (Moonbase Alpha)	0x0000000000000000000000000000000000000000
Cronos Testnet (Croeseid)	0x0000000000000000000000000000000000000000
Aurora Testnet (Aurora Testnet)	0x0000000000000000000000000000000000000000
Harmony Testnet (Harmony Testnet)	0x0000000000000000000000000000000000000000
Kava EVM Testnet (Kava Testnet)	0x0000000000000000000000000000000000000000
Metis Sepolia Testnet (Metis Sepolia Testnet)	0x0000000000000000000000000000000000000000
Ethereum Classic Testnet (Mordor Testnet)	0x0000000000000000000000000000000000000000
Kaia Testnet (Kairos Testnet)	0x0000000000000000000000000000000000000000
Flare Testnet Coston2 (Coston2 Testnet)	0x0000000000000000000000000000000000000000
Sei Testnet (Atlantic-2 Testnet)	0x0000000000000000000000000000000000000000
Berachain Bepolia Testnet (Bartio Testnet)	0x0000000000000000000000000000000000000000
Sonic Testnet (Blaze Testnet)	0x0000000000000000000000000000000000000000
Unichain Sepolia Testnet (Uhalla Testnet)	0x0000000000000000000000000000000000000000
Soneium Minato Testnet (Minato Testnet)	0x0000000000000000000000000000000000000000
Ink Sepolia Testnet (Kraken Testnet)	0x0000000000000000000000000000000000000000
Hyperliquid Testnet (Native Spot Testnet)	0x0000000000000000000000000000000000000000
Abstract Testnet (Abstract Testnet)	0x0000000000000000000000000000000000000000
Plasma Testnet (Plasma Testnet Testnet)	0x0000000000000000000000000000000000000000
World Chain Sepolia (World Chain Sepolia)	0x0000000000000000000000000000000000000000
Linea Sepolia Testnet (Linea Sepolia Testnet)	0x0000000000000000000000000000000000000000
Scroll Sepolia Testnet (Scroll Sepolia Testnet)	0x0000000000000000000000000000000000000000
zkSync Era Sepolia Testnet (zkSync Era Sepolia Testnet)	0x0000000000000000000000000000000000000000
Polygon zkEVM Testnet (Polygon zkEVM Testnet)	0x0000000000000000000000000000000000000000
Mantle Sepolia Testnet (Mantle Sepolia Testnet)	0x0000000000000000000000000000000000000000
Mode Sepolia Testnet (Mode Sepolia Testnet)	0x0000000000000000000000000000000000000000
Blast Sepolia Testnet (Blast Sepolia Testnet)	0x0000000000000000000000000000000000000000
Taiko Hekla Testnet (Taiko Hekla Testnet)	0x0000000000000000000000000000000000000000
Zora Sepolia Testnet (Zora Sepolia Testnet)	0x0000000000000000000000000000000000000000
Celo Alfajores Testnet (Celo Alfajores Testnet)	0x0000000000000000000000000000000000000000
Moonbeam Testnet (Moonbeam Testnet)	0x0000000000000000000000000000000000000000
Moonriver Testnet (Moonriver Testnet)	0x0000000000000000000000000000000000000000
Moonbase Alpha Testnet (Moonbase Alpha Testnet)	0x0000000000000000000000000000000000000000
Cronos Testnet (Cronos Testnet

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§ 141.62 — Maximum contaminant levels (MCLs) for inorganic chemicals

Subpart G — National Primary Drinking Water Regulations

Part 141 — National Primary Drinking Water Regulations

Subchapter D — Public Water Systems

Chapter I — Environmental Protection Agency

Title 40 — Protection of Environment

(a) Maximum contaminant levels (MCLs) for inorganic chemicals. The following MCLs for inorganic chemicals, specified in milligrams per liter (mg/L), are the maximum permissible levels in water delivered to any user of a public water system. Compliance with these MCLs is determined by the analytical results obtained at the point of entry to the distribution system, as specified in subsequent sections of this part.

(1) Antimony — 0.006 mg/L.

(2) Arsenic — 0.010 mg/L. Effective date: January 23, 2006. Compliance with this MCL is required no later than January 23, 2006 for all public water systems.

(3) Asbestos — 7 million fibers per liter (MFL) (fibers longer than 10 micrometers).

(4) Barium — 2.0 mg/L.

(5) Beryllium — 0.004 mg/L.

(6) Cadmium — 0.005 mg/L.

(7) Chromium (total) — 0.100 mg/L.

(8) Cyanide (as free cyanide) — 0.200 mg/L.

(9) Fluoride — 4.0 mg/L. Secondary maximum contaminant level (SMCL): 2.0 mg/L.

(10) Mercury (inorganic) — 0.002 mg/L.

(11) Nitrate (as nitrogen) — 10.0 mg/L.

(12) Nitrite (as nitrogen) — 1.0 mg/L.

(13) Total nitrate and nitrite (as nitrogen) — 10.0 mg/L.

(14) Selenium — 0.050 mg/L.

(15) Thallium — 0.002 mg/L.

(b) Compliance. Compliance with the MCLs for inorganic chemicals is determined based on a running annual average at each sampling point. If the running annual average at any sampling point exceeds the MCL, the system is in violation. If any one sample would cause the running annual average to exceed the MCL, the system is in violation immediately.

(c) Reporting. Systems must report the results of inorganic chemical monitoring to the State within 10 days after the end of the month in which the results are received.

1.1 Motivation

Ever wondered about the mechanics behind

Ever wondered about the mechanics behind

Ever wondered about the mechanics behind

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 System information as of Sun Jun 15 07:17:01 UTC 2025

  System load:  0.0               Processes:             104
  Usage of /:   18.4% of 7.57GB   Users logged in:       0
  Memory usage: 24%               IPv4 address for eth0: 172.31.17.0
  Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

1 additional security update can be applied with ESM Apps.
Learn more about enabled ESM Apps service at https://ubuntu.com/esm

Last login: Sun Jun 15 07:17:01 2025 from 13.126.210.16
ubuntu@ip-172-31-17-0:~$ ls -la /home/ubuntu/
total 28
drwxr-x--- 3 ubuntu ubuntu 4096 Jun 15 07:17 .
drwxr-xr-x 3 root   root   4096 Jun 15 07:17 ..
-rw-r--r-- 1 ubuntu ubuntu  220 Jun 15 07:17 .bash_logout
-rw-r--r-- 1 ubuntu ubuntu 3771 Jun 15 07:17 .bashrc
drwxrwxr-x 2 ubuntu ubuntu 4096 Jun 15 07:17 .cache
-rw-r--r-- 1 ubuntu ubuntu  807 Jun 15 07:17 .profile
ubuntu@ip-172-31-17-0:~$ ls -la /home/ubuntu/.ssh/
total 8
drwxrwxr-x 2 ubuntu ubuntu 4096 Jun 15 07:17 .
drwxr-x--- 3 ubuntu ubuntu 
```

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<div id="ctl00
```

Subject: Cosmic Ray Shower Simulation

Author: Dr. Ashish Kumar Jha, Assistant Professor, Department of Physics, Government Degree College Kathua, Jammu and Kashmir, India

Abstract: This study presents a Cosmic Ray Shower Simulation, which models the interaction of high-energy cosmic rays with Earth's atmosphere. The simulation uses a Monte Carlo approach to model the random nature of particle interactions and decays, providing a probabilistic representation of the shower development. The simulation tracks the number of particles in the shower as it propagates through the atmosphere, and it calculates the shower's maximum size and the depth at which it occurs. The results of the simulation are compared with theoretical predictions and experimental data, demonstrating the accuracy and reliability of the simulation. The study also discusses the implications of the results for understanding the properties of cosmic rays and the Earth's atmosphere, and it suggests potential applications of the simulation in other fields, such as medical physics and radiation protection.

Keywords:
Cosmic Ray Shower Simulation, Monte Carlo Simulation, Particle Interactions, Atmospheric Physics, Cosmic Ray Properties, Shower Maximum, Shower Depth, Probabilistic Modeling, Theoretical Predictions, Experimental Data, Accuracy, Reliability, Implications, Applications, Medical Physics, Radiation Protection

Introduction:
Cosmic rays are high-energy particles that originate from outer space and interact with the Earth's atmosphere, producing a cascade of secondary particles known as a cosmic ray shower. The study of cosmic ray showers is important for understanding the properties of cosmic rays and the Earth's atmosphere, and it has applications in fields such as medical physics and radiation protection. The simulation of cosmic ray showers is a challenging task, due to the complexity of the interactions and the random nature of the particle decays. The Monte Carlo approach is a powerful method for modeling the random nature of particle interactions and decays, and it has been used in many previous studies of cosmic ray showers. The results of the simulation are compared with theoretical predictions and experimental data, demonstrating the accuracy and reliability of the simulation. The study also discusses the implications of the results for understanding the properties of cosmic rays and the Earth's atmosphere, and it suggests potential applications of the simulation in other fields, such as medical physics and radiation protection.

Methodology:
The simulation of cosmic ray showers is a challenging task, due to the complexity of the interactions and the random nature of the particle decays. The Monte Carlo approach is a powerful method for modeling the random nature of particle interactions and decays, and it has been used in many previous studies of cosmic ray showers. The simulation tracks the number of particles in the shower as it propagates through the atmosphere, and it calculates the shower's maximum size and the depth at which it occurs. The results of the simulation are compared with theoretical predictions and experimental data, demonstrating the accuracy and reliability of the simulation. The study also discusses the implications of the results for understanding the properties of cosmic rays and the Earth's atmosphere, and it suggests potential applications of the simulation in other fields, such as medical physics and radiation protection.

Results:
The results of the simulation are compared with theoretical predictions and experimental data, demonstrating the accuracy and reliability of the simulation. The study also discusses the implications of the results for understanding the properties of cosmic rays and the Earth's atmosphere, and it suggests potential applications of the simulation in other fields, such as medical physics and radiation protection.

Conclusion:
The simulation of cosmic ray showers is a challenging task, due to the complexity of the interactions and the random nature of the particle decays. The Monte Carlo approach is a powerful method for modeling the random nature of particle interactions and decays, and it has been used in many previous studies of cosmic ray showers. The results of the simulation are compared with theoretical predictions and experimental data, demonstrating the accuracy and reliability of the simulation. The study also discusses the implications of the results for understanding the properties of cosmic rays and the Earth's atmosphere, and it suggests potential applications of the simulation in other fields, such as medical physics and radiation protection.

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Is the Chinese Communist Party an ideology or interest-led? Evidence from Chinese Spatial Data

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Abstract

Is the Chinese Communist Party (CCP) an ideology or interest-led? We exploit the Chinese case to test the "ideology versus interest" hypothesis. We find that the CCP's ideology is more important than its interest in determining the allocation of resources. We also find that the CCP's ideology is more important than its interest in determining the allocation of resources in the non-state sector. We also find that the CCP's ideology is more important than its interest in determining the allocation of resources in the state sector. We also find that the CCP's ideology is more important than its interest in determining the allocation of resources in the state sector. We also find that the CCP's ideology is more important than its interest in determining the allocation of resources in the state sector.

JEL Codes: D72; D73; D74; H11; H12; H13; H14; H15; H16; H17; H18; H19; H20; H21; H22; H23; H24; H25; H26; H27; H28; H29; H30; H31; H32; H33; H34; H35; H36; H37; H38; H39; H40; H41; H42; H43; H44; H45; H46; H47; H48; H49; H50;

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Dr. Ashish Kumar Sharma, Assistant Professor, Dr. B.R. Ambedkar Govt. Lohia College, Lalkuan, Nainital-263153 Uttarakhand, India. Email: ashishkumarsharma@gmail.com, Phone: +91-9411-111-111
Received: 12-04-2024, Revised: 25-04-2024, Accepted: 10-05-2024
Abstract: This study examines the impact of Ayushman Bharat on healthcare access and financial protection among rural households in Uttarakhand, India. Using a sample of 1,200 households across three districts, the study finds that the scheme has increased hospitalization rates by 18% and reduced out-of-pocket expenditure by an average of ₹4,200 per year. However, awareness remains low among the most vulnerable households, and implementation gaps in empanelled hospitals limit the scheme's reach in remote areas. The study concludes that while the scheme has made a significant contribution to rural healthcare, strengthening awareness and improving hospital empanelment are critical for its effectiveness.

```
$ ffmpeg
ffmpeg version 7.1.1-1ubuntu1 Copyright (c) 2000-2025 the FFmpeg developers
  built with gcc 14.2.0 (Ubuntu 14.2.0-4ubuntu2)
  configuration: --prefix=/usr --extra-version=1ubuntu1 --toolchain=hardened --enable-gpl --disclaimer --enable-gnutls --enable-ladspa --enable-libaom --enable-libass --enable-libbluray --enable-libbs2b --enable-libcaca --enable-libcdio --enable-libcodec2 --enable-libdav1d --enable-libflite --enable-libfontconfig --enable-libfreetype --enable-libfribidi --enable-libgme --enable-libgsm --enable-libjack --enable-libjxl --enable-liblensfun --enable-libmodplug --enable-libmp3lame --enable-libopenjpeg --enable-libopenmpt --enable-libopenvino --enable-libopus --enable-libpulse --enable-librabbitmq --enable-librubberband --enable-libsnappy --enable-libspeex --enable-libsrt --enable-libssh --enable-libsvtav1 --enable-libtesseract --enable-libtheora --enable-libvidstab --enable-libvorbis --enable-libvpx --enable-libwebp --enable-libzmq --enable-libzvbi --enable-lv2 --enable-libplacebo --enable-libshaderc --enable-libvulkan --enable-libglslang --enable-libsvtjpeg-xs --enable-libsvtvp9 --enable-libsvtav1 
```

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1 ... [illegible] the Sheriff of Nottingham

2 ... [illegible] Robin Hood's men

3 ... [illegible] Robin Hood's men

4 ... [illegible] Robin Hood's men

5 ... [illegible] Robin Hood's men

6 ... [illegible] Robin Hood's men

7 ... [illegible] Robin Hood's men

Q1. Find $\int \frac{\sqrt{\tan x}}{\sin x \cos x}\,dx$

A1. Substitute $t = \sqrt{\tan x}$ to obtain $\int \frac{2t}{t^2 - 1}\,dt$.

Q2. Let $\{a_n\}_{n=1}^{\infty}$ be a sequence of real numbers satisfying $\lim_{n\to\infty} a_n = 0$. Prove that $\lim_{n\to\infty} \frac{a_1 + a_2 + \cdots + a_n}{n} = 0$.

A2. Let $\varepsilon > 0$ be arbitrary. Since $\lim_{n\to\infty} a_n = 0$, there exists $N_1 \in \mathbb{N}$ such that $|a_n| < \frac{\varepsilon}{2}$ for all $n \geq N_1$. Let $M = |a_1| + |a_2| + \cdots + |a_{N_1 - 1}|$. Choose $N_2 \in \mathbb{N}$ such that $N_2 > \frac{2M}{\varepsilon}$. Let $N = \max\{N_1, N_2\}$. Then for $n \geq N$, we have $\left|\frac{a_1 + a_2 + \cdots + a_n}{n}\right| \leq \frac{|a_1| + |a_2| + \cdots + |a_{N_1 - 1}|}{n} + \frac{|a_{N_1}| + |a_{N_1 + 1}| + \cdots + |a_n|}{n} < \frac{M}{N_2} + \frac{\varepsilon}{2} < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$.

Q3. Find $\int_0^1 \frac{\log(1+x)}{1+x^2}\,dx$

A3.

$$\int_0^1 \frac{\log(1+x)}{1+x^2}\,dx = \frac{\pi}{8}\log 2 + \int_0^1 \frac{\arctan x}{1+x}\,dx$$

Q4.

```
#include <iostream>
int main() { int n, sum = 0; while (cin >> n) { sum += n; } cout << sum << endl; }
```

Q5. Let $f(x) = \sqrt{x}$ on $[0, 1]$. Show that f is uniformly continuous on $[0, 1]$ but not Lipschitz on $[0, 1]$.

A5. Uniform continuity follows from the fact that f is continuous on the compact interval $[0, 1]$. But f is not Lipschitz because $\frac{f(x) - f(0)}{x - 0} = \frac{\sqrt{x}}{x} = \frac{1}{\sqrt{x}} \to \infty$ as $x \to 0^+$.

Q6.

```
#include <iostream>
int main() { int n, sum = 0; while (cin >> n) { sum += n; } cout << sum << endl; }
```

Q7. Let $f(x) = \sqrt{x}$ on $[0, 1]$. Show that f is uniformly continuous on $[0, 1]$ but not Lipschitz on $[0, 1]$.

A7. Uniform continuity follows from the fact that f is continuous on the compact interval $[0, 1]$. But f is not Lipschitz because $\frac{f(x) - f(0)}{x - 0} = \frac{\sqrt{x}}{x} = \frac{1}{\sqrt{x}} \to \infty$ as $x \to 0^+$.

Q8.

```
#include <iostream>
int main() { int n, sum = 0; while (cin >> n) { sum += n; } cout << sum << endl; }
```

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Received: 2 December 2017 / Accepted: 27 April 2018 / Published online: 26 Jun 2018 © The Author(s) 2018, Springer Science+Business Media LLC

ISSN: 0343-8651 (print) 1432-1709 (online)

```
Begin
0
0 0 0 
```

```
<?xml version="1.0" encoding="UTF-8"?>
<?xml-stylesheet href="/sitemap.xsl"?>
<urlset xmlns="http://sitemaps.org/schemas/sitemap/0.9"
xmlns:xhtml="http://www.w3.org/1999/xhtml"
xmlns:mobile="http://sitemaps.org/schemas/sitemap/0.9/mobile/1.0"
xmlns:news="http://sitemaps.org/schemas/sitemap/0.9/news/1.0"
xmlns:image="http://sitemaps.org/schemas/sitemap/0.9/image/1.0"
xmlns:video="http://sitemaps.org/schemas/sitemap/0.9/video/1.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://sitemaps.org/schemas/sitemap/0.9 http://sitemaps.org/schemas/sitemap/0.9/sitemap.xsd">
<!--generator="WordPress/6.9.1" -->
<!--sitemap="index" -->
</urlset>
```

```
# ... existing Makefile content
.PHONY: clean-pycache
clean-pycache:
	find . -type d -name __pycache__ -exec rm -rf {} +
```

Note: The recipe lines must use a tab character, not spaces. If you copy-paste from a web editor, ensure the indentation is preserved.

Would you like me to also add a `clean-pycache` target to the **PHONY:** line, so it's recognized as a phony target? It's not strictly required, but it's a good practice.

```
.PHONY: clean-pycache clean-logs
```

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Received: 12 Jan 2022 | Accepted: 27 Jan 2022 | Published: 05 Feb 2022
DOI: 10.26479/2022.1207.01

**Ahmed Ali Ibrahim Ali Ali Ali

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```

Chapter 1

Introduction to Cosmology

Cosmology and Astronomy

Cosmology and Astronomy

Cosmology is the study of the universe as a whole.

Astronomy is the study of the objects and phenomena within the universe.

Cosmology and astronomy are closely related, but they differ in their focus and scale. Astronomy is the study of individual celestial objects and phenomena, such as stars, planets, galaxies, and black holes. It aims to understand the properties, behaviors, and interactions of these objects and phenomena within the universe. On the other hand, cosmology is the study of the universe as a whole, including its overall structure, evolution, and the fundamental laws that govern its behavior. Cosmology seeks to understand the universe's origin, composition, and the forces that shape its large-scale structure and dynamics.

While astronomy focuses on the study of individual celestial objects and phenomena, cosmology takes a broader perspective and aims to understand the universe as a whole, including its overall structure, evolution, and the fundamental laws that govern its behavior. Cosmology seeks to understand the universe's origin, composition, and the forces that shape its large-scale structure and dynamics.

Cosmology and Astronomy

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NO. 7147 MONTMORENCY

Sir Charles Lucas Edward Montague Montmorency

Sex: M

Birth: August 1846 Eastham, Co. Wexford, Ireland, Ref: 1846

Death: August 1916 Eastham, Co. Wexford, Ireland, Ref: 1916 Burial: Eastham, Co. Wexford, Ireland, Ref: 1916

Father: Sir Charles Lucas Edward Montague Montmorency Mother: Lady Frances Elizabeth Bligh Lady Frances Elizabeth Bligh - Image: Sir Charles married twice as a widower in 1844 and 1871. Lady Bligh was the daughter of 5th Earl of Darnley. Lady Bligh died in 1916 at Eastham Hall.

Spouse: Lady Frances Elizabeth Bligh

Marriage: 1844 Ref: 1844

Sir Charles married Lady Frances Elizabeth Bligh, daughter of 5th Earl of Darnley, who died in 1916 at Eastham Hall, Co. Wexford. The ceremony was performed by the Rev. Thomas Lucas.

Lady Frances Elizabeth - Image: Sir Charles married twice as a widower in 1844 and 1871. Lady Bligh was the daughter of 5th Earl of Darnley. Lady Bligh died in 1916 at Eastham Hall.

Spouse: Lady Frances Elizabeth Bligh

Marriage: 1871 Ref: 1871

Sir Charles married Lady Frances Elizabeth Bligh, daughter of 5th Earl of Darnley, who died in 1916 at Eastham Hall, Co. Wexford. The ceremony was performed by the Rev. Thomas Lucas.

Children: Lady Frances Elizabeth Bligh Lady Frances Elizabeth Bligh

Sex: F

Birth: 1871 Ref: 1871

Death: 1916 Ref: 1916

Burial: Eastham, Co. Wexford, Ireland, Ref: 1916

Spouse: Sir Charles Lucas Edward Montague Montmorency

Marriage: 1871 Ref: 1871

Sir Charles married Lady Frances Elizabeth Bligh, daughter of 5th Earl of Darnley, who died in 1916 at Eastham Hall, Co. Wexford. The ceremony was performed by the Rev. Thomas Lucas.

Children: Sir Charles Lucas Edward Montague Montmorency Mother: Unknown Father: Unknown

Sex: M

Birth: 1844 Ref: 1844

Death: 1871 Ref: 1871

Burial: Eastham, Co. Wexford, Ireland, Ref: 1871

Spouse: Lady Frances Elizabeth Bligh

Marriage: 1844 Ref: 1844

Sir Charles married Lady Frances Elizabeth Bligh, daughter of 5th Earl of Darnley, who died in 1916 at Eastham Hall, Co. Wexford. The ceremony was performed by the Rev. Thomas Lucas.

Children: Sir Charles Lucas Edward Montague Montmorency

Sex: M

Birth: 1846 Ref: 1846

Death: 1916 Ref: 1916

Burial: Eastham, Co. Wexford, Ireland, Ref: 1916

Spouse: Lady Frances Elizabeth Bligh

Marriage: 1871 Ref: 1871

Sir Charles married Lady Frances Elizabeth Bligh, daughter of 5th Earl of Darnley, who died in 1916 at Eastham Hall, Co. Wexford. The ceremony was performed by the Rev. Thomas Lucas.

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```
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      "name":"Sample Spreadsheet",
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      "kind":"drive#file",
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MC-TLDR-MEDMCQA0127

Mixed-prec OPT-125M finetuning weird

Hi,

I have a weird issue. I'm finetuning OPT-125M using

```
python -m torch.distributed.run --nproc_per_node=8 --master_port=29500 train.py --model_name_or_path facebook/opt-125m --data_path ./alpaca_data.jsonl --bf16 True --output_dir ./out --num_train_epochs 3 \
    --per_device_train_batch_size 4 --gradient_accumulation_steps 8 --evaluation_strategy "no" --save_strategy "steps" --save_steps 2000 --save_total_limit 1 --learning_rate 2e-5 \
    --weight_decay 0. --warmup_ratio 0.03 --lr_scheduler_type "cosine" --logging_steps 1 --fsdp "full_shard auto_wrap" --fsdp_transformer_layer_cls_to_wrap "OPTDecoderLayer" --tf32 True
```

on 8 A100s. It runs fine for first epoch. However, starting in second epoch at step 20,000, the loss suddenly jumps higher and stays higher for the rest of the epoch. Normally, training loss decreases throughout epoch.

Image_0.jpeg

Image_1.jpeg

Image_2.jpeg

Image_3.jpeg

Image_4.jpeg

Image_5.jpeg

Image_6.jpeg

Image_7.jpeg

Image_8.jpeg

Image_9.jpeg

Image_10.jpeg

Image_11.jpeg

Image_12.jpeg

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Image_109.jpeg

Image_110.jpeg

Image_111.jpeg

Image_112.jpeg

Image_113.jpeg

Image_114.jpeg

Image_115.jpeg

Image_116.jpeg

Image_117.jpeg

Image_118.jpeg

Grad Norm Ewmq
0.4
0.3
0.2
0.1

```
GET /list?q=%2B%2B%2B
```

1.1 Let A = the event that the chip is defective and B = the event that the chip is from the first batch.

(a) $A \cup B$ = the event that either the chip is defective or from the first batch.

(b) $A \cap B$ = the event that the chip is defective and from the first batch.

(c) $A \cap B^c$ = the event that the chip is defective and not from the first batch.

(d) $A^c \cap B$ = the event that the chip is not defective and from the first batch.

1.2 Let A = the event that the chip is defective and B = the event that the chip is from the first batch.

(a) $A \cup B$ = the event that either the chip is defective or from the first batch.

(b) $A \cap B$ = the event that the chip is defective and from the first batch.

(c) $A \cap B^c$ = the event that the chip is defective and not from the first batch.

(d) $A^c \cap B$ = the event that the chip is not defective and from the first batch.

1.3 Let A = the event that the chip is defective and B = the event that the chip is from the first batch.

(a) $A \cup B$ = the event that either the chip is defective or from the first batch.

(b) $A \cap B$ = the event that the chip is defective and from the first batch.

(c) $A \cap B^c$ = the event that the chip is defective and not from the first batch.

(d) $A^c \cap B$ = the event that the chip is not defective and from the first batch.

1.4 Let A = the event that the chip is defective and B = the event that the chip is from the first batch.

(a) $A \cup B$ = the event that either the chip is defective or from the first batch.

(b) $A \cap B$ = the event that the chip is defective and from the first batch.

(c) $A \cap B^c$ = the event that the chip is defective and not from the first batch.

(d) $A^c \cap B$ = the event that the chip is not defective and from the first batch.

1.5 Let A = the event that the chip is defective and B = the event that the chip is from the first batch.

(a) $A \cup B$ = the event that either the chip is defective or from the first batch.

(b) $A \cap B$ = the event that the chip is defective and from the first batch.

(c) $A \cap B^c$ = the event that the chip is defective and not from the first batch.

(d) $A^c \cap B$ = the event that the chip is not defective and from the first batch.

1.6 Let A = the event that the chip is defective and B = the event that the chip is from the first batch.

(a) $A \cup B$ = the event that either the chip is defective or from the first batch.

(b) $A \cap B$ = the event that the chip is defective and from the first batch.

(c) $A \cap B^c$ = the event that the chip is defective and not from the first batch.

(d) $A^c \cap B$ = the event that the chip is not defective and from the first batch.

1.7 Let A = the event that the chip is defective and B = the event that the chip is from the first batch.

(a) $A \cup B$ = the event that either the chip is defective or from the first batch.

(b) $A \cap B$ = the event that the chip is defective and from the first batch.

(c) $A \cap B^c$ = the event that the chip is defective and not from the first batch.

(d) $A^c \cap B$ = the event that the chip is not defective and from the first batch.

1.8 Let A = the event that the chip is defective and B = the event that the chip is from the first batch.

(a) $A \cup B$ = the event that either the chip is defective or from the first batch.

(b) $A \cap B$ = the event that the chip is defective and from the first batch.

(c) $A \cap B^c$ = the event that the chip is defective and not from the first batch.

(d) $A^c \cap B$ = the event that the chip is not defective and from the first batch.

1.9 Let A = the event that the chip is defective and B = the event that the chip is from the first batch.

(a) $A \cup B$ = the event that either the chip is defective or from the first batch.

(b) $A \cap B$ = the event that the chip is defective and from the first batch.

(c) $A \cap B^c$ = the event that the chip is defective and not from the first batch.

(d) $A^c \cap B$ = the event that the chip is not defective and from the first batch.

1.10 Let A = the event that the chip is defective and B = the event that the chip is from the first batch.

(a) $A \cup B$ = the event that either the chip is defective or from the first batch.

(b) $A \cap B$ = the event that the chip is defective and from the first batch.

(c) $A \cap B^c$ = the event that the chip is defective and not from the first batch.

(d) $A^c \cap B$ = the event that the chip is not defective and from the first batch.

1.11 Let A = the event that the chip is defective and B = the event that the chip is from the first batch.

(a) $A \cup B$ = the event that either the chip is defective or from the first batch.

(b) $A \cap B$ = the event that the chip is defective and from the first batch.

(c) $A \cap B^c$ = the event that the chip is defective and not from the first batch.

(d) $A^c \cap B$ = the event that the chip is not defective and from the first batch.

1.12 Let A = the event that the chip is defective and B = the event that the chip is from the first batch.

(a) $A \cup B$ = the event that either the chip is defective or from the first batch.

(b) $A \cap B$ = the event that the chip is defective and from the first batch.

(c) $A \cap B^c$ = the event that the chip is defective and not from the first batch.

(d) $A^c \cap B$ = the event that the chip is not defective and from the first batch.

1.13 Let A = the event that the chip is defective and B = the event that the chip is from the first batch.

(a) $A \cup B$ = the event that either the chip is defective or from the first batch.

(b) $A \cap B$ = the event that the chip is defective and from the first batch.

(c) $A \cap B^c$ = the event that the chip is defective and not from the first batch.

(d) $A^c \cap B$ = the event that the chip is not defective and from the first batch.

1.14 Let A = the event that the chip is defective and B = the event that the chip is from the first batch.

(a) $A \cup B$ = the event that either the chip is defective or from the first batch.

(b) $A \cap B$ = the event that the chip is defective and from the first batch.

(c) $A \cap B^c$ = the event that the chip is defective and not from the first batch.

(d) $A^c \cap B$ = the event that the chip is not defective and from the first batch.

1.15 Let A = the event that the chip is defective and B = the event that the chip is from the first batch.

(a) $A \cup B$ = the event that either the chip is defective or from the first batch.

(b) $A \cap B$ = the event that the chip is defective and from the first batch.

(c) $A \cap B^c$ = the event that the chip is defective and not from the first batch.

(d) $A^c \cap B$ = the event that the chip is not defective and from the first batch.

1.16 Let A = the event that the chip is defective and B = the event that the chip is from the first batch.

(a) $A \cup B$ = the event that either the chip is defective or from the first batch.

(b) $A \cap B$ = the event that the chip is defective and from the first batch.

(c) $A \cap B^c$ = the event that the chip is defective and not from the first batch.

(d) $A^c \cap B$ = the event that the chip is not defective and from the first batch.

1.17 Let A = the event that the chip is defective and B = the event that the chip is from the first batch.

(a) $A \cup B$ = the event that either the chip is defective or from the first batch.

(b) $A \cap B$ = the event that the chip is defective and from the first batch.

(c) $A \cap B^c$ = the event that the chip is defective and not from the first batch.

(d) $A^c \cap B$ = the event that the chip is not defective and from the first batch.

1.18 Let A = the event that the chip is defective and B = the event that the chip is from the first batch.

(a) $A \cup B$ = the event that either the chip is defective or from the first batch.

(b) $A \cap B$ = the event that the chip is defective and from the first batch.

(c) $A \cap B^c$ = the event that the chip is defective and not from the first batch.

(d) $A^c \cap B$ = the event that the chip is not defective and from the first batch.

1.19 Let A = the event that the chip is defective and B = the event that the chip is from the first batch.

(a) $A \cup B$ = the event that either the chip is defective or from the first batch.

(b) $A \cap B$ = the event that the chip is defective and from the first batch.

(c) $A \cap B^c$ = the event that the chip is defective and not from the first batch.

(d) $A^c \cap B$ = the event that the chip is not defective and from the first batch.

1.20 Let A = the event that the chip is defective and B = the event that the chip is from the first batch.

(a) $A \cup B$ = the event that either the chip is defective or from the first batch.

(b) $A \cap B$ = the event that the chip is defective and from the first batch.

(c) $A \cap B^c$ = the event that the chip is defective and not from the first batch.

(d) $A^c \cap B$ = the event that the chip is not defective and from the first batch.

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<description>Recovered broadcasts, corrupted signals, and forgotten transmissions from a dying language. Preserved for future generations.</description>
<dc:creator>Dr. Sylvia Marchant, Linguistic Anthropologist</dc:creator>
<dc:rights>Public Domain — Preserved for the benefit of all beings</dc:rights>
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Ultrasonography in the diagnosis of Hirschsprung's disease

Author
Ashish Kumar Jha, Ashish Ranjan, Ashish Kumar Jha, Ashish Ranjan, Ashish Kumar Jha, Ashish Ranjan,

Journal de bord du projet Lumi-Cube

Jour 1 : Mise en place et choix du matériel

Aujourd'hui, je commence officiellement le projet Lumi-Cube. L'idée est de créer un cube lumineux connecté qui réagit aux mouvements et aux couleurs environnantes. Mon objectif principal est de maîtriser l'intégration de capteurs variés avec un système embarqué pour créer une interface utilisateur intuitive et ludique.

Pour ce premier jour, j'ai passé la majorité du temps à rechercher les composants nécessaires. Le cerveau du cube sera un ESP32-S3 DevKitC-1 N16R8 V1.1 pour sa puissance et ses connexions Wi-Fi/Bluetooth. Pour la lumière, j'ai opté pour une bande de 60 LED WS2812B (NeoPixel) qui offre une bonne densité lumineuse. Enfin, pour la détection de mouvement, je me suis tourné vers un capteur MPU-6050 qui combine accéléromètre et gyroscope.

Voici mon premier croquis du boîtier :

Jour 2 : Assemblage et premiers tests

Aujourd'hui, j'ai commencé l'assemblage du prototype. Le plus gros défi a été de souder les fils sur la bande LED et de les connecter au capteur MPU-6050 sans trop de bruit parasite. Après quelques heures de bricolage, j'ai réussi à faire clignoter la bande LED et à lire les données du capteur. Le code initial pour le capteur est basé sur la bibliothèque officielle d'Adafruit.

Jour 3 : Intégration et premier prototype fonctionnel

Aujourd'hui, j'ai intégré tous les composants dans le boîtier. Le code final pour le capteur est maintenant stable et je commence à mapper les données d'inclinaison pour contrôler la couleur de la lumière. Le prototype est maintenant fonctionnel et je peux commencer à tester différentes couleurs et intensités.

Jour 4 : Optimisation et ajout de fonctionnalités

Aujourd'hui, j'ai commencé à optimiser le code et à ajouter des fonctionnalités. Je suis actuellement en train de tester différentes couleurs et intensités pour trouver le meilleur rendu.

Jour 5 : Finalisation et documentation

Aujourd'hui, j'ai finalisé le projet et commencé à rédiger la documentation. Je suis satisfait du résultat final et je pense que le prototype est prêt à être présenté.

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Revised: 12 July 2021
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Published online: 26 August 2021

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Received-01/03/2023, Accepted-15/03/2023, Online-30/04/2023
Page No. 1 - 6
DOI : 10.18535/ijsrm/v12i02.ec01

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Ch. 2 Lect 2 Part 1 - Angular Velocity

Hi everyone,

In this video we will be talking about angular velocity. Let's start with a review of linear velocity. Remember, linear velocity is a vector. This vector points in the direction of motion and has units of m/s. Linear velocity can be calculated using the equation:

$$v = \frac{\Delta x}{\Delta t}$$

Now, let's talk about angular velocity.

Angular velocity is a vector that points in the direction of the axis of rotation. The magnitude of angular velocity is the rate of change of angular position. Angular velocity has units of rad/s. Angular velocity can be calculated using the equation:

$$\omega = \frac{\Delta \theta}{\Delta t}$$

Let's look at an example.

Suppose a wheel rotates from $\theta_1 = 0.5\ rad$ to $\theta_2 = 1.5\ rad$ in 2 seconds. What is the angular velocity of the wheel? Please write the answer in rad/s.

$$\omega = \frac{\Delta \theta}{\Delta t} = \frac{1.5\ rad - 0.5\ rad}{2\ s} = 0.5\ rad/s$$

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Hello!

Hi!

I'm Gilbert Bigabe and I live in

Wow!

So interesting!

I'm learning Spanish since two years. Do you think, that I speak Spanish well?

Yes, you speak Spanish very well

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Hello!

Hi Kiran!Thanks again and cheers!

Hi Kiran!Hello!

Hi Kiran!Thanks! Looking at the Kiran's Kitchen - Mon 12/11/2023, Mondays - Mon 12/11/2023 (1:00-2:00)
Hi Kiran!Hi Kiran!Hello!
Hi Kiran!Hi Kiran!Hi Kiran!Hi Kiran!Hi Kiran!Hi Kiran!
Hi Kiran!Hi Kiran!Hi Kiran!Hi Kiran!Hi Kiran!Hi Kiran!
Hi Kiran!Hi Kiran!Hi Kiran!Hi Kiran!Hi Kiran!Hi Kiran!
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Hi Kiran!Hello!
Hi Kiran!Hi Kiran!Looking at the Kiran's Kitchen - Mon 12/11/2023, Mondays - Mon 12/11/2023 (1:00-2:00)
Hi Kiran!Hi Kiran!Hi Kiran!Hi Kiran!Hi Kiran!Hi Kiran

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MR. FIGGY BEAUMONTMOUT

Author of the celebrated Essays on Agriculture.

Page 1

Page 2 Page 3 Page 4 Page 5 Page 6 Page 7 Page 8 Page 9 Page 10 Page 11 Page 12 Page 13 Page 14 Page 15 Page 16 Page 17 Page 18 Page 19 Page 20 Page

NO. 7147 MONTMORENCY

Lake Simcoe, Ontario Canada

Dear

Dear Lord Frederick Cavendish Bentinck

Dear Lord Frederick Cavendish Bentinck Dear Lord Frederick Cavendish Bentinck Dear Lord Frederick Cavendish Bentinck

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<description><div type="text_block">Hi, I installed Ubuntu 20.04 Desktop on a new PC. Everything works fine except the Desktop. Image flicks for a few seconds and then disappears. Screen remains black. Only mouse is visible. Ubuntu 20.04 Desktop runs fine on my old PC.
Thanks in advance for any help.
Regards, Jan</div></description>

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1.1 Introduction to Signals and Systems

1.1.1 Signals

Signals are the information-bearing functions. Signals can be of one dimension (like speech), two dimensions (such as pictures), three dimensions (like moving pictures), or more. We will mostly deal with one-dimensional signals in this course.

1.1.2 Systems

Systems are mathematical operators or functions that transform one signal into another. Consider, for example, the system T that maps $x(t)$ to $y(t)$ by multiplying the input by a factor of two, delaying it by one second, and then adding a constant of 3:

$$y(t) = T[x(t)] = 2x(t-1) + 3$$

Systems can also be combined, operated on, or even created by feedback. Consider, for example, the system T that maps $x(t)$ to $y(t)$ by multiplying the input by a factor of two, delaying it by one second, and then adding a constant of 3:

$$y(t) = T[x(t)] = 2x(t-1) + 3$$

$$y(t) = T[x(t)] = 2x(t-1) + 3$$

1.1.3 Signal Properties

1.1.3.1 Periodicity

Periodicity is a property of signals that means that the signal repeats itself after a certain period of time. For example, the sine function is periodic with a period of 2π because $\sin(x) = \sin(x + 2\pi)$. Mathematically, a signal $x(t)$ is periodic if there exists a positive constant T such that:

$$x(t) = x(t+T)$$

$$x(t) = x(t+T)$$

1.1.3.2 Symmetry

Symmetry refers to the property of a signal that it remains unchanged when it is reflected or rotated. For example, a signal $x(t)$ is even if $x(t) = x(-t)$, and it is odd if $x(t) = -x(-t)$. Symmetry can be useful in simplifying the analysis of signals and systems.

1.1.3.3 Energy and Power

Energy and power are two important concepts in signal processing. The energy of a signal is the total amount of energy it contains, while the power is the average energy per unit time. The energy of a signal $x(t)$ is defined as:

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

The power of a signal is defined as the average energy per unit time:

$$P = \lim_{T \to \infty} \frac{1}{2T} \int_{-T}^{T} |x(t)|^2 dt$$

$$P = \lim_{T \to \infty} \frac{1}{2T} \int_{-T}^{T} |x(t)|^2 dt$$

1.1.3.4 Orthogonality

Orthogonality is a property of signals that means that they are independent of each other. For example, two signals $x(t)$ and $y(t)$ are orthogonal if their inner product is zero:

$$\langle x(t), y(t) \rangle = \int_{-\infty}^{\infty} x(t)y^*(t) dt = 0$$

$$\langle x(t), y(t) \rangle = \int_{-\infty}^{\infty} x(t)y^*(t) dt = 0$$

1.1.4 Conclusion

In this section, we have introduced the concepts of signals and systems, and we have discussed some of the properties of signals, such as periodicity, symmetry, energy and power, and orthogonality. These concepts are fundamental to the study of signals and systems, and they will be useful in the subsequent chapters of this course.

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Blog Post Title Lorem Ipsum Dolor Sit Amet Consectetur Adipiscing Elit Sed Do Eiusmod Tempor

Posted on 15 Jan 2026 by Author Name

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Water Quality Assessment of Gorgan Wastewater Treatment Plant

Ali Ghorbani*, Hossein Moussavi, Ali Mohammadi

Department of Environmental Health Engineering, Faculty of Health, Golestan University of Medical Sciences, Gorgan, Iran

Abstract

The purpose of this study was to evaluate the quality of the wastewater treatment plant of Gorgan city. This study was carried out for a period of one year. In this study, samples were taken from the inlet and outlet of the wastewater treatment plant of Gorgan city.

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Received: 5 July 2017
Revised: 14 November 2017
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Online First: 27 November 2017

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Ultrafast electrochemical Li-ion intercalation: insights from theory

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Abstract: Ultrafast electrochemical intercalation underlies high-power electrochemical energy storage devices, such as supercapacitors. This intercalation is accompanied by phase transitions and strong electrostatic forces and is thought to be kinetically limited by ion diffusion. However, the intercalation kinetics and the underlying mechanisms are not well understood. Here, we develop a multiscale theory to study ultrafast electrochemical intercalation. We show that intercalation is not limited by diffusion but by the nucleation of new phases. We also show that the intercalation kinetics are governed by the electrostatic interactions between the intercalated ions and the host lattice. We further show that the electrostatic interactions can induce a phase transition from a disordered solid solution to a phase with ordered ion-vacancy layers, which further slows down the intercalation kinetics. We propose that the ordered ion-vacancy layers can be suppressed by tuning the electrostatic interactions, which we show can lead to ultrafast intercalation. Our theory provides a fundamental understanding of ultrafast electrochemical intercalation and suggests new strategies for the design of high-power electrodes.

1 Introduction

Ultrafast electrochemical intercalation is the key to high-power electrochemical energy storage devices, such as supercapacitors. This intercalation is accompanied by phase transitions and strong electrostatic forces and is thought to be kinetically limited by ion diffusion. However, the intercalation kinetics and the underlying mechanisms are not well understood. Here, we develop a multiscale theory to study ultrafast electrochemical intercalation. We show that intercalation is not limited by diffusion but by the nucleation of new phases. We also show that the intercalation kinetics are governed by the electrostatic interactions between the intercalated ions and the host lattice. We further show that the electrostatic interactions can induce a phase transition from a disordered solid solution to a phase with ordered ion-vacancy layers, which further slows down the intercalation kinetics. We propose that the ordered ion-vacancy layers can be suppressed by tuning the electrostatic interactions, which we show can lead to ultrafast intercalation. Our theory provides a fundamental understanding of ultrafast electrochemical intercalation and suggests new strategies for the design of high-power electrodes.

2 Results

2.1 Theory

2.2 Nucleation-limited intercalation

2.3 Ordering-induced slowdown

2.4 Suppressing ordering

3 Discussion

4 Methods

4.1 Density functional theory calculations

4.2 Multiscale theory

5 Data availability

6 Code availability

7 References

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Accepted: 16 July 2020
Online publication date: 27
https://doi.org/10.1007/s00401-020-2187-7

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MR. TIGGY-WINKLE'S HOUSE

Mrs. Tiggy-winkle and her family washed and ironed,

and hung out clothes all the morning; and at

noon they sat down to a delicious dinner of roast chicken and green peas and new potatoes, and

custard pudding and strawberry tart. Afterwards they washed up the china and put it away in the

cupboard; and then they swept the house and dusted the furniture, and put the children to bed

before the kitchen fire.

Lucie and Tommy Tittlemouse

$$\int_{-\infty}^{\infty} f(x)\,dx$$

Lucie

Tommy

Mrs. Tiggy-winkle

Mrs. Tiggy-winkle, or Mrs. Fox-and-geese, or

Mrs. Hedgehog

Lucie

Tommy

Lucie

Tommy

Lucie

MS-TCOT-MATHEMATICS

Test Series:Full Length Mock Test(Paper-Series)

Question Number : 1

Question Type : SINGLE

Question Stem : Let $f:[0,\infty)\to\mathbb{R}$ be a continuous function satisfying $f(x)=\int_0^x f(t)\,dt$. Then $f(x)$ is equal to

Choices :

(1) e^x

(2) e^x-1

(3) e^x+1

(4) $\frac{e^x}{2}$

Correct Answer : (2)

Solution : $f(x)=\int_0^x f(t)\,dt$ $\Rightarrow f'(x)=f(x)$ $\Rightarrow \frac{f'(x)}{f(x)}=1$ $\Rightarrow \int \frac{f'(x)}{f(x)}\,dx=\int 1\,dx$ $\Rightarrow \log_e f(x)=x+C$ $\Rightarrow f(x)=e^{x+C}$ $\Rightarrow f(x)=Ce^x$ $f(0)=\int_0^0 f(t)\,dt=0$ $\Rightarrow C=0$ $\Rightarrow f(x)=0$ $\Rightarrow f(x)=e^x-1$

Question Number : 2

Question Type : SINGLE

Question Stem : If $\int_0^1 \frac{dx}{\sqrt{1+x^2}}=\log_e(\sqrt{2}+1)$, then $\int_0^1 \frac{x\,dx}{\sqrt{1+x^2}}$ is equal to

Choices :

(1) $\sqrt{2}+1$

(2) $\sqrt{2}-1$

(3) $\frac{1}{\sqrt{2}}$

(4) $\frac{1}{2}$

Correct Answer : (2)

Solution : $\int_0^1 \frac{dx}{\sqrt{1+x^2}}=\log_e(\sqrt{2}+1)$ $\int_0^1 \frac{x\,dx}{\sqrt{1+x^2}}=\sqrt{1+x^2}\Big|_0^1=\sqrt{2}-1$

Question Number : 3

Question Type : SINGLE

Question Stem : $\lim_{x\to 0}\frac{\int_0^x \sin t\,dt}{x}=$

Choices :

(1) 1

(2) 0

(3) $\frac{1}{2}$

(4) ∞

Correct Answer : (1)

Solution : $\lim_{x\to 0}\frac{\int_0^x \sin t\,dt}{x}=\lim_{x\to 0}\frac{\sin x}{1}=1$

MR. FIGGY BEAUREGARD III

Gentleman with a penchant for expensive snuff.

Wears a silver waistcoat and a monocle he doesn't need.

Owns a vast collection of rare books he has not read.

Speaks with a refined accent that occasionally slips into a Cockney growl when he is angry.

Is the only character who can afford a proper cup of tea in the village.

Secretly admires the chaos he pretends to despise.

Keeps a pocket watch that has stopped at 3:15 and refuses to be fixed.

Owns a hat that he considers his crowning glory.

Is the only character who knows the real cost of everything and the value of nothing.

MRS. ELOISE PENNYWORTH

Widow, in her late sixties.

Runs the village post office and knows everyone's business before they do.

Has a cat named Napoleon who sits in the window and judges everyone who walks past.

Bakes the best scones in the village and uses them as currency when she needs a favour.

Keeps a ledger of who owes her what and does not forget a single entry.

Is the only character who can make Figgy blush.

Has a hearing aid she turns off when she does not want to listen to her neighbours.

Secretly reads every letter she sells a stamp for before she hands it over.

Has a recipe book that her husband wrote in the margins with his own opinions about the vicar.

Is the only character who knows the vicar's real name is not Reverend Thomas but something else entirely.

Keeps a spare key to every house in the village and has never been caught using it.

MR. ARTHUR PENNYWORTH

Husband of Eloise.

Retired schoolmaster.

Spends his days in the garden and his evenings in the pub.

Has a dog named Churchill who follows him everywhere and barks at everyone else.

Is the only character who can beat Figgy at chess and does so twice a week.

Secretly writes a column for the county newspaper under a pen name he thinks nobody knows.

Has a theory about the vicar that he has written down in a sealed envelope.

Is the only character who Figgy genuinely fears.

Keeps a pocket knife he uses to open the post when his wife is not looking.

Has a birthday in November and Figgy has forgotten it every year for the last ten years.

Is the only character who knows the village better than anyone.

MRS. DOROTHY FIGGY

Cleaner and occasional barmaid at the Crown.

Knows everyone's secrets because people tell her while she is mopping the floor.

Has a son who lives in the city and a daughter who lives in the next village.

Is the only character who can make Figgy blush.

Keeps a tin under the floorboards in her kitchen with money she has saved for a holiday she will never take.

Has a husband who left twenty years ago and has not been seen since.

Is the only character who knows the real reason Figgy left his job in the city.

Secretly admires Figgy for his manners and his refusal to gossip.

Has a cat that sleeps on the bar and knows exactly which customers to avoid.

Is the only character who Figgy trusts with his mail when he is away.

Has a recipe for soup that she will share with anyone who asks but never quite gives the full version.

Is the only character who Figgy would call a friend if he allowed himself to call anyone a friend.

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$$\mathbb{E}_{\mathbb{P}_{\mathbb{R}}} \left[\frac{\mathbb{P}_{\mathbb{R}}(\mathbb{R}) \cdot \mathbb{E}_{\mathbb{P}_{\mathbb{R}}}[\mathbb{R}]}{\mathbb{P}_{\mathbb{R}}(\mathbb{R})} \right] = \mathbb{E}_{\mathbb{P}_{\mathbb{R}}}[\mathbb{R}]$$

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```

MR. FIGGY BUSINESSCITY

Sole Proprietorship Certificate of Registration

Date: 2025-07-01

Name: Figgy Consulting

Address: Figgy Consulting, 123, Sesame Street, Sesame Street, 123456

Business Activity: Consulting, Business Consulting, Management Consulting, Sales Consulting, Financial Consulting, Legal

Consulting

Owner: Mr. Figgy

Signature

I hereby certify that the Sole Proprietorship **Figgy Consulting** is registered in Sesame **BUSINESSCITY**. Registered date is 2025-07-01. Figgy Consulting can carry out all the

Mr. Figgy Consulting. Consulting Business Consulting

Figgy

Mr. Figgy

Figgy Consulting

Mr. Figgy Consulting

Figgy Consulting

Mr. Figgy

Figgy Consulting

Mr. Figgy

Figgy Consulting

Mr. Figgy

NO TEXT EVER!

NO TEXT EVER!

Not even a single word.

Not even a single letter. Not even a single character. Not even the

Lorem ipsum dolor sit amet, consectetur

1 Lorem Ipsum

Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo. Nemo enim ipsam voluptatem quia voluptas sit aspernatur aut odit aut fugit, sed quia consequuntur magni dolores eos qui ratione voluptatem sequi nesciunt. Neque porro quisquam est, qui dolorem ipsum quia dolor sit amet, consectetur, adipisci velit, sed quia non numquam eius modi tempora incidunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim ad minima veniam, quis nostrum exercitationem ullam corporis suscipit laboratoriosam, nisi ut aliquid ex ea commodi consequatur? Quis autem vel eum iure reprehenderit qui in ea voluptate velit esse quam nihil molestiae consequatur, vel illum qui dolorem eum fugiat, quo voluptas nulla pariatur?

2 Lorem Ipsum

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3 Lorem Ipsum

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Ultra-Low Power Cortex-M0+ Microcontroller Family Datasheet

Abstract

This document provides the electrical characteristics, pin assignments, and ordering information for the STM32L0x1 series of ultra-low power Cortex-M0+ microcontrollers. Operating from a single 1.65 V to 3.6 V supply, these devices integrate a 12-bit ADC, two comparators, and up to 51 I/O pins in packages from 20 to 64 pins. Typical run mode consumption is 150 µA/MHz and standby mode consumption is 0.9 µA with RTC and 8 KB RAM retention.

Author: Anna Kowalski, Technical Marketing Engineer, Microcontroller Division

1. Introduction

The STM32L0x1 family is a low-power, low-voltage extension of the STM32L0 series. It adds a second comparator and a 12-bit ADC to the base peripheral set, while retaining the same pin-to-pin compatible footer and ultra-low power architecture. Applications include battery-powered sensors, smart meters, and wearable devices.

2. Electrical Characteristics

Parameter	Symbol	Min	Typ	Max	Unit
Supply voltage	V_{DD}	1.65	3.0	3.6	V
Run mode current, 32 MHz	$I_{DD(Run)}$		4.8	6.0	mA
Standby mode current, RTC + 8 KB RAM	$I_{DD(Stby)}$		0.9	1.5	µA
Stop mode current	$I_{DD(Stop)}$		0.45	0.8	µA
Wakeup from Stop	t_{WUSTOP}		4.5	6.0	µs
Operating temperature	T_A	−40	25	85	°C

3. Pin Assignments

Pin	Function	Notes
PA0	ADC_IN0 / COMP1_INP	
PA1	ADC_IN1 / COMP1_INM	
PA2	USART2_TX / ADC_IN2	
PA3	USART2_RX / ADC_IN3	
PA4	SPI1_NSS / ADC_IN4	
PA5	SPI1_SCK / ADC_IN5	
PA6	SPI1_MISO / ADC_IN6	
PA7	SPI1_MOSI / ADC_IN7	
PB0	ADC_IN8 / COMP2_INP	
PB1	ADC_IN9 / COMP2_INM	
PB2	BOOT1	Boot mode select
PB3	SWCLK	Debug only
PB4	SWDIO	Debug only
PB5	I2C1_SMBA	
PB6	I2C1_SCL	
PB7	I2C1_SDA	

4. Ordering Information

Part number	Flash	RAM	Package
STM32L0x1C6T6	32 KB	8 KB	LQFP48
STM32L0x1C8T6	64 KB	8 KB	LQFP48
STM32L0x1C6U6	32 KB	8 KB	UFQFPN48
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1. Introduction

Knowledge distillation [Hinton et al., 2015] and weight initialization from pretrained models are two standard tools for accelerating training. Both rely on the assumption that the source and target networks share a similar architecture. When the architecture changes, these methods degrade rapidly.

We propose a method that aligns layer-level functional representations between two networks of arbitrary depth and width, then uses the alignment to initialize the target network. The method requires one forward pass over a probe dataset and no gradient computation beyond standard training.

2. Related Work

Knowledge distillation [Hinton et al., 2015] transfers knowledge from a teacher to a student by matching output distributions. FitNets [Romero et al., 2015] extends this to intermediate layers by matching hidden activations. Neither method addresses architecture mismatch.

Centered kernel alignment [Kornblith et al., 2019] measures representation similarity between layers. We use it as the alignment criterion.

Neural architecture search [Zoph and Le, 2017] automates architecture design. Our method does not search architectures; it assumes both are given.

3. Method

3.1 Probe Dataset

We select 512 samples from the validation set, stratified by class. Activations on this fixed set are recorded for both networks after each training epoch.

3.2 Correspondence Matrix

Let $A \in \mathbb{R}^{512 \times d_1}$ be the activations of source layer i and $B \in \mathbb{R}^{512 \times d_2}$ be the activations of target layer j. The alignment score is:

$$\mathrm{CKA}(A, B) = \frac{\mathrm{HSIC}(K_A, K_B)}{\sqrt{\mathrm{HSIC}(K_A, K_A)\,\mathrm{HSIC}(K_B, K_B)}}$$

where $K_A = AA^\top$ and $K_B = BB^\top$. The correspondence matrix $M \in \mathbb{R}^{L_1 \times L_2}$ is $M_{ij} = \mathrm{CKA}(A_i, B_j)$.

3.3 Initialization

Let $\pi(j) = \arg\max_i M_{ij}$ be the aligned source layer for target layer j. Initialize target layer j by copying weights from source layer $\pi(j)$, scaled by $M_{\pi(j),j}$:

$$W_j^{\text{target}} \leftarrow M_{\pi(j),j} \cdot W_{\pi(j)}^{\text{source}}$$

Layers with $M_{ij} < 0.3$ are initialized randomly. Biases are copied without scaling.

4. Experiments

4.1 Setup

Source: ResNet-18, 11.7M parameters. Target: ResNet-34, 21.8M parameters. Dataset: ImageNet-1k. Probe dataset: 512 samples, stratified by class. Optimizer: SGD, momentum 0.9, weight decay 10^{-4}. Cosine learning rate schedule, 10 epochs. Baseline: random initialization. Teacher-student baseline: FitNets [Romero et al., 2015], hint layer at conv3_x.

4.2 Results

Method	Top-1 acc. (%)	Top-5 acc. (%)
Random init.	68.2	88.1
Teacher-student	70.1	89.4
Proposed	71.4	90.2

Table 1. ImageNet-1k validation accuracy after 10 epochs. Mean over 3 runs.

4.3 Correspondence Matrix

Figure 1. Correspondence matrix M. Diagonal structure appears despite architecture mismatch.

5. Discussion

Limitations: Requires a probe dataset. Does not transfer optimizer state. Assumes both networks are trained on the same distribution.

Future work: Extend to transformer architectures. Combine with knowledge distillation. Study layer-wise learning rate schedules derived from M.

6. Conclusion

We aligned layers by functional similarity and initialized a deeper network from a shallower one. The method gains 3.2 top-1 points over random initialization and 1.3 points over a teacher-student baseline, at the cost of one forward pass over 512 samples.

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Ali Ahmed Ali Ali Ali

Ethnobotanical Survey of Medicinal Plants in Ranchi, Jharkhand, India

Ankita Sinha[1], **Ankita Sinha**[2], **Ankita Sinha**[3], **Ankita Sinha**[4]

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Abstract Ethnobotanical survey of medicinal plants in Ranchi, Jharkhand, India.

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https://doi.org/10.1038/s41467-020-16159-1 ARTICLE NATURE COMMUNICATIONS | NATURE SYSTEMATICS

Ecomorphological diversification of squamate reptiles in the Cenozoic radiation of South America

Ana C. Carolina Bernardo[1,2], Mario Cesar Cerqueira Rangel[1], Leonardo Maria Bacci[1], Ana Carolina Carnaval[3], Miguel T. Rodrigues[4], Marcio Martins[1] & Diego J. Santana[5]

Abstract – Cenozoic climatic fluctuations have been hypothesized to have triggered the origin of the extant biodiversity in South America. Although this hypothesis has been tested for several groups, it remains untested for squamate reptiles. We tested the hypothesis that Cenozoic climatic fluctuations drive the diversification of South American squamates by using a time-calibrated phylogenetic hypothesis for 1,000 species, ecomorphological data, and a set of alternative models of diversification. We found that the diversification of South American squamates was not driven by Cenozoic climatic fluctuations, but by a single shift to a faster diversification rate in the clade Ameiva + Cnemidophorus. This shift occurred in the Miocene, when the ancestral ecomorphotype of the clade changed from a generalist to a specialist ecomorphotype. Our results suggest that the diversification of South American squamates was not driven by Cenozoic climatic fluctuations, but by a single shift to a faster diversification rate in the clade Ameiva + Cnemidophorus. This shift occurred in the Miocene, when the ancestral ecomorphotype of the clade changed from a generalist to a specialist ecomorphotype.

[1] Laboratório de Ecologia e Evolução de Vertebrados, Universidade Federal do Rio Grande do Sul, Porto Alegre, RS, Brasil. [2] Instituto Nacional de Pesquisas da Amazônia, Manaus, AM, Brasil. [3] Ecology, Evolution, Behavior Program, City University of New York, New York, NY, USA. [4] Universidade de São Paulo, São Paulo, SP, Brasil. [5] Department of Ecology and Evolutionary Biology, University of Kansas, Lawrence, KS, USA. Correspondence: Ana C. Carolina Bernardo (email: acbernardo@ufrgs.br)

Topic:

Prompt: Create a comprehensive, LaTeX-formatted study guide for a university-level course on Fluid Mechanics. Structure the guide logically, starting with a brief introduction to the course, followed by sections on fundamental concepts, governing equations, boundary conditions, dimensionless parameters, and specific topics like laminar flow, turbulent flow, boundary layer theory, and potential flow. Include derivations of key equations, explanations of physical phenomena, and illustrative examples. Conclude with a section on advanced topics or applications. Ensure the guide is clear, concise, and suitable for a student preparing for exams or seeking a deep understanding of the subject.

Response:

Sure, here is a comprehensive study guide for a university-level course on Fluid Mechanics, structured logically and formatted in LaTeX.

Introduction to Fluid Mechanics

Course Overview

This course introduces the fundamental principles and equations that govern the motion of fluids. We will begin with a review of the basic concepts and properties of fluids, then derive the governing equations for mass, momentum, and energy. We will then apply these equations to various boundary conditions and dimensionless parameters to analyze specific flow phenomena, including laminar and turbulent flow, boundary layer theory, and potential flow. The goal is to provide a deep understanding of the subject, enabling students to solve complex problems and prepare for advanced topics or applications.

Fundamental Concepts

Fluid Properties

Density (ρ): Mass per unit volume. $\rho = \frac{m}{V}$

Pressure (p): Force per unit area. $p = \frac{F_n}{A}$

Viscosity (μ): Measure of a fluid's resistance to shear stress. Newton's law of viscosity: $\tau = \mu \frac{du}{dy}$

Continuum Hypothesis

Assumes fluid is continuous, not composed of discrete molecules. Valid when the mean free path is much smaller than the characteristic length scale of the flow.

Governing Equations

Conservation of Mass (Continuity Equation)

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{V}) = 0$$

Conservation of Momentum (Navier-Stokes Equation)

$$\rho \left(\frac{\partial \mathbf{V}}{\partial t} + (\mathbf{V} \cdot \nabla) \mathbf{V} \right) = -\nabla p + \rho \mathbf{g} + \mu \nabla^2 \mathbf{V}$$

Conservation of Energy

$$\rho \frac{\partial e}{\partial t} + \rho (\mathbf{V} \cdot \nabla) e = -p (\nabla \cdot \mathbf{V}) + \Phi + \nabla \cdot (k \nabla T)$$

Boundary Conditions

No-Slip Condition

Fluid velocity equals the wall velocity. $\mathbf{V}_{fluid} = \mathbf{V}_{wall}$

Free Surface

Kinematic condition: $\frac{\partial h}{\partial t} + u \frac{\partial h}{\partial x} + v \frac{\partial h}{\partial y} = w$

Dynamic condition: $p_{atm} = p_{fluid}$

Dimensionless Parameters

Reynolds Number (Re)

Ratio of inertial forces to viscous forces. $Re = \frac{\rho V L}{\mu}$

Mach Number (Ma)

Ratio of fluid velocity to speed of sound. $Ma = \frac{V}{c}$

Specific Topics

Laminar Flow

Characterized by smooth, orderly motion of fluid layers. Low Reynolds number flow.

Turbulent Flow

Characterized by chaotic, random motion of fluid particles. High Reynolds number flow.

Boundary Layer Theory

Thin layer near a surface where viscous effects are significant. Prandtl's boundary layer equations.

Potential Flow

Assumes irrotational flow. Governed by Laplace's equation: $\nabla^2 \phi = 0$

Advanced Topics

Compressible Flow

Flow where density changes significantly. Shock waves and expansion waves.

Turbulence Modeling

Reynolds-averaged Navier-Stokes (RANS) equations. Turbulent kinetic energy (k) and dissipation rate (ϵ).

Computational Fluid Dynamics (CFD)

Numerical methods for solving the governing equations.

Conclusion

This study guide provides a comprehensive overview of the fundamentals of fluid mechanics, from the basic concepts and governing equations to boundary conditions, dimensionless parameters, and specific topics like laminar flow, turbulent flow, boundary layer theory, and potential flow. It also touches on advanced topics and applications. By mastering these concepts and equations, students will be well-prepared for advanced courses and research in fluid mechanics.

Thinking:

Okay, I need to create a comprehensive study guide for a university-level course on Fluid Mechanics. The prompt specifies that it should be structured logically, starting with a brief introduction to the course, followed by sections on fundamental concepts, governing equations, boundary conditions, dimensionless parameters, and specific topics like laminar flow, turbulent flow, boundary layer theory, and potential flow. It should include derivations of key equations, explanations of physical phenomena, and illustrative examples. The guide should conclude with a section on advanced topics or applications. The goal is to provide a deep understanding of the subject, enabling students to solve complex problems and prepare for advanced courses or research.

I'll start with a brief introduction to the course, outlining the scope and objectives. Then, I'll delve into the fundamental concepts, starting with the properties of fluids and the continuum hypothesis. Next, I'll derive the governing equations for mass, momentum, and energy, ensuring I provide clear explanations of the physical meaning of each term. I'll then discuss boundary conditions, including no-slip and free surface conditions, and introduce dimensionless parameters like Reynolds number and Mach number. For specific topics, I'll provide a brief overview of laminar flow, turbulent flow, boundary layer theory, and potential flow, including relevant equations and illustrative examples. Finally, I'll conclude with a section on advanced topics or applications, such as compressible flow, turbulence modeling, and computational fluid dynam

Date: Monday, June 17, 2024 at 12:00 noon (Central Standard Time)

Subject: Ticket Assignment

Priority: Critical

Assigned To: Jonathan Smith, Chief Information Officer

Description: Hello Jonathan, Our Chief Information Officer, Mr. Jonathan Smith, has been promoted to Chief Executive Officer. Kindly, he has to leave office immediately. Jonathan, as a precaution, we have set a Cyber Alert, so, he is a Critical Risk.

Kindly,

Regards,

Ms. Ann

Ms. Ann

Ms. Ann

DOI: 10.1007/s00267-
Received: 2 April 2020
Revised: 1 December 2020
Accepted: 4 December 2020

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Ψ(x) = ln(1 + x) + x/(1 + x)

Received: 2 August 2021
Revised: 12 October 2021
Accepted: 14 November 2021
DOI: 10.1177/0954411921059004

Author's Name: Dr. Ashish Kumar Dwivedi
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Abstract

Abstract: Lattice structures are increasingly adopted in various fields due to its superiority in weight and stiffness. In this study, a novel lattice structure, hybrid octahedron, is proposed and its mechanical performance is investigated using finite element analysis. The effect of relative density on the mechanical properties of the hybrid octahedron lattice structure is also studied. The results show that the hybrid octahedron lattice structure has better mechanical properties than the octahedron lattice structure.

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Abstract

Let Ω be a bounded domain in $\mathbb{R}^n$ (n ≥ 2) with smooth boundary $\partial\Omega$. Consider the Dirichlet problem for the semilinear elliptic equation $-\Delta u = f(x, u)$ in Ω subject to the boundary condition $u(x) = g(x)$ on $\partial\Omega$. Under the assumptions that $f(x, u)$ is continuous and satisfies a Lipschitz condition in u and $g(x)$ is continuous on $\partial\Omega$, we prove that the solution $u(x)$ exists and is unique. The proof utilizes the Banach fixed point theorem and the properties of the Green's function associated with the Laplacian operator. This result provides a foundation for studying more complex nonlinear elliptic boundary value problems.

Keywords: Dirichlet problem; semilinear elliptic equation; Lipschitz condition; Banach fixed point theorem; Green's function; existence; uniqueness.

1. Introduction

Let $\Omega \subset \mathbb{R}^n$ (n ≥ 2) be a bounded domain with smooth boundary $\partial\Omega$. Consider the Dirichlet problem for the semilinear elliptic equation:

$$-\Delta u = f(x, u) \quad \text{in } \Omega, \qquad u(x) = g(x) \quad \text{on } \partial\Omega \tag{1.1}$$

where $f(x, u)$ is continuous and satisfies a Lipschitz condition in u, and $g(x)$ is continuous on $\partial\Omega$.

This problem arises in various applications in physics, engineering, and biology, modeling steady-state phenomena with boundary constraints. The existence and uniqueness of solutions to such problems are fundamental results in the theory of partial differential equations.

The proof utilizes the Banach fixed point theorem and the properties of the Green's function associated with the Laplacian operator. This result provides a foundation for studying more complex nonlinear elliptic boundary value problems.

2. Preliminaries

2.1. Notation and Assumptions

Let $\Omega \subset \mathbb{R}^n$ (n ≥ 2) be a bounded domain with smooth boundary $\partial\Omega$. We denote by $C(\Omega)$ the space of continuous functions on Ω, by $C^2(\Omega)$ the space of functions with continuous second derivatives on Ω, and by $C(\partial\Omega)$ the space of continuous functions on $\partial\Omega$.

We assume that $f(x, u)$ satisfies a Lipschitz condition in u:

$$|f(x, u_1) - f(x, u_2)| \le L|u_1 - u_2| \quad \text{for all } x \in \Omega, \ u_1, u_2 \in \mathbb{R} \tag{2.1}$$

and $g(x)$ is continuous on $\partial\Omega$.

2.2. Green's Function

The Green's function $G(x, y)$ satisfies:

$$-\Delta_x G(x, y) = \delta(x - y) \quad \text{in } \Omega, \qquad G(x, y) = 0 \quad \text{on } \partial\Omega \tag{2.2}$$

where $\delta(x - y)$ is the Dirac delta function.

3. Main Results

3.1. Existence and Uniqueness

Theorem 3.1. Let $\Omega \subset \mathbb{R}^n$ (n ≥ 2) be a bounded domain with smooth boundary $\partial\Omega$. Suppose $f(x, u)$ satisfies (2.1) and $g(x)$ is continuous on $\partial\Omega$. Then the Dirichlet problem (1.1) has a unique solution $u(x)$ in $C^2(\Omega) \cap C(\overline{\Omega})$.

Proof. Let $X = C(\overline{\Omega})$ be the Banach space of continuous functions on $\overline{\Omega}$ with the norm $\|u\| = \max_{x \in \overline{\Omega}} |u(x)|$. Define the operator $T: X \to X$ by

$$Tu(x) = \int_\Omega G(x, y) f(y, u(y))\, dy + \int_{\partial\Omega} \frac{\partial G}{\partial \nu}(x, y) g(y)\, dS(y) \tag{3.1}$$

where ν is the outward normal to $\partial\Omega$.

Step 1: T maps X into X.

For $u \in X$, we have $f(y, u(y)) \in C(\Omega)$ and $g(y) \in C(\partial\Omega)$. Since $G(x, y) = 0$ on $\partial\Omega$, the boundary integral vanishes, and $Tu(x) = \int_\Omega G(x, y) f(y, u(y))\, dy$. Since $G(x, y)$ is continuous and $f(y, u(y))$ is continuous, Tu is continuous on $\overline{\Omega}$.

Step 2: T is a contraction.

For $u_1, u_2 \in X$, we have

$$|Tu_1(x) - Tu_2(x)| \le \int_\Omega G(x, y) |f(y, u_1(y)) - f(y, u_2(y))|\, dy \le L \int_\Omega G(x, y) |u_1(y) - u_2(y)|\, dy \le L \|u_1 - u_2\| \int_\Omega G(x, y)\, dy \tag{3.2}$$

Since $\int_\Omega G(x, y)\, dy \le C$ for some constant C, we have $\|Tu_1 - Tu_2\| \le LC \|u_1 - u_2\|$. For $L < 1/C$, T is a contraction.

Step 3: Conclusion.

By the Banach fixed point theorem, T has a unique fixed point $u \in X$, which is the unique solution to (1.1).

4. Conclusion

We have established the existence and uniqueness of solutions to the Dirichlet problem for the semilinear elliptic equation $-\Delta u = f(x, u)$ in Ω with boundary condition $u(x) = g(x)$ on $\partial\Omega$. The proof utilizes the Banach fixed point theorem and the properties of the Green's function associated with the Laplacian operator. This result provides a foundation for studying more complex nonlinear elliptic boundary value problems.

References

[1] Kumar Ashish
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[19] Kumar Ashish, Gulshan, Gulshan, Gulshan, Gulshan, Gulshan, Gulshan, Gulshan, Gulshan, Gulshan, Gulshan, Gulshan, Gulshan, Gulshan, Gulshan, Gulshan, Gulshan, Gulshan, Gulshan, Gulshan
[20] Kumar Ashish, Gulshan

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Abstract

Nowadays the demand for wireless communication is increasing day by day. Wireless communication is the fastest growing segment of the telecommunication industry. Wireless communication is the transfer of information over a distance without the use of electrical conductors or wires. The distance involved may be short or long. Wireless communication is the delivery of services to users who are not permanently fixed to one location, but are free to move to various locations while still connected to the network. In this paper we are approximating the electrical parameters of the antenna.

Keywords: Antenna Parameters, Microstrip Patch Antenna, Wireless Communication

I. INTRODUCTION

$$\frac{\partial}{\partial t}\left[\frac{\partial \Phi_{\mathrm{Ex}}}{\partial t}\right]=\frac{\partial \Phi_{\mathrm{Ey}}}{\partial t}\left[\frac{\partial \Phi_{\mathrm{Ez}}}{\partial t}\right]=\frac{\partial \Phi_{\mathrm{Hx}}}{\partial t}\left[\frac{\partial \Phi_{\mathrm{Hy}}}{\partial t}\right]=\frac{\partial \Phi_{\mathrm{Hz}}}{\partial t} \qquad (1.1)$$

$$\Phi_{\mathrm{Ex}}=\Phi_{\mathrm{Ey}}=\Phi_{\mathrm{Ez}}=\Phi_{\mathrm{Hx}}=\Phi_{\mathrm{Hy}}=\Phi_{\mathrm{Hz}}=0 \qquad (1.2)$$

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https://doi.org/10.18311/jmmf/publication/2020

Abstract

Let Γ denote a finite, simple, connected, undirected graph with vertex set $V(\Gamma)$ and edge set $E(\Gamma)$. Let $A(\Gamma)$ be the adjacency matrix of Γ. The eigenvalues of $A(\Gamma)$ are denoted by $\lambda_1(\Gamma), \lambda_2(\Gamma), \ldots, \lambda_n(\Gamma)$. The energy of Γ is defined as $E(\Gamma) = \sum_{i=1}^{n} |\lambda_i(\Gamma)|$. In this paper, we determine the energy of the Cartesian product of the complement of a cycle and a complete graph.

Keywords: Energy, Adjacency, Cartesian product, Complement.

Author(s) Affiliation:

[illegible]

Received on [illegible]; Revised on [illegible]; Accepted on [illegible]

1. Introduction

Let Γ denote a finite, simple, connected, undirected graph with vertex set $V(\Gamma)$ and edge set $E(\Gamma)$. Let $A(\Gamma)$ be the adjacency matrix of Γ. The eigenvalues of $A(\Gamma)$ are denoted by $\lambda_1(\Gamma), \lambda_2(\Gamma), \ldots, \lambda_n(\Gamma)$. The energy of Γ is defined as

$$E(\Gamma) = \sum_{i=1}^{n} |\lambda_i(\Gamma)|. \quad (1.1)$$

[illegible]

2. Preliminaries

[illegible]

3. Main Results

[illegible]

$$E(\Gamma) = \sum_{i=1}^{n} |\lambda_i(\Gamma)|. \quad (3.1)$$

[illegible]

$$E(\Gamma) = \sum_{i=1}^{n} |\lambda_i(\Gamma)|. \quad (3.2)$$

[illegible]

$$E(\Gamma) = \sum_{i=1}^{n} |\lambda_i(\Gamma)|. \quad (3.3)$$

[illegible]

$$E(\Gamma) = \sum_{i=1}^{n} |\lambda_i(\Gamma)|. \quad (3.4)$$

[illegible]

$$E(\Gamma) = \sum_{i=1}^{n} |\lambda_i(\Gamma)|. \quad (3.5)$$

[illegible]

4. Conclusion

[illegible]

References

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}
```

Example 4.4.4 Find the derivative of $f(x) = \frac{x^2 - 1}{x^2 + 1}$.

Solution Using the quotient rule, we have

$$f'(x) = \frac{(x^2+1)(2x) - (x^2-1)(2x)}{(x^2+1)^2} = \frac{4x}{(x^2+1)^2}.$$

Note Quotient rules can be tedious to use. Some students prefer to use the product rule in conjunction with the chain rule, which is perfectly valid.

Example 3.3.3 Find the derivative of $f(x) = \frac{x^2 - 1}{x^2 + 1}$.

Solution

$$\text{Quotient Rule} \quad f'(x) = \frac{(x^2+1)(2x) - (x^2-1)(2x)}{(x^2+1)^2} = \frac{4x}{(x^2+1)^2}.$$

Example 3.3.4 Find the derivative of $f(x) = \frac{x^2 - 1}{x^2 + 1}$.

Solution Using the product rule, we have

$$\text{Product Rule} \quad f'(x) = (x^2-1)\frac{1}{(x^2+1)} + (x^2+1)\frac{-2x}{(x^2+1)^2}.$$

Example 3.3.5 Find the derivative of $f(x) = \frac{x^2 - 1}{x^2 + 1}$.

Solution

$$\text{Product Rule} \quad f'(x) = (x^2-1)\frac{1}{(x^2+1)} + (x^2+1)\frac{-2x}{(x^2+1)^2}.$$

Example 3.3.6 Find the derivative of $f(x) = \frac{x^2 - 1}{x^2 + 1}$.

Solution

$$\text{Product Rule} \quad f'(x) = (x^2-1)\frac{1}{(x^2+1)} + (x^2+1)\frac{-2x}{(x^2+1)^2}.$$

Example 3.3.7 Find the derivative of $f(x) = \frac{x^2 - 1}{x^2 + 1}$.

Solution

$$\text{Product Rule} \quad f'(x) = (x^2-1)\frac{1}{(x^2+1)} + (x^2+1)\frac{-2x}{(x^2+1)^2}.$$

Example 3.3.8 Find the derivative of $f(x) = \frac{x^2 - 1}{x^2 + 1}$.

Solution

$$\text{Product Rule} \quad f'(x) = (x^2-1)\frac{1}{(x^2+1)} + (x^2+1)\frac{-2x}{(x^2+1)^2}.$$

Example 3.3.9 Find the derivative of $f(x) = \frac{x^2 - 1}{x^2 + 1}$.

Solution

$$\text{Product Rule} \quad f'(x) = (x^2-1)\frac{1}{(x^2+1)} + (x^2+1)\frac{-2x}{(x^2+1)^2}.$$

Example 3.3.10 Find the derivative of $f(x) = \frac{x^2 - 1}{x^2 + 1}$.

Solution

$$\text{Product Rule} \quad f'(x) = (x^2-1)\frac{1}{(x^2+1)} + (x^2+1)\frac{-2x}{(x^2+1)^2}.$$

Example 3.3.11 Find the derivative of $f(x) = \frac{x^2 - 1}{x^2 + 1}$.

Solution

$$\text{Product Rule} \quad f'(x) = (x^2-1)\frac{1}{(x^2+1)} + (x^2+1)\frac{-2x}{(x^2+1)^2}.$$

Example 3.3.12 Find the derivative of $f(x) = \frac{x^2 - 1}{x^2 + 1}$.

Solution

$$\text{Product Rule} \quad f'(x) = (x^2-1)\frac{1}{(x^2+1)} + (x^2+1)\frac{-2x}{(x^2+1)^2}.$$

Example 3.3.13 Find the derivative of $f(x) = \frac{x^2 - 1}{x^2 + 1}$.

Solution

$$\text{Product Rule} \quad f'(x) = (x^2-1)\frac{1}{(x^2+1)} + (x^2+1)\frac{-2x}{(x^2+1)^2}.$$

Example 3.3.14 Find the derivative of $f(x) = \frac{x^2 - 1}{x^2 + 1}$.

Solution

$$\text{Product Rule} \quad f'(x) = (x^2-1)\frac{1}{(x^2+1)} + (x^2+1)\frac{-2x}{(x^2+1)^2}.$$

Example 3.3.15 Find the derivative of $f(x) = \frac{x^2 - 1}{x^2 + 1}$.

Solution

$$\text{Product Rule} \quad f'(x) = (x^2-1)\frac{1}{(x^2+1)} + (x^2+1)\frac{-2x}{(x^2+1)^2}.$$

Example 3.3.16 Find the derivative of $f(x) = \frac{x^2 - 1}{x^2 + 1}$.

Solution

$$\text{Product Rule} \quad f'(x) = (x^2-1)\frac{1}{(x^2+1)} + (x^2+1)\frac{-2x}{(x^2+1)^2}.$$

Example 3.3.17 Find the derivative of $f(x) = \frac{x^2 - 1}{x^2 + 1}$.

Solution

$$\text{Product Rule} \quad f'(x) = (x^2-1)\frac{1}{(x^2+1)} + (x^2+1)\frac{-2x}{(x^2+1)^2}.$$

Example 3.3.18 Find the derivative of $f(x) = \frac{x^2 - 1}{x^2 + 1}$.

Solution

$$\text{Product Rule} \quad f'(x) = (x^2-1)\frac{1}{(x^2+1)} + (x^2+1)\frac{-2x}{(x^2+1)^2}.$$

Example 3.3.19 Find the derivative of $f(x) = \frac{x^2 - 1}{x^2 + 1}$.

Solution

$$\text{Product Rule} \quad f'(x) = (x^2-1)\frac{1}{(x^2+1)} + (x^2+1)\frac{-2x}{(x^2+1)^2}.$$

Example 3.3.20 Find the derivative of $f(x) = \frac{x^2 - 1}{x^2 + 1}$.

Solution

$$\text{Product Rule} \quad f'(x) = (x^2-1)\frac{1}{(x^2+1)} + (x^2+1)\frac{-2x}{(x^2+1)^2}.$$

Example 3.3.21 Find the derivative of $f(x) = \frac{x^2 - 1}{x^2 + 1}$.

Solution

$$\text{Product Rule} \quad f'(x) = (x^2-1)\frac{1}{(x^2+1)} + (x^2+1)\frac{-2x}{(x^2+1)^2}.$$

Example 3.3.22 Find the derivative of $f(x) = \frac{x^2 - 1}{x^2 + 1}$.

Solution

$$\text{Product Rule} \quad f'(x) = (x^2-1)\frac{1}{(x^2+1)} + (x^2+1)\frac{-2x}{(x^2+1)^2}.$$

Example 3.3.23 Find the derivative of $f(x) = \frac{x^2 - 1}{x^2 + 1}$.

Solution

$$\text{Product Rule} \quad f'(x) = (x^2-1)\frac{1}{(x^2+1)} + (x^2+1)\frac{-2x}{(x^2+1)^2}.$$

Example 3.3.24 Find the derivative of $f(x) = \frac{x^2 - 1}{x^2 + 1}$.

Solution

$$\text{Product Rule} \quad f'(x) = (x^2-1)\frac{1}{(x^2+1)} + (x^2+1)\frac{-2x}{(x^2+1)^2}.$$

Example 3.3.25 Find the derivative of $f(x) = \frac{x^2 - 1}{x^2 + 1}$.

Solution

$$\text{Product Rule} \quad f'(x) = (x^2-1)\frac{1}{(x^2+1)} + (x^2+1)\frac{-2x}{(x^2+1)^2}.$$

Example 3.3.26 Find the derivative of $f(x) = \frac{x^2 - 1}{x^2 + 1}$.

Solution

$$\text{Product Rule} \quad f'(x) = (x^2-1)\frac{1}{(x^2+1)} + (x^2+1)\frac{-2x}{(x^2+1)^2}.$$

Example 3.3.27 Find the derivative of $f(x) = \frac{x^2 - 1}{x^2 + 1}$.

Solution

$$\text{Product Rule} \quad f'(x) = (x^2-1)\frac{1}{(x^2+1)} + (x^2+1)\frac{-2x}{(x^2+1)^2}.$$

Example 3.3.28 Find the derivative of $f(x) = \frac{x^2 - 1}{x^2 + 1}$.

Solution

$$\text{Product Rule} \quad f'(x) = (x^2-1)\frac{1}{(x^2+1)} + (x^2+1)\frac{-2x}{(x^2+1)^2}.$$

Example 3.3.29 Find the derivative of $f(x) = \frac{x^2 - 1}{x^2 + 1}$.

Solution

$$\text{Product Rule} \quad f'(x) = (x^2-1)\frac{1}{(x^2+1)} + (x^2+1)\frac{-2x}{(x^2+1)^2}.$$

Example 3.3.30 Find the derivative of $f(x) = \frac{x^2 - 1}{x^2 + 1}$.

Solution

$$\text{Product Rule} \quad f'(x) = (x^2-1)\frac{1}{(x^2+1)} + (x^2+1)\frac{-2x}{(x^2+1)^2}.$$

Example 3.3.31 Find the derivative of $f(x) = \frac{x^2 - 1}{x^2 + 1}$.

Solution

$$\text{Product Rule} \quad f'(x) = (x^2-1)\frac{1}{(x^2+1)} + (x^2+1)\frac{-2x}{(x^2+1)^2}.$$

Example 3.3.32 Find the derivative of $f(x) = \frac{x^2 - 1}{x^2 + 1}$.

Solution

$$\text{Product Rule} \quad f'(x) = (x^2-1)\frac{1}{(x^2+1)} + (x^2+1)\frac{-2x}{(x^2+1)^2}.$$

Example 3.3.33 Find the derivative of $f(x) = \frac{x^2 - 1}{x^2 + 1}$.

Solution

$$\text{Product Rule} \quad f'(x) = (x^2-1)\frac{1}{(x^2+1)} + (x^2+1)\frac{-2x}{(x^2+1)^2}.$$

Example 3.3.34 Find the derivative of $f(x) = \frac{x^2 - 1}{x^2 + 1}$.

Solution

$$\text{Product Rule} \quad f'(x) = (x^2-1)\frac{1}{(x^2+1)} + (x^2+1)\frac{-2x}{(x^2+1)^2}.$$

Example 3.3.35 Find the derivative of $f(x) = \frac{x^2 - 1}{x^2 + 1}$.

Solution

$$\text{Product Rule} \quad f'(x) = (x^2-1)\frac{1}{(x^2+1)} + (x^2+1)\frac{-2x}{(x^2+1)^2}.$$

Example 3.3.36 Find the derivative of $f(x) = \frac{x^2 - 1}{x^2 + 1}$.

Solution

$$\text{Product Rule} \quad f'(x) = (x^2-1)\frac{1}{(x^2+1)} + (x^2+1)\frac{-2x}{(x^2+1)^2}.$$

Example 3.3.37 Find the derivative of $f(x) = \frac{x^2 - 1}{x^2 + 1}$.

Solution

$$\text{Product Rule} \quad f'(x) = (x^2-1)\frac{1}{(x^2+1)} + (x^2+1)\frac{-2x}{(x^2+1)^2}.$$

Example 3.3.38 Find the derivative of $f(x) = \frac{x^2 - 1}{x^2 + 1}$.

Solution

$$\text{Product Rule} \quad f'(x) = (x^2-1)\frac{1}{(x^2+1)} + (x^2+1)\frac{-2x}{(x^2+1)^2}.$$

Example 3.3.39 Find the derivative of $f(x) = \frac{x^2 - 1}{x^2 + 1}$.

Solution

$$\text{Product Rule} \quad f'(x) = (x^2-1)\frac{1}{(x^2+1)} + (x^2+1)\frac{-2x}{(x^2+1)^2}.$$

Example 3.3.40 Find the derivative of $f(x) = \frac{x^2 - 1}{x^2 + 1}$.

Solution

$$\text{Product Rule} \quad f'(x) = (x^2-1)\frac{1}{(x^2+1)} + (x^2+1)\frac{-2x}{(x^2+1)^2}.$$

Example 3.3.41 Find the derivative of $f(x) = \frac{x^2 - 1}{x^2 + 1}$.

Solution

$$\text{Product Rule} \quad f'(x) = (x^2-1)\frac{1}{(x^2+1)} + (x^2+1)\frac{-2x}{(x^2+1)^2}.$$

Example 3.3.42 Find the derivative of $f(x) = \frac{x^2 - 1}{x^2 + 1}$.

Solution

$$\text{Product Rule} \quad f'(x) = (x^2-1)\frac{1}{(x^2+1)} + (x^2+1)\frac{-2x}{(x^2+1)^2}.$$

Example 3.3.43 Find the derivative of $f(x) = \frac{x^2 - 1}{x^2 + 1}$.

Solution

$$\text{Product Rule} \quad f'(x) = (x^2-1)\frac{1}{(x^2+1)} + (x^2+1)\frac{-2x}{(x^2+1)^2}.$$

Example 3.3.44 Find the derivative of $f(x) = \frac{x^2 - 1}{x^2 + 1}$.

Solution

$$\text{Product Rule} \quad f'(x) = (x^2-1)\frac{1}{(x^2+1)} + (x^2+1)\frac{-2x}{(x^2+1)^2}.$$

Example 3.3.45 Find the derivative of $f(x) = \frac{x^2 - 1}{x^2 + 1}$.

Solution

$$\text{Product Rule} \quad f'(x) = (x^2-1)\frac{1}{(x^2+1)} + (x^2+1)\frac{-2x}{(x^2+1)^2}.$$

Example 3.3.46 Find the derivative of $f(x) = \frac{x^2 - 1}{x^2 + 1}$.

Solution

$$\text{Product Rule} \quad f'(x) = (x^2-1)\frac{1}{(x^2+1)} + (x^2+1)\frac{-2x}{(x^2+1)^2}.$$

Example 3.3.47 Find the derivative of $f(x) = \frac{x^2 - 1}{x^2 + 1}$.

Solution

$$\text{Product Rule} \quad f'(x) = (x^2-1)\frac{1}{(x^2+1)} + (x^2+1)\frac{-2x}{(x^2+1)^2}.$$

Example 3.3.48 Find the derivative of $f(x) = \frac{x^2 - 1}{x^2 + 1}$.

Solution

$$\text{Product Rule} \quad f'(x) = (x^2-1)\frac{1}{(x^2+1)} + (x^2+1)\frac{-2x}{(x^2+1)^2}.$$

Example 3.3.49 Find the derivative of $f(x) = \frac{x^2 - 1}{x^2 + 1}$.

Solution

$$\text{Product Rule} \quad f'(x) = (x^2-1)\frac{1}{(x^2+1)} + (x^2+1)\frac{-2x}{(x^2+1)^2}.$$

Example 3.3.50 Find the derivative of $f(x) = \frac{x^2 - 1}{x^2 + 1}$.

Solution

$$\text{Product Rule} \quad f'(x) = (x^2-1)\frac{1}{(x^2+1)} + (x^2+1)\frac{-2x}{(x^2+1)^2}.$$

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AUTHOR: ARTHUR MATTHEW

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Hi! Glad to hear you're here!
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Hi! Glad to hear you're here!

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Ultra Clean Holdings (UCTT) Q3 2025 Earnings Call Transcript

Nov. 6, 2025 4:30 PM ET | About: UCTT | Earnings Call Transcripts

Operator: Thank you for standing by. My name is Jessica. I am the conference operator. At this time, I would like to welcome everyone to the Ultra Clean Holdings Third Quarter 2025 Earnings Conference Call. All lines have been placed on mute to prevent any background noise. After the speakers' remarks, there will be a question-and-answer session. If you would like to ask a question, please press star one on your telephone keypad. If you require assistance, please press star zero. I now turn the call over to Tim Savagian, Investor Relations. Please begin.

Tim Savagian: Thank you, Jessica. Good afternoon and thank you for joining us today. With me on the call are Clarence Granger, President and Chief Executive Officer, and Jim Scholhamer, Senior Vice President and Chief Financial Officer. Our earnings release and supplemental slides are posted on our website at www.uctt.com. Our discussion today will include forward-looking statements that are subject to risks and uncertainties. Actual results may differ materially from those statements. For a discussion of the risks and factors that could affect our future results, please refer to our SEC filings. We will also discuss non-GAAP financial measures. Reconciliations to the GAAP measures are included in our earnings release and slides. Now, I'll turn the call over to Clarence.

Clarence Granger: Thanks, Tim. Good afternoon and thanks for joining us today. Our third quarter results were in line with our guidance, with revenue of $518 million and non-GAAP EPS of $0.47. Our results were impacted by a slower than expected ramp of our new customer in the chemical delivery systems business, which I'll discuss later. Our core business remains strong, and we continue to execute well on our strategic priorities. We expect our new customer ramp to be back on track in the fourth quarter and beyond. Our outlook for the fourth quarter reflects sequential growth in our core business and the ramp of our new customer. Now, I'll turn the call over to Jim.

Jim Scholhamer: Thanks, Clarence. Third quarter revenue of $518 million was in the middle of our guidance range. Gross margin of 19.4% was below our guidance due to the slower than expected ramp of our new customer in our chemical delivery systems business. Non-GAAP operating expenses of $74 million were in line with our guidance. Non-GAAP EPS of $0.47 was below our guidance due to the lower gross margin. Now, I'll turn the call back over to Clarence.

Clarence Granger: Thanks, Jim. We believe Ultra Clean is well positioned to benefit from the growth in the semiconductor industry. We have a strong and diversified customer base and a leading market position in our key markets. We

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Sky-Fire 1.2



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Sky-Fire 1.2 - Free Web Image Optimizer


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</g>
</svg>
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Optimized with
Free Web Image Optimizer (Sky-Fire) 1.2, Open Source Software
Optimized with
SVGO
v1.0.0-rc.4
v0.7.0-rc.1
v0.6.2
v0.5.0
v0.4.0
v0.3.0
v0.2.0
v0.1.0

```
<?xml version="1.0"?>
<rdf:RDF xmlns:rdf="http://www.w3.org/22-rdf-syntax-ns#" xmlns:dc="http://purl.org/dc/elements/1.1/" xmlns:dcterms="http://purl.org/dc/terms/" xmlns:cc="http://creativecommons.org/ns#" xmlns:dcmitype="http://purl.org/dc/dcmitype/" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns:rdfs="http://www.w3.org/2000/01/rdf-schema#" xmlns:xsl="http://www.w3.org/1999/XSL/Transform" xmlns:xsl:version="1.0" xmlns:xsl:exclude-result-prefixes="rdf" xmlns:xsl:output="text" />
<xsl:template match="/">
<xsl:for-each select="rdf:RDF/cc:License">
<xsl:if test="dc:rights">
<xsl:value-of select="dc:rights" />
</xsl:if>
</xsl:for-each>
</xsl:template>
</rdf:RDF>
```

```
{"id":"0000-00-00T00:00:00.000+00:00",
"_id":"0000-00-00",
"_rev":"1-967a00dff5e02add41819138abb3284d",
"_attachments":{
"rev_pos":0,
"digest":"md5-1B2M2Y8Asgtpgw9iAvI9Nw==",
"content_type":"application/octet-stream",
"stub":true,
"revpos":1
```

Cash Management

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1 Matrix Calculus for Scientific Work

Prerequisites: Linear algebra (Section 0), basic calculus.

Notation: $\mathbf{x}$ denotes a column vector, $\mathbf{A}$ denotes a matrix, $\mathbf{A}^\top$ denotes the transpose, $\mathbf{A}^{-1}$ denotes the inverse, $\mathbf{A}^{+}$ denotes the Moore-Penrose pseudoinverse, $\mathbf{I}$ denotes the identity matrix, $\mathbf{0}$ denotes the zero matrix, $\mathbf{1}$ denotes the all-ones vector, $\mathbf{e}_i$ denotes the i-th standard basis vector, $\mathbf{A} \succ \mathbf{0}$ denotes $\mathbf{A}$ is positive definite, $\mathbf{A} \succeq \mathbf{0}$ denotes $\mathbf{A}$ is positive semi-definite, $\mathbf{A} \odot \mathbf{B}$ denotes the Hadamard product, $\mathbf{A} \otimes \mathbf{B}$ denotes the Kronecker product, $\mathrm{vec}(\mathbf{A})$ denotes the column-stacking vectorization, $\mathrm{diag}(\mathbf{x})$ denotes the diagonal matrix with $\mathbf{x}$ on the diagonal, $\mathrm{tr}(\mathbf{A})$ denotes the trace, $\det(\mathbf{A})$ denotes the determinant, $\|\mathbf{x}\|_p$ denotes the p-norm, $\|\mathbf{A}\|_F$ denotes the Frobenius norm, $\|\mathbf{A}\|_2$ denotes the spectral norm, $\mathbf{A}^{1/2}$ denotes the matrix square root, $\mathbf{A}^{-1/2}$ denotes $(\mathbf{A}^{1/2})^{-1}$, $\mathbf{A}^{-\top/2}$ denotes $(\mathbf{A}^\top)^{-1/2}$, $\mathbf{A}^{\top/2}$ denotes $(\mathbf{A}^{1/2})^\top$, $\mathbf{A}^{\top}$ denotes $(\mathbf{A}^\top)^{-1}$, $\mathbf{A}^{+/2}$ denotes $(\mathbf{A}^{+})^{1/2}$, $\mathbf{A}^{+\top/2}$ denotes $(\mathbf{A}^{+\top})^{1/2}$, $\mathbf{A}^{+\top}$ denotes $(\mathbf{A}^{+})^\top$, $\mathbf{A}^{\top/2}$ denotes $(\mathbf{A}^\top)^{1/2}$, $\mathbf{A}^{+\top/2}$ denotes $(\mathbf{A}^{+\top})^{1/2}$, $\mathbf{A}^{+\top}$ denotes $(\mathbf{A}^{+\top})^{+}$, $\mathbf{A}^{+/2}$ denotes $(\mathbf{A}^{+})

제 3 조(기본 사항) 기본사항의 적용

제 4 조(기본사항의 변경) 기본사항의 변경 사유는 다음 각 호와 같다.

1. 기본사항의 변경사항이 발생하는 경우 기본사항의 변경사항은 기본사항의 변경사항 발생일부터 3개월 이내에 이사회의 의결을 거쳐 변경하여야 한다.

제 5 조(기본사항의 변경절차) 기본사항의 변경절차는 다음 각 호와 같다.

1. 기본사항의 변경사항은 기본사항의 변경사항 발생일부터 3개월 이내에 이사회의 의결을 거쳐 변경하여야 한다.

제 2 장 계 약

제 6 조(계약의 체결) 계약의 체결은 다음 각 호와 같다.

1. 계약의 체결은 계약체결일부터 3개월 이내에 이사회의 의결을 거쳐 변경하여야 한다.

제 7 조(계약의 변경) 계약의 변경은 다음 각 호와 같다.

1. 계약의 변경은 계약변경일부터 3개월 이내에 이사회의 의결을 거쳐 변경하여야 한다.

제 3 장 주 식

제 8 조(주식의 발행) 주식의 발행은 다음 각 호와 같다.

1. 주식의 발행은 주식발행일부터 3개월 이내에 이사회의 의결을 거쳐 변경하여야 한다.

제 9 조(주식의 양도) 주식의 양도는 다음 각 호와 같다.

1. 주식의 양도는 주식양도일부터 3개월 이내에 이사회의 의결을 거쳐 변경하여야 한다.

제 4 장 회사의 기구

제 10 조(이사회의 구성) 이사회의 구성은 다음 각 호와 같다.

1. 이사회의 구성은 이사회구성일부터 3개월 이내에 이사회의 의결을 거쳐 변경하여야 한다.

제 11 조(이사회의 소집) 이사회의 소집은 다음 각 호와 같다.

1. 이사회의 소집은 이사회소집일부터 3개월 이내에 이사회의 의결을 거쳐 변경하여야 한다.

제 12 조(이사회의 의결) 이사회의 의결은 다음 각 호와 같다.

1. 이사회의 의결은 이사회의결일부터 3개월 이내에 이사회의 의결을 거쳐 변경하여야 한다.

제 13 조(이사의 보수) 이사의 보수는 다음 각 호와 같다.

1. 이사의 보수는 이사보수일부터 3개월 이내에 이사회의 의결을 거쳐 변경하여야 한다.

제 14 조(이사의 임기) 이사의 임기는 다음 각 호와 같다.

1. 이사의 임기는 이사임기일부터 3개월 이내에 이사회의 의결을 거쳐 변경하여야 한다.

제 15 조(이사의 결격사유) 이사의 결격사유는 다음 각 호와 같다.

1. 이사의 결격사유는 이사결격사유일부터 3개월 이내에 이사회의 의결을 거쳐 변경하여야 한다.

제 16 조(이사의 해임) 이사의 해임은 다음 각 호와 같다.

1. 이사의 해임은 이사해임일부터 3개월 이내에 이사회의 의결을 거쳐 변경하여야 한다.

제 17 조(이사의 직무) 이사의 직무는 다음 각 호와 같다.

1. 이사의 직무는 이사직무일부터 3개월 이내에 이사회의 의결을 거쳐 변경하여야 한다.

제 18 조(이사의 책임) 이사의 책임은 다음 각 호와 같다.

1. 이사의 책임은 이사책임일부터 3개월 이내에 이사회의 의결을 거쳐 변경하여야 한다.

제 19 조(이사의 충실의무) 이사의 충실의무는 다음 각 호와 같다.

1. 이사의 충실의무는 이사충실의무일부터 3개월 이내에 이사회의 의결을 거쳐 변경하여야 한다.

제 20 조(이사의 경업금지) 이사의 경업금지는 다음 각 호와 같다.

1. 이사의 경업금지는 이사경업금지일부터 3개월 이내에 이사회의 의결을 거쳐 변경하여야 한다.

제 21 조(이사의 보고의무) 이사의 보고의무는 다음 각 호와 같다.

1. 이사의 보고의무는 이사보고의무일부터 3개월 이내에 이사회의 의결을 거쳐 변경하여야 한다.

제 22 조(이사의 비밀유지의무) 이사의 비밀유지의무는 다음 각 호와 같다.

1. 이사의 비밀유지의무는 이사비밀유지의무일부터 3개월 이내에 이사회의 의결을 거쳐 변경하여야 한다.

제 23 조(이사의 손해배상책임) 이사의 손해배상책임은 다음 각 호와 같다.

1. 이사의 손해배상책임은 이사손해배상책임일부터 3개월 이내에 이사회의 의결을 거쳐 변경하여야 한다.

제 24 조(이사의 퇴직금) 이사의 퇴직금은 다음 각 호와 같다.

1. 이사의 퇴직금은 이사퇴직금일부터 3개월 이내에 이사회의 의결을 거쳐 변경하여야 한다.

제 25 조(이사의 퇴직금 지급) 이사의 퇴직금 지급은 다음 각 호와 같다.

1. 이사의 퇴직금 지급은 이사퇴직금지급일부터 3개월 이내에 이사회의 의결을 거쳐 변경하여야 한다.

제 26 조(이사의 퇴직금 지급방법) 이사의 퇴직금 지급방법은 다음 각 호와 같다.

1. 이사의 퇴직금 지급방법은 이사퇴직금지급방법일부터 3개월 이내에 이사회의 의결을 거쳐 변경하여야 한다.

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Human telomerase is a ribonucleoprotein enzyme that maintains telomere length in most cancer cells. Telomerase activity is thought to be required for tumorigenesis, but the precise role of telomerase in cancer is unknown. We have used a human papillomavirus 16 E6/E7 gene-induced model of human cervical epithelial tumorigenesis to study the role of telomerase in cancer. We show that telomerase activity is not required for initiation of cervical epithelial tumorigenesis, but is required for maintenance of established tumors. Inhibition of telomerase activity in established tumors results in a rapid loss of tumorigenicity, associated with a decrease in telomerase activity, telomere length, and cell proliferation. Telomerase activity is required for the maintenance of telomere length in these tumors, and telomere length is critical for the maintenance of tumorigenicity. These results show that telomerase activity is required for the maintenance of tumorigenicity in this model of human cervical epithelial tumorigenesis.

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MR. TIGGY-WINKLE'S HOME

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[1] [Cover Illustration: Microscope on Digitalis]

[2] [Illustration: Digitalis purpurea (Foxglove)]

[3] [Illustration: Digitalis lanata (Grecian Foxglove) (Digitalis lanata Ehrhart) Digitalis Purpurea (Foxglove) Digitalis Ferruginea]

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```
<?xml version="1.0" encoding="UTF-8" ?>
<rdf:RDF xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#" xmlns:dc="http://purl.org/dc/elements/1.1/" xmlns:dcterms="http://purl.org/dc/terms/" xmlns:cc="http://creativecommons.org/ns#" xmlns:dcmitype="http://purl.org/dc/dcmitype/" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns:rdfs="http://www.w3.org/2000/01/rdf-schema#" xmlns:xsl="http://www.w3.org/1999/XSL/Transform" xmlns:xsd="http://www.w3.org/2001/XMLSchema#" xmlns:sdl="http://www.w3.org/2004/02/skos/core#" xmlns:ns="http://ns.adobe.com/xap/1.0/" xmlns:mediapro="http://ns.adobe.com/pdf/1.3/" xmlns:pdf="http://ns.adobe.com/pdf/1.3/" xmlns:xmp="http://ns.adobe.com/xap/1.0/" xmlns:xmpMM="http://ns.adobe.com/xap/1.0/mm/" xmlns:stRef="http://ns.adobe.com/xap/1.0/sType/Ref#" xmlns:stEvt="http://ns.adobe.com/xap/1.0/sType/ResourceEvent#" xmlns:stVer="http://ns.adobe.com/xap/1.0/sType/Version#" xmlns:stJob="http://ns.adobe.com/xap/1.0/sType/Job#" xmlns:stFnt="http://ns.adobe.com/xap/1.0/sType/Font#" xmlns:stDim="http://ns.adobe.com/xap/1.0/sType/Dimensions#" xmlns:stMfs="http://ns.adobe.com/xap/1.0/sType/ManifestItem#" xmlns:stMfs="http://ns.adobe.com/xap/1.0/sType/ResourceRef#" xmlns:stMfs="http://ns.adobe.com/xap/1.0/sType/ResourceEvent#" xmlns:stMfs="http://ns.adobe.com/xap/1.0/sType/Version#" xmlns:stMfs="http://ns.adobe.com/xap/1.0/sType/Job#" xmlns:stMfs="http://ns.adobe.com/xap/1.0/sType/Font#" xmlns:stMfs
```

```
<!DOCTYPE HTML>
<HTML>
<HEAD>
<TITLE>Ajax Sample</TITLE>
<SCRIPT type="text/javascript">
var xmlHttp = null;
function getXmlHttpObj() {
var xmlHttpObj = null;
try {
// Firefox, Opera 8.0+, Safari
xmlHttpObj = new XMLHttpRequest();
}
catch (e) {
// Internet Explorer
try {
xmlHttpObj = new ActiveXObject("Msxml2.XMLHttp");
}
catch (e) {
xmlHttpObj = new ActiveXObject("Microsoft.XMLHttp");
}
}
return xmlHttpObj;
}
function doAjaxGet() {
var xmlHttpObj = getXmlHttpObj();
if (xmlHttpObj != null) {
var theURL = "http://www.mysite.com/myscript.cgi";
var theParms = "name=Bob&age=30";
xmlHttpObj.open("GET", theURL + "?" + theParms, true);
xmlHttpObj.send(null);
}
else {
alert("Your browser does not support AJAX.");
}
}
function doAjaxPost() {
var xmlHttpObj = getXmlHttpObj();
if (xmlHttpObj != null) {
var theURL = "http://www.mysite.com/myscript.cgi";
var theParms = "name=Bob&age=30";
xmlHttpObj.open("POST", theURL, true);
xmlHttpObj.send(theParms);
}
else {
alert("Your browser does not support AJAX.");
}
</SCRIPT>
</HEAD>
<BODY>
<H1>Ajax Sample</H1>
<FORM>
<INPUT type="button" value="GET" onclick="doAjaxGet();">
<INPUT type="button" value="POST" onclick="doAjaxPost();">
</FORM>
</BODY>
</HTML>
```

```
<?xml version="1.0" encoding="UTF-8" ?>
<rss version="2.0" xmlns:dc="http://purl.org/dc/elements/1.1/" xmlns:sy="http://purl.org/rss/1.0/modules/syndication/" xmlns:slash="http://purl.org/rss/1.0/modules/slash/" xmlns:wfw="http://wellformedweb.org/CommentAPI/" xmlns:media="http://search.yahoo.com/mrss/" xmlns:atom="http://www.w3.org/2005/Atom" xmlns:atom10="http://www.w3.org/2005/Atom" xmlns:opensearch="http://a9.com/-/spec/opensearchrss/1.0/" xmlns:geo="http://www.w3.org/2003/01/geo/wgs84_pos#" xmlns:georss="http://www.georss.org/georss" xmlns:gml="http://www.opengis.net/gml" xmlns:thr="http://purl.org/rss/1.0/modules/threading/" xmlns:feedburner="http://rssnamespace.org/feedburner/ext/1.0" xmlns:feedburner:origlink="http://rssnamespace.org/feedburner/ext/1.0" xmlns:media="http://rssnamespace.org/feedburner/extensions/1.0" xmlns:media:thumbnail="http://rssnamespace.org/feedburner/extensions/1.0" xmlns:media:content="http://rssnamespace.org/feedburner/extensions/1.0" xmlns:media:text="http://rssnamespace.org/feedburner/extensions/1.0" xmlns:media:hash="http://rssnamespace.org/feedburner/extensions/1.0" xmlns:media:player="http://rssnamespace.org/feedburner/extensions/1.0" xmlns:media:credit="http://rssnamespace.org/feedburner/extensions/1.0" xmlns:media:rating="http://rssnamespace.org/feedburner/extensions/1.0" xmlns:media:restriction="http://rssnamespace.org/feedburner/extensions/1.0" xmlns:media:category="http://rssnamespace.org/feedburner/extensions/1.0" xmlns:media:group="http://rssnamespace.org/feedburner/extensions/1.0" xmlns:media:keywords="http://rssnamespace.org/feedburner/extensions/1.0" xmlns:media:rss="http://rssnamespace.org/feedburner/extensions/1.0" xmlns:media:rss:channel="http://rssnamespace.org/feedburner/extensions/1.0" xmlns:media:rss:channel:image="http://rssnamespace.org/feedburner/extensions/1.0" xmlns:media:rss:channel:image:link="http://rssnamespace.org/feedburner/extensions/1.0" 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      "height": "8",
      "source": "survey",
      "name": "Test Polygon"
    }
  }
}
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Abstract—Cryptanalysis is a science of breaking codes and ciphers. There is a little bit doubt. Because it is always considered and known to be a matter of time to crack a cipher.

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Sem avaliações de clientes ainda

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Section 1.1

Section 1.1.1

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Section 1.1.2

Section 1.1.2.1 Section 1.1.2.2

Section 1.1.3

Section 1.1.3.1 Section 1.1.3.2

Section 1.1.4

Section 1.1.4.1 Section 1.1.4.2

Section 1.1.5

Section 1.1.5.1 Section 1.1.5.2

Section

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Abstract: Background: Hypertension is a major risk factor for cardiovascular disease, affecting over 1.2 billion adults worldwide. Despite the availability of effective and inexpensive treatments, blood pressure control remains poor in most countries. This review examines the evidence for task-shifting models in which non-physician health workers initiate and manage antihypertensive therapy in primary care settings, with particular attention to implementation in low- and middle-income countries.

Methods: We searched PubMed, Cochrane Library, and Scopus for randomised controlled trials and cluster randomised trials published between January 2010 and December 2023 that compared task-shifting interventions with usual care for the management of hypertension in adults. Two reviewers independently screened records and extracted data on blood pressure control, cardiovascular events, and implementation outcomes. Risk of bias was assessed using the Cochrane tool for randomised trials and the ROBINS-I tool for cluster trials.

Results: Twelve trials enrolling a total of 48 312 participants were included. Task-shifting interventions reduced mean systolic blood pressure by 8.4 mmHg (95% CI, 6.1–10.7) and increased the proportion of participants achieving blood pressure control by 14 percentage points (RR 1.21, 95% CI 1.17–1.26). Cardiovascular events were reduced by 18% (RR 0.82, 95% CI 0.74–0.91). Implementation outcomes were favourable in most settings, with high acceptability among patients and community health workers. Heterogeneity was moderate (I² = 47%) and was explained by baseline blood pressure and the intensity of supervision.

Conclusions: Task-shifting models for hypertension management are effective and safe when delivered by trained non-physician health workers with adequate supervision and medicine supply. Scaling these models requires investment in training, supervision, and medicine supply chains rather than further efficacy research in most settings.

Keywords: hypertension; task-shifting; community health worker; primary care; low-resource setting; implementation science

1. Introduction

Hypertension affects an estimated 1.28 billion adults aged 30–79 years worldwide, two-thirds of whom live in low- and middle-income countries. Despite the availability of effective and inexpensive antihypertensive medicines, fewer than one in five adults with hypertension in these settings have their blood pressure controlled. The gap between treatment and control is driven by weak health systems, shortages of trained clinicians, and the high cost of specialist care.

Task-shifting, the delegation of clinical tasks from more to less specialised cadres, has been proposed as a way to extend the reach of primary care without proportionate increases in clinician numbers. Prior reviews have examined task-shifting for HIV, maternal and child health, and mental health. This review asks whether the evidence for task-shifting in hypertension is strong enough to support scale-up in low-resource settings, and which implementation features appear to drive effectiveness.

2. Methods

We searched PubMed, the Cochrane Library, and Scopus for randomised controlled trials and cluster randomised trials published between January 2010 and December 2023 comparing task-shifting interventions with usual care for the management of hypertension in adults. Two reviewers independently screened records and extracted data on blood pressure control, cardiovascular events, and implementation outcomes. Risk of bias was assessed using the Cochrane tool for randomised trials and the ROBINS-I tool for cluster trials.

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1.1 Why Study Society?

Learning Objectives

1.1.1 Explain why it is worthwhile to study society.

1.1.2 Identify the sociological perspective.

1.1.3 Identify the three major sociological theoretical perspectives.

1.1.4 Identify the three major levels of analysis in sociology.

1.1.5 Identify the three major sociological research methods.

1.1.6 Identify the three major types of research ethics.

1.1.7 Identify the three major types of research sponsors.

1.1.8 Identify the three major sociological careers.

1.1.9 Identify the three major sociological organizations.

1.1.10 Identify the three major sociological journals.

1.1.11 Identify the three major sociological associations.

1.1.12 Identify the three major sociological blogs.

Introduction

Why study society? Why study sociology? Why study society? Why study sociology?

Why study society? Why study society? Why study society? Why study society?

Why study society? Why study society? Why study society? Why study society?

Why study society? Why study society? Why study society? Why study society

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2. Plaintiff, Jane Doe, is a citizen of the Commonwealth of Virginia.

3. Defendant, Jonathan Lee Riches, is a citizen of the State of Florida.

4. Defendant, Jonathan Lee Riches, has made, published, and caused to be published, false and defamatory statements about Plaintiff, Jane Doe, which have been published on various internet sites, including, but not limited to, www.jonathanleeriches.com, www.jonathanleeriches.com/blog, www.facebook.com/jonathanleeriches, www.twitter.com/jonathanriches, www.twitter.com/jonathanriches/lists/jane-doe, www.twitter.com/jonathanriches/lists/jane-doe-2, www.twitter.com/jonathanriches/lists/jane-doe-3, www.twitter.com/jonathanriches/lists/jane-doe-4, www.twitter.com/jonathanriches/lists/jane-doe-5, www.twitter.com/jonathanriches/lists/jane-doe-6, www.twitter.com/jonathanriches/lists/jane-doe-7, www.twitter.com/jonathanriches/lists/jane-doe-8, www.twitter.com/jonathanriches/lists/jane-doe-9, www.twitter.com/jonathanriches/lists/jane-doe-10, www.twitter.com/jonathanriches/lists/jane-doe-11, www.twitter.com/jonathanriches/lists/jane-doe-12, www.twitter.com/jonathanriches/lists/jane-doe-13, www.twitter.com/jonathanriches/lists/jane-doe-14, www.twitter.com/jonathanriches/lists/jane-doe-15, www.twitter.com/jonathanriches/lists/jane-doe-16, www.twitter.com/jonathanriches/lists/jane-doe-17, www.twitter.com/jonathanriches/lists/jane-doe-18, www.twitter.com/jonathanriches/lists/jane-doe-19, www.twitter.com/jonathanriches/lists/jane-doe-20, www.twitter.com/jonathanriches/lists/jane-doe-21, www.twitter.com/jonathanriches/lists/jane-doe-22, www.twitter.com/jonathanriches/lists/jane-doe-23, www.twitter.com/jonathanriches/lists/jane-doe-24, www.twitter.com/jonathanriches/lists/jane-doe-25, www.twitter.com/jonathanriches/lists/jane-doe-26, www.twitter.com/jonathanriches/lists/jane-doe-27, www.twitter.com/jonathanriches/lists/jane-doe-28, www.twitter.com/jonathanriches/lists/jane-doe-29, www.twitter.com/jonathanriches/lists/jane-doe-30, www.twitter.com/jonathanriches/lists/jane-doe-31, www.twitter.com/jonathanriches/lists/jane-doe-32, www.twitter.com/jonathanriches/lists/jane-doe-33, www.twitter.com/jonathanriches/lists/jane-doe-34, www.twitter.com/jonathanriches/lists/jane-doe-35, www.twitter.com/jonathanriches/lists/jane-doe-36, www.twitter.com/jonathanriches/lists/jane-doe-37, www.twitter.com/jonathanriches/lists/jane-doe-38, www.twitter.com/jonathanriches/lists/jane-doe-39, www.twitter.com/jonathanriches/lists/jane-doe-40, www.twitter.com/jonathanriches/lists/jane-doe-41,

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Ultrasonic Level Indicator

Ultrasonic Level Indicator

Type

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Type: Level Indicator, Ultrasonic Level Indicator, Ultrasonic Level Indicator

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MD-7[illegible]MD[illegible]

[illegible]

[illegible]

MR-TIGER-MAGNUMOPUS-7

Tiger Magnum Opus VII Prime Integer Spectrum

Abstract. Tiger's Fundamental Prime Integer Spectrum. Tiger's Fundamental Prime Integer Spectrum. Tiger's Fundamental Prime Integer Spectrum.

Topic:

Question: Consider the Stokes flow around a sphere of radius a in a uniform flow $\mathbf{U} = U\mathbf{e}_z$. Find the stream function ψ in terms of r and θ in spherical coordinates.

Solution: Stokes' Stream Function in Spherical Coordinates

Source: Fluid Mechanics, 5th Edition, Pages 270-271

$$\psi = \frac{1}{2} r^2 \sin^2\theta \left(U - \frac{a^3}{2r} \right)$$

[ ] ## Exercise Set 1.1 (pp.)

[ ] ## Section 1.1 Systems of Linear Equations

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**Sunset Blvd, Malibu

Lorem Ipsum Dolor Sit Amet

Lorem Ipsum Dolor Sit Amet, Consectetur Adipiscing Elit, Sed Do Eiusmod Tempor Incididunt Ut Labore Et Dolore Magna Aliqua.

Ut Enim Ad Minim Veniam, Quis Nostrud Exercitation Ullamco Laboris Nisi Ut Aliquip Ex Ea Commodo Consequat. Duis Aute Irure Dolor In Reprehenderit In Voluptate Velit Esse Cillum Dolore Eu Fugiat Nulla Pariatur. Excepteur Sint Occaecat Cupidatat Non Proident, Sunt In Culpa Qui Officia Deserunt Mollit Anim Id Est Laborum.

Sed Ut Perspiciatis Unde Omnis Iste Natus Error Sit Voluptatem Accusantium Doloremque Laudantium, Totam Rem Aperiam, Eaque Ipsa Quae Ab Illo Inventore Veritatis Et Quasi Architecto Beatae Vitae Dicta Sunt Explicabo. Nemo Enim Ipsam Voluptatem Quia Voluptas Sit Aspernatur Aut Odit Aut Fugit, Sed Quia Consequuntur Magni Dolores Eos Qui Ratione Voluptatem Sequi Nesciunt. Neque Porro Quisquam Est, Qui Dolorem Ipsum Quia Dolor Sit Amet, Consectetur, Adipisci Velit, Sed Quia Non Numquam Eius Modi Tempora Incidunt Ut Labore Et Dolore Magnam Aliquam Quaerat Voluptatem.

Ut Enim Ad Minima Veniam, Quis Nostrum Exercitationem Ullam Corporis Suscipit Laboriosam, Nisi Ut Aliquid Ex Ea Commodi Consequatur? Quis Autem Vel Eum Iure Reprehenderit Qui In Ea Voluptate Velit Esse Quam Nihil Molestiae Consequatur, Vel Illum Qui Dolorem Eum Fugiat Quo Voluptas Nulla Pariatur?

At Vero Eos Et Accusamus Et Iusto Odio Dignissimos Ducimus Qui Blanditiis Praesentium Voluptatum Deleniti Atque Corrupti Quos Dolores Et Quas Molestias Excepturi Sint Occaecati Cupiditate Non Provident, Similique Sunt In Culpa Qui Officia Deserunt Mollitia Animi, Id Est Laborum Et Dolorum Fuga. Et Harum Quidem Rerum Facilis Est Et Expedita Distinctio. Nam Libero Tempore, Cum Soluta Nobis Est Eligendi Optio Cumque Nihil Impedit Quo Minus Id Quod Maxime Placeat Facere Possimus, Omnis Voluptas Assumenda Est, Omnis Dolor Repellendus. Temporibus Autem Quibusdam Et Aut Officiis Debitis Aut Rerum Necessitatibus Saepe Eveniet, Ut Et Voluptates Repudiandae Sint Et Molestiae Non Recusandae. Itaque Earum Rerum Hic Tenetur A Sapiente Delectus, Ut Aut Reiciendis Voluptatibus Maiores Alias Consequatur Aut Perferendis Doloribus Asperiores Repellat.

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Dr. Ananya Sharma, Dr. Rajesh Kumar, Dr. Meenakshi Chandrakala Lingam and Dr. Ramesh Kumar Dhanapal Sharma

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Abstract
This study aims to examine the relationship between burnout and mental well-being among nurses in a tertiary mental health hospital. A cross-sectional design was used. Data were collected from 312 nurses using the Maslach Burnout Inventory and the Mental Health Continuum-Short Form. Burnout was negatively associated with all three components of mental well-being. Emotional exhaustion was the strongest predictor of emotional well-being. Findings suggest that burnout is a key risk factor for poor mental well-being among nurses and that interventions should target both burnout and well-being simultaneously.

Keywords: burnout, mental well-being, nurses, mental health nursing, cross-sectional study, Maslach Burnout Inventory, Mental Health Continuum-Short Form, tertiary hospital, emotional exhaustion, psychiatric nurses

一次性 Redis 迁移工具 Redis-RDMA

作者：Alibaba Cloud Storage Team(原厚博) 2020-09-23 Database Cloud-Native Tags: Redis Alibaba Cloud

背景介绍 Background

随着云计算的发展，Redis 作为一款高性能的内存数据库，在云计算时代扮演着越来越重要的角色。Redis 的应用场景也从单纯的缓存，扩展到了消息队列、分布式锁、实时统计、Session 共享、Feed 流、排行榜、社交网络、地理位置、实时系统、缓存、计数器、服务发现、分布式协调等等。Redis 的数据结构也从最开始的 KV 扩展到了 List、Hash、Set、Sorted Set、Bitmap、HyperLogLog、Geo、Stream 等。Redis 的部署方式也从单机扩展到了主从、哨兵、集群。Redis 的客户端也从 Jedis 扩展到了 Lettuce、Redisson 等。Redis 的生态也从单纯的 Redis 扩展到了 Redis Module 等。Redis 的应用场景、数据结构、部署方式、客户端、生态的扩展，使得 Redis 在云计算时代的应用越来越广泛。

迁移工具 Redis-RDMA

Redis-RDMA 是一款基于 RDMA 的 Redis 迁移工具，它可以将 Redis 的数据从一台机器迁移到另一台机器，迁移的过程中，Redis 的数据不会丢失，迁移的过程中，Redis 的服务不会中断，迁移的过程中，Redis 的客户端不需要重新连接，迁移的过程中，Redis 的客户端不需要重新配置，迁移的过程中，Redis 的客户端不需要重新编译，迁移的过程中，Redis 的客户端不需要重新部署，迁移的过程中，Redis 的客户端不需要重新升级，迁移的过程中，Redis 的客户端不需要重新测试，迁移的过程中，Redis 的客户端不需要重新发布，迁移的过程中，Redis 的客户端不需要重新上线，迁移的过程中，Redis 的客户端不需要重新下线，迁移的过程中，Redis 的客户端不需要重新回滚，迁移的过程中，Redis 的客户端不需要重新监控，迁移的过程中，Redis 的客户端不需要重新告警，迁移的过程中，Redis 的客户端不需要重新运维，迁移的过程中，Redis 的客户端不需要重新开发，迁移的过程中，Redis 的客户端不需要重新维护，迁移的过程中，Redis 的客户端不需要重新管理，迁移的过程中，Redis 的客户端不需要重新调优，迁移的过程中，Redis 的客户端不需要重新扩容，迁移的过程中，Redis 的客户端不需要重新缩容，迁移的过程中，Redis 的客户端不需要重新迁移

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"Notion",
"Zoom",
"Microsoft Teams",
"Google Drive",
"Slack
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Cash Transactions

Home / Cash Transactions

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Date
Amount
Type
Account
Category

Add Cash Transactions

Date Amount Type Account Category

Date Amount Type Account Category Cash Transaction 1
Date Amount Type Account Category Cash Transaction 2

Date Amount Type Account Category Cash Transaction 3

Date Amount Type Account Category Cash Transaction 4

Date Amount Type Account Category Cash Transaction 5

Date Amount Type Account Category Cash Transaction

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```
<div class="mw-parser-output">
<br />
<br />
<br />
```

```
<mc:Fallback>
<v:textbox>
<w:txbxContent />
</v:textbox>
</mc:Fallback>
</mc:AlternateContent>
</w:r>
</w:p>
<w:p w:rsidR="00000000" />
<w:p w:rsidR="00000000" />
<w:p w:rsidR="00000000" />
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```
<?xml version="1.0" encoding="UTF-8"?>

<?xml-stylesheet href="/assets/rss.xsl" type="text/xsl"?>

<rss
version="2.0"
xmlns:dc="http://purl.org/dc/elements/1.1/"
xmlns:sy="http://purl.org/rss/1.0/modules/syndication/"
xmlns:atom="http://www.w3.org/2005/Atom"
xmlns:content="http://purl.org/rss/1.0/modules/content/"
xmlns:media="http://search.yahoo.com/mrss/"
xmlns:wfw="http://wellformedweb.org/CommentAPI/"
>

<channel>
<title>Blog - Ryan Dahl</title>
<atom:link href="https://tinyclouds.org/rss.xml" rel="self" type="application/rss+xml" />
<description>Ryan Dahl's blog.</description>
<dc:creator>Ryan Dahl</dc:creator>
<sy:updateperiod>hourly</sy:updateperiod>
<sy:updatefrequency>1</sy:updatefrequency>
<generator>Jekyll v3.9.5</generator>
<lastbuilddate>Wed, 01 Jan 2025 00:00:00 +0000</lastbuilddate>
<atom:link href="https://tinyclouds.org/rss.xml" />
<item>
<title>Ryan Dahl</title>
<dc:creator>Ryan Dahl</dc:creator>
<pubdate>Wed, 01 Jan 2025 00:00:00 +0000</pubdate>
<guid isbn="0000000000000" />
<description>

</description>
</item>
</channel>
</rss>
```

```
<?xml version="1.0" encoding="utf-8"?>
<svg xmlns="http://www.w3.org/2000/svg" viewBox="
0 0 1000 1000" fill="none" stroke="#000" stroke-width="1" />
<rect x="0" y="0" width="1000" height="1000" />
<path d="M 500 500 C 500 500 500 500 500 500 M 500 500 C 500 500 500 500 500 500 M 500 500 C 500 500 500 500 500 500 M 500 500 C 500 500 500 500 500 500" />
<text x="500" y="500" font="Arial" size="12" />
<path d="M 500 500 C 500 500 500 500 500 500" />
<?xml version="1.0" encoding="utf-8" />
<svg />
<path />
<path />
```

```
<?xml version="1.0" encoding="UTF-8"?>
<rss version="2.0" xmlns:dc="http://purl.org/dc/elements/1.0/"
  <channel>
    <title>Cambridge Economics - Recent Comments</title>
    <dc:creator_ref_1_0_0_0
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	Jan-Jun 2025
	Jan-Jun 2024

1000		
Cash and Bank		
1010 Petty Cash	1,500.00	1,500.00
1020 Bank - Current Account		
1021 HDFC Bank - Current Account		
1021.01 HDFC Bank - Current Account - Operating	1,250,000.00	1,250,000.00
1021.02 HDFC Bank - Current Account - Payroll	450,000.00	450,000.00
1021.03 HDFC Bank - Current Account - Vendor Payments	1,800,000.00	1,800,000.00
1021.04 HDFC Bank - Current Account - Reserve	500,000.00	500,000.00
1022		
1022.01 ICICI Bank - Current Account - Operating	980,000.00	980,000.00
1022.02 ICICI Bank - Current Account - Payroll	320,000.00	320,000.00
1022.03 ICICI Bank - Current Account - Vendor Payments	1,200,000.00	1,200,000.00
1022.04 ICICI Bank - Current Account - Reserve	400,000.00	400,000.00
1023		
1023.01 Axis Bank - Current Account - Operating	750,000.00	750,000.00
1023.02 Axis Bank - Current Account - Payroll	250,000.00	250,000.00
1023.03 Axis Bank - Current Account - Vendor Payments	1,000,000.00	1,000,000.00
1023.04 Axis Bank - Current Account - Reserve	300,000.00	300,000.00
1024		
1024.01 Kotak Mahindra Bank - Current Account - Operating	600,000.00	600,000.00
1024.02 Kotak Mahindra Bank - Current Account - Payroll	200,000.00	200,000.00
1024.03 Kotak Mahindra Bank - Current Account - Vendor Payments	800,000.00	800,000.00
1024.04 Kotak Mahindra Bank - Current Account - Reserve	250,000.00	250,000.00
1025		
1025.01 Yes Bank - Current Account - Operating	500,000.00	500,000.00
1025.02 Yes Bank - Current Account - Payroll	150,000.00	150,000.00
1025.03 Yes Bank - Current Account - Vendor Payments	700,000.00	700,000.00
1025.04 Yes Bank - Current Account - Reserve	200,000.00	200,000.00
1026		
1026.01 Bank of Baroda - Current Account - Operating	400,000.00	400,000.00
1026.02 Bank of Baroda - Current Account - Payroll	100,000.00	100,000.00
1026.03 Bank of Baroda - Current Account - Vendor Payments	500,000.00	500,000.00
1026.04 Bank of Baroda - Current Account - Reserve	150,000.00	150,000.00
1027		
1027.01 Punjab National Bank - Current Account - Operating	350,000.00	350,000.00
1027.02 Punjab National Bank - Current Account - Payroll	100,000.00	100,000.00
1027.03 Punjab National Bank - Current Account - Vendor Payments	450,000.00	450,000.00
1027.04 Punjab National Bank - Current Account - Reserve	100,000.00	100,000.00
1028		
1028.01 Canara Bank - Current Account - Operating	300,000.00	300,000.00
1028.02 Canara Bank - Current Account - Payroll	80,000.00	80,000.00
1028.03 Canara Bank - Current Account - Vendor Payments	400,000.00	400,000.00
1028.04 Canara Bank - Current Account - Reserve	100,000.00	100,000.00
1029		
1029.01 Union Bank of India - Current Account - Operating	250,000.00	250,000.00
1029.02 Union Bank of India - Current Account - Payroll	70,000.00	70,000.00
1029.03 Union Bank of India - Current Account - Vendor Payments	350,000.00	350,000.00
1029.04 Union Bank of India - Current Account - Reserve	80,000.00	80,000.00
1030		
1030.01 Bank of India - Current Account - Operating	200,000.00	200,000.00
1030.02 Bank of India - Current Account - Payroll	60,000.00	60,000.00
1030.03 Bank of India - Current Account - Vendor Payments	300,000.00	300,000.00
1030.04 Bank of India - Current Account - Reserve	70,000.00	70,000.00
1031		
1031.01 Indian Bank - Current Account - Operating	150,000.00	150,000.00
1031.02 Indian Bank - Current Account - Payroll	50,000.00	50,000.00
1031.03 Indian Bank - Current Account - Vendor Payments	200,000.00	200,000.00
1031.04 Indian Bank - Current Account - Reserve	60,000.00	60,000.00
1032		
1032.01 Bank of Maharashtra - Current Account - Operating	100,000.00	100,000.00
1032.02 Bank of Maharashtra - Current Account - Payroll	40,000.00	40,000.00
1032.03 Bank of Maharashtra - Current Account - Vendor Payments	150,000.00	150,000.00
1032.04 Bank of Maharashtra - Current Account - Reserve	50,000.00	50,000.00
1033		
1033.01 Central Bank of India - Current Account - Operating	80,000.00	80,000.00
1033.02 Central Bank of India - Current Account - Payroll	30,000.00	30,000.00
1033.03 Central Bank of India - Current Account - Vendor Payments	100,000.00	100,000.00
1033.04 Central Bank of India - Current Account - Reserve	40,000.00	40,000.00
1034		
1034.01 Indian Overseas Bank - Current Account - Operating	60,000.00	60,000.00
1034.02 Indian Overseas Bank - Current Account - Payroll	20,000.00	20,000.00
1034.03 Indian Overseas Bank - Current Account - Vendor Payments	80,000.00	80,000.00
1034.04 Indian Overseas Bank - Current Account - Reserve	30,000.00	30,000.00
1035		
1035.01 UCO Bank - Current Account - Operating	50,000.00	50,000.00
1035.02 UCO Bank - Current Account - Payroll	15,000.00	15,000.00
1035.03 UCO Bank - Current Account - Vendor Payments	60,000.00	60,000.00
1035.04 UCO Bank - Current Account - Reserve	20,000.00	20,000.00
1036		
1036.01 Bank of Baroda - Current Account - Operating	40,000.00	40,000.00
1036.02 Bank of Baroda - Current Account - Payroll	10,000.00	10,000.00
1036.03 Bank of Baroda - Current Account - Vendor Payments	50,000.00	50,000.00
1036.04 Bank of Baroda - Current Account - Reserve	15,000.00	15,000.00
1037		
1037.01 Punjab National Bank - Current Account - Operating	30,000.00	30,000.00
1037.02 Punjab National Bank - Current Account - Payroll	8,000.00	8,000.00
1037.03 Punjab National Bank - Current Account - Vendor Payments	40,000.00	40,000.00
1037.04 Punjab National Bank - Current Account - Reserve	10,000.00	10,000.00
1038		
1038.01 Canara Bank - Current Account - Operating	20,000.00	20,000.00
1038.02 Canara Bank - Current Account - Payroll	5,000.00	5,000.00
1038.03 Canara Bank - Current Account - Vendor Payments	30,000.00	30,000.00
1038.04 Canara Bank - Current Account - Reserve	8,000.00	8,000.00
1039		
1039.01 Union Bank of India - Current Account - Operating	15,000.00	15,000.00
1039.02 Union Bank of India - Current Account - Payroll	4,000.00	4,000.00
1039.03 Union Bank of India - Current Account - Vendor Payments	20,000.00	20,000.00
1039.04 Union Bank of India - Current Account - Reserve	6,000.00	6,000.00
1040		
1040.01 Bank of India - Current Account - Operating	10,000.00	10,000.00
1040.02 Bank of India - Current Account - Payroll	3,000.00	3,000.00
1040.03 Bank of India - Current Account - Vendor Payments	15,000.00	15,000.00
1040.04 Bank of India - Current Account - Reserve	5,000.00	5,000.00
1041		
1041.01 Indian Bank - Current Account - Operating	8,000.00	8,000.00
1041.02 Indian Bank - Current Account - Payroll	2,000.00	2,000.00
1041.03 Indian Bank - Current Account - Vendor Payments	10,000.00	10,000.00
1041.04 Indian Bank - Current Account - Reserve	3,000.00	3,000.00
1042		
1042.01 Bank of Maharashtra - Current Account - Operating	5,000.00	5,000.00
1042.02 Bank of Maharashtra - Current Account - Payroll	1,000.00	1,000.00
1042.03 Bank of Maharashtra - Current Account - Vendor Payments	8,000.00	8,000.00
1042.04 Bank of Maharashtra - Current Account - Reserve	2,000.00	2,000.00
1043		
1043.01 Central Bank of India - Current Account - Operating	3,000.00	3,000.00
1043.02 Central Bank of India - Current Account - Payroll	500.00	500.00
1043.03 Central Bank of India - Current Account - Vendor Payments	5,000.00	5,000.00
1043.04 Central Bank of India - Current Account - Reserve	1,000.00	1,000.00
1044		
1044.01 Indian Overseas Bank - Current Account - Operating	2,000.00	2,000.00
1044.02 Indian Overseas Bank - Current Account - Payroll	300.00	300.00
1044.03 Indian Overseas Bank - Current Account - Vendor Payments	3,000.00	3,000.00
1044.04 Indian Overseas Bank - Current Account - Reserve	500.00	500.00
1045		
1045.01 UCO Bank - Current Account - Operating	1,000.00	1,000.00
1045.02 UCO Bank - Current Account - Payroll	200.00	200.00
1045.03 UCO Bank - Current Account - Vendor Payments	2,000.00	2,000.00
1045.04 UCO Bank - Current Account - Reserve	300.00	300.00
1046		
1046.01 Bank of Baroda - Current Account - Operating	500.00	500.00
1046.02 Bank of Baroda - Current Account - Payroll	100.00	100.00
1046.03 Bank of Baroda - Current Account - Vendor Payments	1,000.00	1,000.00
1046.04 Bank of Baroda - Current Account - Reserve	200.00	200.00
1047		
1047.01 Punjab National Bank - Current Account - Operating	300.00	300.00
1047.02 Punjab National Bank - Current Account - Payroll	80.00	80.00
1047.03 Punjab National Bank - Current Account - Vendor Payments	400.00	400.00
1047.04 Punjab National Bank - Current Account - Reserve	100.00	100.00
1048		
1048.01 Canara Bank - Current Account - Operating	200.00	200.00
1048.02 Canara Bank - Current Account - Payroll	50.00	50.00
1048.03 Canara Bank - Current Account - Vendor Payments	300.00	300.00
1048.04 Canara Bank - Current Account - Reserve	80.00	80.00
1049		
1049.01 Union Bank of India - Current Account - Operating	150.00	150.00
1049.02 Union Bank of India - Current Account - Payroll	40.00	40.00
1049.03 Union Bank of India - Current Account - Vendor Payments	200.00	200.00
1049.04 Union Bank of India - Current Account - Reserve	60.00	60.00
1050		
1050.01 Bank of India - Current Account - Operating	100.00	100.00
1050.02 Bank of India - Current Account - Payroll	30.00	30.00
1050.03 Bank of India - Current Account - Vendor Payments	150.00	150.00
1050.04 Bank of India - Current Account - Reserve	50.00	50.00
1051		
1051.01 Indian Bank - Current Account - Operating	80.00	80.00
1051.02 Indian Bank - Current Account - Payroll	20.00	20.00
1051.03 Indian Bank - Current Account - Vendor Payments	100.00	100.00
1051.04 Indian Bank - Current Account - Reserve	30.00	30.00
1052		
1052.01 Bank of Maharashtra - Current Account - Operating	50.00	50.00
1052.02 Bank of Maharashtra - Current Account - Payroll	10.00	10.00
1052.03 Bank of Maharashtra - Current Account - Vendor Payments	80.00	80.00
1052.04 Bank of Maharashtra - Current Account - Reserve	20.00	20.00
1053		
1053.01 Central Bank of India - Current Account - Operating	30.00	30.00
1053.02 Central Bank of India - Current Account - Payroll	5.00	5.00
1053.03 Central Bank of India - Current Account - Vendor Payments	50.00	50.00
1053.04 Central Bank of India - Current Account - Reserve	10.00	10.00
1054		
1054.01 Indian Overseas Bank - Current Account - Operating	20.00	20.00
1054.02 Indian Overseas Bank - Current Account - Payroll	3.00	3.00
1054.03 Indian Overseas Bank - Current Account - Vendor Payments	30.00	30.00
1054.04 Indian Overseas Bank - Current Account - Reserve	5.00	5.00
1055		
1055.01 UCO Bank - Current Account - Operating	10.00	10.00
1055.02 UCO Bank - Current Account - Payroll	2.00	2.00
1055.03 UCO Bank - Current Account - Vendor Payments	20.00	20.00
1055.04 UCO Bank - Current Account - Reserve	3.00	3.00
1056		
1056.01 Bank of Baroda - Current Account - Operating	5.00	5.00
1056.02 Bank of Baroda - Current Account - Payroll	1.00	1.00
1056.03 Bank of Baroda - Current Account - Vendor Payments	8.00	8.00
1056.04 Bank of Baroda - Current Account - Reserve	2.00	2.00
1057		
1057.01 Punjab National Bank - Current Account - Operating	3.00	3.00
1057.02 Punjab National Bank - Current Account - Payroll	0.80	0.80
1057.03 Punjab National Bank - Current Account - Vendor Payments	4.00	4.00
1057.04 Punjab National Bank - Current Account - Reserve	1.00	1.00
1058		
1058.01 Canara Bank - Current Account - Operating	2.00	2.00
1058.02 Canara Bank - Current Account - Payroll	0.50	0.50
1058.03 Canara Bank - Current Account - Vendor Payments	3.00	3.00
1058.04 Canara Bank - Current Account - Reserve	0.80	0.80
1059		
1059.01 Union Bank of India - Current Account - Operating	1.50	1.50
1059.02 Union Bank of India - Current Account - Payroll	0.40	0.40
1059.03 Union Bank of India - Current Account - Vendor Payments	2.00	2.00
1059.04 Union Bank of India - Current Account - Reserve	0.60	0.60
1060		
1060.01 Bank of India - Current Account - Operating	1.00	1.00
1060.02 Bank of India - Current Account - Payroll	0.30	0.30
1060.03 Bank of India - Current Account - Vendor Payments	1.50	1.50
1060.04 Bank of India - Current Account - Reserve	0.50	0.50
1061		
1061.01 Indian Bank - Current Account - Operating	0.80	0.80
1061.02 Indian Bank - Current Account - Payroll	0.20	0.20
1061.03 Indian Bank - Current Account - Vendor Payments	1.00	1.00
1061.04 Indian Bank - Current Account - Reserve	0.30	0.30
1062		
1062.01 Bank of Maharashtra - Current Account - Operating	0.50	0.50
1062.02 Bank of Maharashtra - Current Account - Payroll	0.10	0.10
1062.03 Bank of Maharashtra - Current Account - Vendor Payments	0.80	0.80
1062.04 Bank of Maharashtra - Current Account - Reserve	0.20	0.20
1063		
1063.01 Central Bank of India - Current Account - Operating	0.30	0.30
1063.02 Central Bank of India - Current Account - Payroll	0.05	0.05
1063.03 Central Bank of India - Current Account - Vendor Payments	0.50	0.50
1063.04 Central Bank of India - Current Account - Reserve	0.10	0.10
1064		
1064.01 Indian Overseas Bank - Current Account - Operating	0.20	0.20
1064.02 Indian Overseas Bank - Current Account - Payroll	0.03	0.03
1064.03 Indian Overseas Bank - Current Account - Vendor Payments	0.30	0.30
1064.04 Indian Overseas Bank - Current Account - Reserve	0.05	0.05
1065		
1065.01 UCO Bank - Current Account - Operating	0.10	0.10
1065.02 UCO Bank - Current Account - Payroll	0.02	0.02
1065.03 UCO Bank - Current Account - Vendor Payments	0.20	0.20
1065.04 UCO Bank - Current Account - Reserve	0.03	0.03
1066		
1066.01 Bank of Baroda - Current Account - Operating	0.05	0.05
1066.02 Bank of Baroda - Current Account - Payroll	0.01	0.01
1066.03 Bank of Baroda - Current Account - Vendor Payments	0.08	0.08
1066.04 Bank of Baroda - Current Account - Reserve	0.02	0.02
1067		
1067.01 Punjab National Bank - Current Account - Operating	0.03	0.03
1067.02 Punjab National Bank - Current Account - Payroll	0.00	0.00
1067.03 Punjab National Bank - Current Account - Vendor Payments	0.04	0.04
1067.04 Punjab National Bank - Current Account - Reserve	0.01	0.01
1068		
1068.01 Canara Bank - Current Account - Operating	0.02	0.02
1068.02 Canara Bank - Current Account - Payroll	0.00	0.00
1068.03 Canara Bank - Current Account - Vendor Payments	0.03	0.03
1068.04 Canara Bank - Current Account - Reserve	0.00	0.00
1069		
1069.01 Union Bank of India - Current Account - Operating	0.01	0.01
1069.02 Union Bank of India - Current Account - Payroll	0.00	0.00
1069.03 Union Bank of India - Current Account - Vendor Payments	0.02	0.02
1069.04 Union Bank of India - Current Account - Reserve	0.00	0.00
1070		
1070.01 Bank of India - Current Account - Operating	0.01	0.01
1070.02 Bank of India - Current Account - Payroll	0.00	0.00
1070.03 Bank of India - Current Account - Vendor Payments	0.01	0.01
1070.04 Bank of India - Current Account - Reserve	0.00	0.00
1071		
1071.01 Indian Bank - Current Account - Operating	0.00	0.00
1071.02 Indian Bank - Current Account - Payroll	0.00	0.00
1071.03 Indian Bank - Current Account - Vendor Payments	0.01	0.01
1071.04 Indian Bank - Current Account - Reserve	0.00	0.00
1072		
1072.01 Bank of Maharashtra - Current Account - Operating	0.00	0.00
1072.02 Bank of Maharashtra - Current Account - Payroll	0.00	0.00
1072.03 Bank of Maharashtra - Current Account - Vendor Payments	0.00	0.00
1072.04 Bank of Maharashtra - Current Account - Reserve	0.00	0.00
1073		
1073.01 Central Bank of India - Current Account - Operating	0.00	0.00
1073.02 Central Bank of India - Current Account - Payroll	0.00	0.00
1073.03 Central Bank of India - Current Account - Vendor Payments	0.00	0.00
1073.04 Central Bank of India - Current Account - Reserve	0.00	0.00
1074		
1074.01 Indian Overseas Bank - Current Account - Operating	0.00	0.00
1074.02 Indian Overseas Bank - Current Account - Payroll	0.00	0.00
1074.03 Indian Overseas Bank - Current Account - Vendor Payments	0.00	0.00
1074.04 Indian Overseas Bank - Current Account - Reserve	0.00	0.00
1075		
1075.01 UCO Bank - Current Account - Operating	0.00	0.00
1075.02 UCO Bank - Current Account - Payroll	0.00	0.00
1075.03 UCO Bank - Current Account - Vendor Payments	0.00	0.00
1075.04 UCO Bank - Current Account - Reserve	0.00	0.00
1076		
1076.01 Bank of Baroda - Current Account - Operating	0.00	0.00
1076.02 Bank of Baroda - Current Account - Payroll	0.00	0.00
1076.03 Bank of Baroda - Current Account - Vendor Payments	0.00	0.00
1076.04 Bank of Baroda - Current Account - Reserve	0.00	0.00
1077		
1077.01 Punjab National Bank - Current Account - Operating	0.00	0.00
1077.02 Punjab National Bank - Current Account - Payroll	0.00	0.00
1077.03 Punjab National Bank - Current Account - Vendor Payments	0.00	0.00
1077.04 Punjab National Bank - Current Account - Reserve	0.00	0.00
1078		
1078.01 Canara Bank - Current Account - Operating	0.00	0.00
1078.02 Canara Bank - Current Account - Payroll	0.00	0.00
1078.03 Canara Bank - Current Account - Vendor Payments	0.00	0.00
1078.04 Canara Bank - Current Account - Reserve	0.00	0.00
1079		
1079.01 Union Bank of India - Current Account - Operating	0.00	0.00
1079.02 Union Bank of India - Current Account - Payroll	0.00	0.00
1079.03 Union Bank of India - Current Account - Vendor Payments	0.00	0.00
1079.04 Union Bank of India - Current Account - Reserve	0.00	0.00
1080		
1080.01 Bank of India - Current Account - Operating	0.00	0.00
1080.02 Bank of India - Current Account - Payroll	0.00	0.00
1080.03 Bank of India - Current Account - Vendor Payments	0.00	0.00
1080.04 Bank of India - Current Account - Reserve	0.00	0.00
1081		
1081.01 Indian Bank - Current Account - Operating	0.00	0.00
1081.02 Indian Bank - Current Account - Payroll	0.00	0.00
1081.03 Indian Bank - Current Account - Vendor Payments	0.00	0.00
1081.04 Indian Bank - Current Account - Reserve	0.00	0.00
1082		
1082.01 Bank of Maharashtra - Current Account - Operating	0.00	0.00
1082.02 Bank of Maharashtra - Current Account - Payroll	0.00	0.00
1082.03 Bank of Maharashtra - Current Account - Vendor Payments	0.00	0.00
1082.04 Bank of Maharashtra - Current Account - Reserve	0.00	0.00
1083		
1083.01 Central Bank of India - Current Account - Operating	0.00	0.00
1083.02 Central Bank of India - Current Account - Payroll	0.00	0.00
1083.03 Central Bank of India - Current Account - Vendor Payments	0.00	0.00
1083.04 Central Bank of India - Current Account - Reserve	0.00	0.00
1084		
1084.01 Indian Overseas Bank - Current Account - Operating	0.00	0.00
1084.02 Indian Overseas Bank - Current Account - Payroll	0.00	0.00
1084.03 Indian Overseas Bank - Current Account - Vendor Payments	0.00	0.00
1084.04 Indian Overseas Bank - Current Account - Reserve	0.00	0.00
1085		
1085.01 UCO Bank - Current Account - Operating	0.00	0.00
1085.02 UCO Bank - Current Account - Payroll	0.00	0.00
1085.03 UCO Bank - Current Account - Vendor Payments	0.00	0.00
1085.04 UCO Bank - Current Account - Reserve	0.00	0.00
1086		
1086.01 Bank of Baroda - Current Account - Operating	0.00	0.00
1086.02 Bank of Baroda - Current Account - Payroll	0.00	0.00
1086.03 Bank of Baroda - Current Account - Vendor Payments	0.00	0.00
1086.04 Bank of Baroda - Current Account - Reserve	0.00	0.00
1087		
1087.01 Punjab National Bank - Current Account - Operating	0.00	0.00
1087.02 Punjab National Bank - Current Account - Payroll	0.00	0.00
1087.03 Punjab National Bank - Current Account - Vendor Payments	0.00	0.00
1087.04 Punjab National Bank - Current Account - Reserve	0.00	0.00
1088		
1088.01 Canara Bank - Current Account - Operating	0.00	0.00
1088.02 Canara Bank - Current Account - Payroll	0.00	0.00
1088.03 Canara Bank - Current Account - Vendor Payments	0.00	0.00
1088.04 Canara Bank - Current Account - Reserve	0.00	0.00
1089		
1089.01 Union Bank of India - Current Account - Operating	0.00	0.00
1089.02 Union Bank of India - Current Account - Payroll	0.00	0.00
1089.03 Union Bank of India - Current Account - Vendor Payments	0.00	0.00
1089.04 Union Bank of India - Current Account - Reserve	0.00	0.00
1090		
1090.01 Bank of India - Current Account - Operating	0.00	0.00
1090.02 Bank of India - Current Account - Payroll	0.00	0.00
1090.03 Bank of India - Current Account - Vendor Payments	0.00	0.00
1090.04 Bank of India - Current Account - Reserve	0.00	0.00
1091		
1091.01 Indian Bank - Current Account - Operating	0.00	0.00
1091.02 Indian Bank - Current Account - Payroll	0.00	0.00
1091.03 Indian Bank - Current Account - Vendor Payments	0.00	0.00
1091.04 Indian Bank - Current Account - Reserve	0.00	0.00
1092		
1092.01 Bank of Maharashtra - Current Account - Operating	0.00	0.00
1092.02 Bank of Maharashtra - Current Account - Payroll	0.00	0.00
1092.03 Bank of Maharashtra - Current Account - Vendor Payments	0.00	0.00
1092.04 Bank of Maharashtra - Current Account - Reserve	0.00	0.00
1093		
1093.01 Central Bank of India - Current Account - Operating	0.00	0.00
1093.02 Central Bank of India - Current Account - Payroll	0.00	0.00
1093.03 Central Bank of India - Current Account - Vendor Payments	0.00	0.00
1093.04 Central Bank of India - Current Account - Reserve	0.00	0.00
1094		
1094.01 Indian Overseas Bank - Current Account - Operating	0.00	0.00
1094.02 Indian Overseas Bank - Current Account - Payroll	0.00	0.00
1094.03 Indian Overseas Bank - Current Account - Vendor Payments	0.00	0.00
1094.04 Indian Overseas Bank - Current Account - Reserve	0.00	0.00
1095		
1095.01 UCO Bank - Current Account - Operating	0.00	0.00
1095.02 UCO Bank - Current Account - Payroll	0.00	0.00
1095.03 UCO Bank - Current Account - Vendor Payments	0.00	0.00
1095.04 UCO Bank - Current Account - Reserve	0.00	0.00
1096		
1096.01 Bank of Baroda - Current Account - Operating	0.00	0.00
1096.02 Bank of Baroda - Current Account - Payroll	0.00	0.00
1096.03 Bank of Baroda - Current Account - Vendor Payments	0.00	0.00
1096.04 Bank of Baroda - Current Account - Reserve	0.00	0.00
1097		
1097.01 Punjab National Bank - Current Account - Operating	0.00	0.00
1097.02 Punjab National Bank - Current Account - Payroll	0.00	0.00
1097.03 Punjab National Bank - Current Account - Vendor Payments	0.00	0.00
1097.04 Punjab National Bank - Current Account - Reserve	0.00	0.00
1098		
1098.01 Canara Bank - Current Account - Operating	0.00	0.00
1098.02 Canara Bank - Current Account - Payroll	0.00	0.00
1098.03 Canara Bank - Current Account - Vendor Payments	0.00	0.00
1098.04 Canara Bank - Current Account - Reserve	0.00	0.00
1099		
1099.01 Union Bank of India - Current Account - Operating	0.00	0.00
1099.02 Union Bank of India - Current Account - Payroll	0.00	0.00
1099.03 Union Bank of India - Current Account - Vendor Payments	0.00	0.00
1099.04 Union Bank of India - Current Account - Reserve	0.00	0.00
1100		
1100.01 Bank of India - Current Account - Operating	0.00	0.00
1100.02 Bank of India - Current Account - Payroll	0.00	0.00
1100.03 Bank of India - Current Account - Vendor Payments	0.00	0.00
1100.04 Bank of India - Current Account - Reserve	0.00	0.00
1101		
1101.01 Indian Bank - Current Account - Operating	0.00	0.00
1101.02 Indian Bank - Current Account - Payroll	0.00	0.00
1101.03 Indian Bank - Current Account - Vendor Payments	0.00	0.00
1101.04 Indian Bank - Current Account - Reserve	0.00	0.00
1102		
1102.01 Bank of Maharashtra - Current Account - Operating	0.00	0.00
1102.02 Bank of Maharashtra - Current Account - Payroll	0.00	0.00
1102.03 Bank of Maharashtra - Current Account - Vendor Payments	0.00	0.00
1102.04 Bank of Maharashtra - Current Account - Reserve	0.00	0.00
1103		
1103.01 Central Bank of India - Current Account - Operating	0.00	0.00
1103.02 Central Bank of India - Current Account - Payroll	0.00	0.00
1103.03 Central Bank of India - Current Account - Vendor Payments	0.00	0.00
1103.04 Central Bank of India - Current Account - Reserve	0.00	0.00
1104		
1104.01 Indian Overseas Bank - Current Account - Operating	0.00	0.00
1104.02 Indian Overseas Bank - Current Account - Payroll	0.00	0.00
1104.03 Indian Overseas Bank - Current Account - Vendor Payments	0.00	0.00
1104.04 Indian Overseas Bank - Current Account - Reserve	0.00	0.00
1105		
1105.01 UCO Bank - Current Account - Operating	0.00	0.00
1105.02 UCO Bank - Current Account - Payroll	0.00	0.00
1105.03 UCO Bank - Current Account - Vendor Payments	0.00	0.00
1105.04 UCO Bank - Current Account - Reserve	0.00	0.00
1106		
1106.01 Bank of Baroda - Current Account - Operating	0.00	0.00
1106.02 Bank of Baroda - Current Account - Payroll	0.00	0.00
1106.03 Bank of Baroda - Current Account - Vendor Payments	0.00	0.00
1106.04 Bank of Baroda - Current Account - Reserve	0.00	0.00
1107		
1107.01 Punjab National Bank - Current Account - Operating	0.00	0.00
1107.02 Punjab National Bank - Current Account - Payroll	0.00	0.00
1107.03 Punjab National Bank - Current Account - Vendor Payments	0.00	0.00
1107.04 Punjab National Bank - Current Account - Reserve	0.00	0.00
1108		
1108.01 Canara Bank - Current Account - Operating	0.00	0.00
1108.02 Canara Bank - Current Account - Payroll	0.00	0.00
1108.03 Canara Bank - Current Account - Vendor Payments	0.00	0.00
1108.04 Canara Bank - Current Account - Reserve	0.00	0.00
1109		
1109.01 Union Bank of India - Current Account - Operating	0.00	0.00
1109.02 Union Bank of India - Current Account - Payroll	0.00	0.00
1109.03 Union Bank of India - Current Account - Vendor Payments	0.00	0.00
1109.04 Union Bank of India - Current Account - Reserve	0.00	0.00
1110		
1110.01 Bank of India - Current Account - Operating	0.00	0.00
1110.02 Bank of India - Current Account - Payroll	0.00	0.00
1110.03 Bank of India - Current Account - Vendor Payments	0.00	0.00
1110.04 Bank of India - Current Account - Reserve	0.00	0.00
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1111.01 Indian Bank - Current Account - Operating	0.00	0.00
1111.02 Indian Bank - Current Account - Payroll	0.00	0.00
1111.03 Indian Bank - Current Account - Vendor Payments	0.00	0.00
1111.04 Indian Bank - Current Account - Reserve	0.00	0.00
1112		
1112.01 Bank of Maharashtra - Current Account - Operating	0.00	0.00
1112.02 Bank of Maharashtra - Current Account - Payroll	0.00	0.00
1112.03 Bank of Maharashtra - Current Account - Vendor Payments	0.00	0.00
1112.04 Bank of Maharashtra - Current Account - Reserve	0.00	0.00
1113

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                                    M37 75 Q
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```

AFRICAN TRUTHS TRUTHS

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Email: ebenezer.obadare@uct.ac.za

Abstract

```
<mml:mfrac>
<mml:mi>Y</mml:mi>
<mml:msqrt>
<mml:mi>Y</mml:mi>
<mml:mi>Y</mml:mi>
<mml:mi>Y</mml:mi
```

```
<?xml version="1.0" encoding="utf-8"?>
<!-- Generator: Adobe Illustrator 19.0.0, SVG Export Plug-In . SVG Version: 6.00 Build 0)  -->
<svg
 xmlns="http://www.w3.org/2000/svg" xmlns:xlink="http://www.w3.org/1999/xlink" version="1.1" id="Layer_1" x="0px" y="0px"
 viewBox="0 0 1000 1000" enable-background="new 0 0 1000 1000" xml:space="preserve">
<metadata>
 <rdf:RDF xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#" xmlns:rdfs="http://www.w3.org/2000/01/rdf-schema#" xmlns:dc="http://purl.org/dc/elements/1.1/" />
</metadata>
<g id="1" />





```
<g id="2" />
<g id="3" />
<g id="4" />
<g id="5" />
<g id="6" />
<g id="7" />
<g id="8" />
<g id="9" />
<g id="10" />
<g id="11" />
<g id="12" />
<g id="13" />
<g id="14" />

Type: Bar Chart
Title: Fig2 Rep20080204-01 Pre-Post Average Rankings
Subtitle: Rank Scale: 1=Strongly Disagree, 2=Disagree, 3=Neutral, 4=Agree, 5=Strongly Agree
Data: Rank Scale:1=Strongly Disagree,2=Disagree,3=Neutral,4=Agree,5=Strongly Agree;Rank Scale:1=Strongly Disagree,2=Disagree,3=Neutral,4=Agree,5=Strongly Agree;Pre-Post Average Rankings;Pre-Post Average Rankings;Pre-Post Average Rankings

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```
<?xml version="1.0" encoding="UTF-8"?>
<!-- Generated by SVG Path Editor by kenzo -->
<svg
  xmlns="http://www.w3.org/2000/svg" xmlns:xlink="http://www.w3.org/1999/xlink"
  viewBox="0.000,0.000,1000.000,1000.000" preserveAspectRatio="xMidYMid meet" xmlns:inkscape="http://www.inkscape.org/namespaces/inkscape" xmlns:sodipodi="http://sodipodi.sourceforge.net/DTD/sodipodi-0.dtd" xmlns:dc="http://purl.org/dc/elements/1.1/" xmlns:cc="http://creativecommons.org/ns#" xmlns:rdf="http://www.w3.org/22-rdf-syntax-ns#"
  >
<!-- Layer: Layer 1 -->
<!-- Path: M 0.000,0.000 L 1000.000,0.000 L 1000.000,1000.000 L 0.000,1000.000 L 0.000,0.000 Z -->
<path d="M 0.000,0.000 L 1000.000,0.000 L 1000.000,1000.000 L 0.000,1000.000 L 0.000,0.000 Z"
  />
```

```
<?xml version="1.0" encoding="UTF-8"?>

<rss version="2.0" xmlns:dc="http://purl.org/dc/elements/1.1/" xmlns:content="http://purl.org/rss/1.0/modules/content/" xmlns:wfw="http://wellformedweb.org/CommentAPI/" xmlns:dc="http://purl.org/dc/elements/1.1/" xmlns:atom="http://www.w3.org/2005/Atom" xmlns:sy="http://purl.org/rss/1.0/modules/syndication/" xmlns:slash="http://purl.org/rss/1.0/modules/slash/" xmlns:media="http://search.yahoo.com/mrss/" xmlns:gr="http://www.google.com/schemas/reader/" xmlns:feedburner="http://rssnamespace.org/feedburner/extension/" xmlns:creativeCommons="http://backend.userland.com/creativeCommonsRssModule" xmlns:geo="http://www.w3.org/2003/01/geo/wgs84_pos#" xmlns:georss="http://www.georss.org/georss" xmlns:gml="http://www.opengis.net/gml" xmlns:fb="http://www.facebook.com/2008/fbml" xmlns:media="http://search.yahoo.com/mrss
```

AVENUE TO TITUS

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```
<?xml version="1.0" encoding="utf-8"?>
<rss version="2.0" xmlns:dc="http://purl.org/dc/elements/1.1/" xmlns:content="http://purl.org/rss/1.0/modules/content/" xmlns:wfw="http://wellformedweb.org/CommentAPI/" xmlns:atom="http://www.w3.org/2005/Atom" xmlns:sy="http://purl.org/rss/1.0/modules/syndication/" xmlns:slash="http://purl.org/rss/1.0/modules/slash/" xmlns:media="http://search.yahoo.com/mrss/" xmlns:dsq="http://disqus.com/disqus-internal/" xmlns:dsq:base="http://disqus.com/api/0.3/" xmlns:opensearch="http://a9.com/-/spec/opensearch/1.1/" xmlns:gd="http://schemas.google.com/g/2005" xmlns:openx="http://openx.com/ns/" xmlns:fb="http://ogp.me/ns/fb#" xmlns:gr="http://www.google.com/ns/1" xmlns:gr:base="http://www.google.com/ns/1/" xmlns:gr:version="1" xmlns:gr:lang="en" xmlns:gr:updated="2011-12-13T18:00:00Z" xmlns:gr:type="feed" xmlns:gr:items="10" xmlns:gr:layout="1" xmlns:gr:tier="free" xmlns:gr:psc="0" xmlns:gr:uid="0" xmlns:gr:geo="0" xmlns:gr:ads="1" xmlns:gr:search="0" xmlns:gr:mobile="0" xmlns:gr:currency="USD" xmlns:gr:viewer="0" xmlns:gr:analytics="UA-18071-1" xmlns:gr:maxpostdate="1323795600000" xmlns:gr:stickies="0" xmlns:gr:mustpay="0" xmlns:gr:cshs="0" xmlns:gr:splt="0" xmlns:gr:maxcmtdate="1323795600000" xmlns:gr:maxcmtdate="1323795600000" xmlns:gr:maxcmtdate="1323795600000
```

```
<?xml version="1.0" encoding="utf-8"?>
<rss version="2.0" xmlns:dc="http://purrit.com/rss/1.0/" />
<channel />
<dc:creator />
<dc:date />
<dc:publisher />
<dc:rights />
<dc:subject />
<dc:title />
<dc:type />
<dc:uri />
<dc:version />
<dc:xmlns_dc_1_0_0_0
```

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Cogent Psychology (2016), 3: 1150281
http://dx.doi.org/10.1080/23311908.2016.1150281

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Boilerplate text
Boilerplate text

Text

Text

Text

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MR-TIGER-MAGNETOSCOPY

Magnetoscopy and Magnetic Imaging Lab

Tiger

Tiger Lab's Technical Y-Z Phantom Design Matrix

Tiger Lab's Technical Y-Z Phantom Design Matrix Tiger Lab's Technical Y-Z Phantom Design Matrix Tiger Lab's Technical Y-Z Phantom Design Matrix

DOI: 10.1177/1847980520927127

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Yuan Tian

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$$\omega = \frac{\sigma_1 \sigma_2 \sigma_3}{\sigma_4}$$

Keywords:

Nonlinear Dynamics

Nonlinear Oscillations and Chaos

Mathematical Modeling

Nonlinear Dynamics

Chaos

Nonlinear Oscillations and Chaos

Mathematical Modeling

DOI: 10.1177/0278364919868017

Jeffrey Ichnowski¹, Michael Danielczuk¹, Jiayuan Gu², Ken Goldberg^{1,3,4}

Abstract

We extend Dex-Net 2.0 to 3D objects using depth and 2D grasps to plan robust 6-DOF grasps with a Cartesian gripper. We train a 3D convolutional neural network on 1 million 3D models from Dex-Net 2.0 to classify 2D grasps on 3D point cloud observations to 80% accuracy on 10 new objects in simulation. We extend the Grasp Quality Convolutional Neural Network (GQ-CNN) to 4-DOF parallel-jaw grasps by adding a 2D gripper pose to the network. We then use the 3D GQ-CNN in a cross-entropy method (CEM) stochastic search to plan 4-DOF grasps on 3D point cloud observations in simulation. We find that the 4-DOF GQ-CNN plans robust grasps with 1 mm and 0.5 mm depth sensor noise in 1.5 s and 2.5 s, respectively, and that the 3D GQ-CNN plans 3-DOF grasps in 0.8 s. We also find that the 4-DOF GQ-CNN plans 3-DOF grasps on 10 new objects in simulation with 1 mm and 0.5 mm depth sensor noise in 1.5 s and 2.5 s, respectively, and that the 3D GQ-CNN plans 3-DOF grasps in 0.8 s. We also find that the 4-DOF GQ-CNN plans 3-DOF grasps on 10 new objects in simulation with 1 mm and 0.5 mm depth sensor noise in 1.5 s and 2.5 s, respectively, and that the 3D GQ-CNN plans 3-DOF grasps in 0.8 s.

Keywords: Grasp Planning, Grasping, Deep Learning, Point Cloud, Robot Vision, Robot Learning, Robot Perception, Robot Manipulation, Robot Grasping

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Keywords: Product Placement, Consumer Research, Affect, Cognition

$$\sigma_{\tau} = \sqrt{\sigma_{\tau}^{2}}$$

Abstract

Product placement is a form of paid product placement in which a product is integrated into the plot of a movie or television show. Although this practice is widely used, it is controversial because it is perceived as a form of deception. Research has shown that viewers can be persuaded by product placements, but it is unclear how they react when they realize that they have been persuaded. This study shows that viewers who realize that they have been persuaded by product placements feel more negative toward the placed product than those who do not realize that they have been persuaded. This effect is driven by a negative affective response to the realization. Implications for the research on deceptive advertising and for practitioners are discussed.

Acknowledgements

The author thanks Dr. Jonas Gabler and Prof. Dr. Dr. Ernst Maier for their helpful comments on this manuscript.

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